

NeuroPLC: A Type Theory of Certifiable Neural Architectures, with Industrial PLC Instantiation

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Abstract

We present NeuroPLC: a **type theory of certifiable neural architectures**, in which well-typed architectures admit design-time correctness certification under fixed-point compilation. Three results anchor the theory. First, the **Compilable Frontier Characterization** (Theorem A) proves that the SVNN conditions are *necessary and sufficient* for dimension-optimal affine certification—a $\Theta(d)$ per-layer gap separating certifiable architectures from standard MLPs, verified through a sharp adversarial construction ($\|W\|_{1,\infty} = \sqrt{d}$, $d = 4\text{--}256$). Second, **Doubleton Arithmetic** (DA) is the **Galois-optimal** abstract domain for C^2 univariate functions (Theorem B)—the first Galois connection defined for neural network activation functions—with **semantically tight** bounds (Theorem C: $f^*(x) = M_2x^2/2$ attains $M_2h^2/8$ at every LUT midpoint, 1,000/1,000 exact). Third, an IEC 61131-3 REAL arithmetic semantics extends the SVNN guarantee to **any** IEC-compliant controller worldwide (Lemma 3). The type system is **sound** (Theorem 2), **compositional** (Theorem 8: algebraic monoid), and deployed with **non-interference** guarantees (Theorem E). Compiler correctness is **type soundness** (Theorem D).

We instantiate this theory in NeuroPLC, the first verified PyTorch-to-IEC 61131-3 SCL compiler. Generated SCL compiles to **0e 0w** in TIA Portal V21 (45.2 KB, 90.4% S7-1200 budget, WCET 22.67 ms, $4.4\times$ scan-cycle margin). The compiler auto-generates a companion safety monitor (Algorithm 3). Five C^2 -BV architectures are validated: B-spline KAN reaches 99.93% CWRU accuracy (91.7% XJTU-SY, 512/512 Z3 preserved); FourierKAN and WaveletKAN each achieve 100% CWRU and 100% XJTU-SY domain-shift accuracy with 512/512 Z3 preserved (E62). In total, 68 experiments spanning 5 architectures and 3 datasets validate the framework.

Index Terms

Type Theory, Neural Architecture Certification, Compilable Frontier, Galois Connection, Optimal Abstract Domain, IEC 61131-3, Kolmogorov-Arnold Networks, Non-Interference, Formal Semantics.

I. INTRODUCTION

This paper introduces a **type theory for certifiable neural architectures**. The central insight is that the question “can a neural network compiler provide design-time correctness guarantees?” is, at its core, a **type-theoretic question**: the answer depends on whether the architecture is *well-typed* under a type system whose rules encode the structural preconditions for certifiable compilation.

We define the **Structurally Verifiable Neural Network** (SVNN) type system. Its types classify operations as purely linear (*linear*), purely element-wise (*elemwise*), or SVNN-composite (*svnn*). Its typing rules enforce two conditions: operation-type closure (Condition 1: linear and element-wise operations are never coupled in a single graph node) and C^2 boundedness (Condition 2: every element-wise function has a computable curvature bound). The **Type Soundness Theorem** (Theorem 2) guarantees that well-typed architectures have computable, finite per-output error bounds under fixed-point compilation—a property that ill-typed architectures (standard MLPs with $\sigma(Wx + b)$) provably lack (Proposition 1).

The SVNN type system is distinguished by three properties:

- (i) **Completeness (Theorem A)**: Within affine abstract interpretation, SVNN well-typedness is *necessary and sufficient* for dimension-optimal certification. A sharp adversarial MLP ($W = (1/\sqrt{d}) \cdot J_d$) achieves $\|W\|_{1,\infty} = \sqrt{d}$, against KAN’s d -independent $\gamma < 1$ (per-layer ratio: $22\times$ at $d=16$, $62\times$ at $d=128$). The Compilable Frontier is a mathematical boundary—not an implementation choice.
- (ii) **Optimality (Theorem 9 + Theorem B + Theorem C)**: DA is the **Galois-optimal** abstract domain for C^2 univariate functions—the first Galois connection for neural network activations—and its bound $M_2h^2/8$ is **semantically tight** (attained by $f^*(x) = M_2x^2/2$ at every LUT midpoint, 1,000/1,000 exact). Any tighter coefficient violates soundness on 32.1% of random C^3 instances.
- (iii) **Compositionality (Theorem 8)**. The SVNN type system forms an **algebraic monoid** under serial composition, enabling modular certification without re-analysis across independently-certified subnetworks.

We give the NeuroPLC IR a **formal operational semantics** and prove **Type Soundness** (Theorem D): compiler correctness (Theorem 1) is equivalent to the statement that well-typed IR programs compile to SCL with bounded semantic deviation. For industrial deployment, we prove a **Non-Interference Theorem** (Theorem E): inserting a NeuroPLC inference block into a certified PLC program preserves all pre-existing safety properties, directly addressing IEC 61508 independence requirements—the first such guarantee for neural-network-on-PLC deployment. An IEC 61131-3 REAL arithmetic semantics (Lemma 3) extends the SVNN guarantee to any IEC-compliant controller worldwide—replacing per-platform validation with a one-time arithmetic-standard-based proof. The compiler provides a parameterized WCET guarantee (Theorem 10: 22.67 ms, $4.4\times$ scan-cycle margin) and auto-generates a companion safety monitor (Algorithm 3).

The industrial instantiation of this type theory targets **programmable logic controllers (PLCs)**—the workhorses of factory automation—which execute IEC 61131-3 Structured Control Language (SCL) in deterministic scan cycles with severe memory constraints (50 KB for Siemens S7-1200 CPU 1211C). Existing ML→PLC tools (RTNNigen [1], MLconverter [2], ICSML [3]) provide only empirical validation; general-purpose compilers cannot target IEC 61131-3.

We instantiate the SVNN type system in NeuroPLC, the first IR-based compiler from PyTorch to Siemens S7-1200/1500 SCL. The compiler is the *proof of concept* for the type theory—not its primary contribution. Its 6-op IR has a formal syntax, operational semantics, and typing rules (§III-Q). Three verification mechanisms compose: (1) one-time Z3 SMT template proofs; (2) compositional per-function certificates (512/512 verified); and (3) DA—the optimal first-order abstract domain for C^2 functions—achieving $2.2\times$ tighter error bounds than interval arithmetic ($8.5\times$ safety factor, up to $3.1\times$ vs. the $\sqrt{d}$ random-weights baseline, §IV-B9). KAN’s B-spline decomposition is *structural necessity*: identically-sized KAN vs. MLP yield 512/512 vs. 0/48 Z3-verifiable components (MLP [28, 32, 16, 4], E41).

Prior ML→PLC tools established that compilation is *possible* but did not address whether it can be *certified*. RTNNigen [1] converts Keras models to IEC 61131-3 Structured Text; MLconverter [2] targets scikit-learn for ABB controllers. Both provide only empirical test-set validation—a correctness model inherited from ML benchmarking that is insufficient for safety-critical PLC deployment. The sole prior KAN-on-PLC demonstration (Corrêa [4], USP 2024) relied on distributed PC-to-PLC communication via Snap7 rather than native SCL compilation, and provided no correctness guarantees. The gap these tools leave is not an implementation gap—it is an *architectural* gap. Our central finding is that no amount of compiler engineering can close it for architectures violating the SVNN conditions (Theorem 5). NeuroPLC is the first compiler that compiles trained PyTorch models to IEC 61131-3 SCL for Siemens PLCs *and* provides a formalism for when such compilation can be certified.

Scope. NeuroPLC compiles the classifier (KAN/MLP) to SCL. Frequency-domain features are assumed pre-extracted via DSP and transmitted over PROFINET/OPC UA. An on-PLC time-domain subset is also compiled (E51). This paper contributes:

- 1) **Theory (Type-Theoretic Foundation):** The SVNN framework constitutes a **type system** for neural computation graphs (§III-Q), whose soundness (Theorem 2) guarantees that well-typed architectures have computable error bounds. The IR is given a formal operational semantics and typing rules; compiler correctness (Theorem 1) is recast as **Type Soundness** (Theorem D). The **Compilable Frontier Characterization** (Theorem A) proves SVNN Conditions 1–2 are *necessary and sufficient* for dimension-optimal affine certification ($O(d \cdot r^2)$ per layer), while ill-typed architectures degrade to $\Omega(d^2 \cdot r^2)$ —a $\Theta(d)$ gap verified on networks up to $d = 128$ (MATLAB). **DA Optimality** (Theorem 9) proves Doubleton Arithmetic is the tightest sound first-order abstract domain for C^2 functions (32.1% of C^3 instances reject any tighter linear coefficient). **SVNN Compositionality** (Theorem 8) proves the SVNN class forms an algebraic monoid, enabling modular certification.
- 2) **Theory (Abstract Interpretation Foundation):** **DA Galois Connection** (Theorem B) proves that Doubleton Arithmetic forms a canonical instance of the Cousot abstract interpretation framework with a defined Galois connection (α, γ) , making NeuroPLC the first neural network compiler to be formally grounded in abstract interpretation theory. **DA Tightness** (Theorem C) proves the DA bound is *semantically tight*—the de Boor bound $M_2 \cdot h^2/8$ is *attained* by $f^*(x) = M_2/2 \cdot x^2$ at every LUT interval midpoint (1,000/1,000 exact, MATLAB-verified). **Universal IEC 61131-3 Guarantee** (Lemma 3) extends the SVNN certification from Siemens TIA Portal to *any* IEC 61131-3:2025 compliant controller worldwide, using the standard’s IEEE 754 normative reference.
- 3) **Theory (Deployment Safety):** The **Non-Interference Theorem** (Theorem E) proves that inserting a NeuroPLC inference block into a certified PLC program preserves all pre-existing safety properties—the first such guarantee for neural-network-on-PLC deployment, directly addressing IEC 61508 independence requirements. The **WCET Theorem** (Theorem 10) provides parameterized worst-case execution time bounds (22.67 ms for KAN [28, 16, 4], $4.4\times$ scan-cycle margin).
- 4) **System (Compiler):** NeuroPLC, the first PyTorch-to-Siemens SCL compiler, uses a 6-op IR matching SVNN decomposition of KAN (§IV-B1). Generated SCL compiles with **0e 0w** in TIA Portal V21 on S7-1200 and S7-1500 (§IV-D).
- 5) **Algorithm (Verification & Optimization):** Three novel algorithms: (i) **sign-structural affine arithmetic (DA)**, achieving a $2.2\times$ tighter error bound than interval arithmetic ($3.1\times$ vs. the $\sqrt{d}$ random-weights baseline) by preserving weight-matrix sign structure (§IV-B9); (ii) **Segment-Aware de Boor Bounds**, exploiting the piecewise-linear structure of cubic B-spline second derivatives for $6.0\times$ per-segment tightening (composing with DA for a combined safety factor of $\sim 11.9\times$); and (iii) **Adaptive Mixed-Precision LUT Allocation**, a greedy max-heap algorithm reducing worst-case LUT error by 71.6% at equal storage, with **Theorem 3** proving its minimax optimality via an exchange argument (§IV-B11).
- 6) **Theory (Completeness):** Beyond sufficiency, we establish **computational necessity** (Theorem 5, §III-G): violating Condition 1 makes tight bound computation NP-hard, proving the SVNN conditions cannot be relaxed. Condition 3 yields a depth-improving **generalization bound** $\Delta_{\text{gen}} \leq O(\gamma^L/\sqrt{n})$ (Theorem 6, §III-H). The framework is extended to **ChebyKAN** (Chebyshev polynomial KANs): 496/512 Z3-verifiable via polynomial NRA, 100.0% CWRU accuracy (Proposition 2). **E56** (3-layer KAN: 608/608 Z3, 99.60% acc) and **E57** (NeuroPLC DA $85\times$ tighter than CROWN-IBP) provide empirical depth-scaling and verifier-comparison evidence. **DA segment-exactness** (Proposition 3, §III-J) proves Doubleton Arithmetic achieves segment-exactness on affine B-spline segments, explaining the mechanism behind DA’s tightening over IA. **Theorem 7** establishes the sufficient condition for Z3 NRA verifiability ($M_2 \cdot h^2/8 \leq C(k)$), providing the theoretical basis for the B-spline 512/512 and Chebyshev 496/512 verification rates.
- 7) **Theory (Correctness):** **Theorem 1** (§IV) guarantees by structural induction on the IR graph that generated SCL preserves

PyTorch semantics to within a computable, architecture-aware error bound for all 6 IR operation types. **Theorem 4** generalizes the guarantee to L -layer KANs via martingale concentration, proving error grows only linearly with depth under the random-sign model.

- 8) **Theory (Algebraic Completeness): Theorem 8 (§III)** proves the SVNN class is closed under sequential composition—forming an algebraic monoid—enabling modular certification where independently-certified subnetworks compose their guarantees without re-analysis. **Theorem 9** proves DA is the *tightest possible* sound first-order abstract domain for C^2 univariate functions: any bound tighter than DA’s violates soundness on 62.4% of random C^3 instances (MATLAB symbolic verification). **Proposition 4** formalizes the **Operation Separation Principle**—the C^2 -BV architecture family (B-spline, Fourier, Wavelet, RBF, Chebyshev) all admit SVNN guarantees; standard MLPs are structurally excluded.
- 9) **Theory (Deployment Completeness): Theorem 10 (§IV-G)** provides the worst-case execution time guarantee: for any SVNN network N with E edges and N_{LUT} LUT points on Siemens S7-1200, $\text{WCET}(N) \leq E \cdot (C_{\text{LUT}} + C_{\text{matmul}}) + C_{\text{softmax}}$. For KAN [28, 16, 4] with $N_{\text{LUT}} = 15$: $\text{WCET} = 22.67$ ms ($4.4\times$ margin within 100 ms scan cycle, 22.7% utilization).
- 10) **Algorithm (Safety Monitor): Algorithm 3 (§IV-H)**—the compiler automatically generates a companion SCL safety monitor implementing domain violation detection, confidence fallback, and output range bounds. The monitor adds $\leq 5\%$ code overhead and ~ 66 μs execution time, providing a run-time safety net for the compiled inference.
- 11) **System (Multi-Architecture Validation): E60–E61** extend SVNN validation to FourierKAN and WaveletKAN architectures: FourierKAN [28, 16, 4] (6 harmonics, $\omega = 0.4$, 6,676 params) achieves 100.0% CWRU accuracy with 512/512 edges $M_2 \cdot h^2/8 \leq 0.063 < 0.182$ (100% Z3-equivalent, $2.9\times$ margin); WaveletKAN [28, 16, 4] (8 Mexican hat scales, M2-regularized, $\sim 4,600$ params) achieves 100.0% CWRU accuracy with 512/512 edges verified (100% Z3-equivalent, $5.6\times$ margin), confirming the C^2 -BV generalization (Proposition 5).

We demonstrate the pipeline through a bearing fault diagnosis case study: a 6,148-parameter KAN student distilled via VRM-KD [5] from a 48K-parameter 1D-CNN teacher achieves 99.93% CWRU test accuracy with $7.9\times$ compression. Fine-tuning on XJTU-SY run-to-failure data (natural bearing degradation, 2020) reaches 91.7% accuracy with 512/512 Z3 verification preserved post-adaptation. FourierKAN and WaveletKAN achieve 100% CWRU and 100% XJTU-SY domain-shift accuracy (E62), with 512/512 Z3-equivalent verification preserved, confirming the C^2 -BV family’s cross-architecture fine-tuning stability. In total, 68 experiments (E1–E58 + E60–E62 + V1–V7) spanning compression, generalization, correctness bounds, scalability, cross-dataset transfer, cross-architecture verification (FourierKAN E60, WaveletKAN E61), and domain-shift SVNN preservation (E62) validate the compiler. We further discuss the SVNN framework’s relationship to emerging AI functional safety standards (ISO/IEC TS 22440) and formal verification tools for IEC 61131-3 (ESBMC-PLC+), positioning NeuroPLC within the broader industrial verification ecosystem.

II. RELATED WORK

A. Deep Learning for Bearing Fault Diagnosis

Three lines of work have pushed CWRU fault diagnosis accuracy above 99%. WDCNN [6] demonstrated that wide first-layer convolutional kernels capture the long-period fault signatures that narrow kernels miss. Zhang et al. [7] extended residual architectures to 1-D signals, confirming that depth helps when the receptive field is sufficiently wide. Together, these works established a near-solved benchmark. But Rauber et al. [8] showed that many reported near-100% results contain train/test leakage from overlapping recording-level windows—the problem is better solved than the literature reports, yet the deployment problem is worse: all methods assume GPU-equipped industrial PCs. **The gap:** 99.9% accuracy on a lab GPU contributes nothing if the inference cannot run on the PLC that governs the monitored machinery.

B. Kolmogorov-Arnold Networks

Liu et al. [9] introduced KANs by replacing MLPs’ fixed activations with learnable univariate B-splines—a structural change whose implications for compiler design have been underestimated. **What the literature has established:** Rigas et al. [10], CNN-1D-KAN [11], and AdaptPolyKAN [12] independently confirmed that KANs match or exceed MLP accuracy on bearing fault diagnosis while using far fewer parameters. **What the literature has not addressed:** none of these works consider whether the B-spline decomposition that makes KANs parameter-efficient also makes them *verifiable at compile time*—the question this paper answers.

KAN hardware deployment. Four independent lines of work demonstrate that KAN’s univariate B-spline structure is uniquely amenable to LUT-based hardware evaluation—a property absent from MLPs. On FPGA, KANÉLÉ [13] (ISFPGA 2026 Best Paper) achieves $2,700\times$ speedup via LUT-based evaluation with quantized B-splines. On microcontrollers, LUT-KAN [14] replaces Cox-de Boor recursion with precomputed LUT operations ($7\text{--}25\times$ speedup, as little as 68 bytes/edge); KAN-SoC [15] achieved $10\times$ memory reduction vs. MLP (693 bytes vs. 6,807 bytes) for EV battery estimation. On PLCs, Corrêa [4] (USP 2024) provided the sole prior KAN-on-PLC demonstration: a KAN [7, 5, 6] deployed via Snap7 distributed execution (PC computes inference, PLC receives results over Ethernet), achieving 88.5% accuracy on a bearing classification task.

These four works collectively establish that KAN's B-spline structure compiles to LUT primitives across hardware platforms. But *none* answer the question this paper asks: can the compilation itself be *certified*? KANELE and LUT-KAN validate the LUT paradigm empirically; we formalize *why* it works (Theorem 2) and *when* it fails (Theorem 5, Proposition 1). Corrêa demonstrates KAN-on-PLC feasibility but provides no native SCL compilation, no correctness guarantee, and no compiler toolchain. NeuroPLC is the first to provide all three.

KAN verification. Schwartz et al. [16] proposed PWA abstraction with MILP encoding for KAN verification—operating at SVNN Level 3 (exact certification within PWA abstraction error). Tankman [17] proved layer-wise Lipschitz $P(\text{KAN}) \leq 1$ as a structural property. KAN4CBC [18] demonstrated SMT-based safe controller synthesis. GloroKAN [19] and PolyKAN [20] provide complementary robustness and compression. These are synergistic with NeuroPLC: Schwartz et al.'s MILP verification can be applied to the KAN compiled by NeuroPLC for SIL 3+ certification, while NeuroPLC's Level 2 arithmetic bounds provide fast, no-solver-required pre-certification for CI/CD integration. Tankman's Lipschitz result provides independent theoretical support for Condition 3 (contractivity).

a) *FPGA Deployment.*: KANELE [13] (ISFPGA 2026 Best Paper) maps KAN inference to LUT-based FPGA evaluation with $2,700\times$ speedup over prior FPGA KAN implementations. KANELE's LUT paradigm independently validates the same design principle NeuroPLC applies to PLCs—B-spline activations are inherently discretizable across hardware platforms. No prior work compiles KANs to IEC 61131-3 SCL for PLC execution.

C. Knowledge Distillation

Hinton et al. [21] introduced temperature-scaled knowledge distillation as a mechanism for transferring dark knowledge from ensembles to compact models. VRM [5] (ICCV 2025) revitalized relation-based distillation through virtual relation matching with dynamic affinity graph pruning—directly applicable to KAN students because KAN's univariate activation structure produces naturally sparse inter-sample relations. Ren et al. [22] applied multi-stage pruning-distillation for industrial fault diagnosis, establishing the practical viability of compressed models for bearing monitoring. We contribute the first VRM-KD to a KAN student: the distillation is the *training mechanism*, not the contribution; the contribution is that KAN's $7.9\times$ compression does not compromise verifiability.

D. Neural Network Compilation for Industrial Controllers

RTNNigen [1] (IECON 2024) converts Keras models to IEC 61131-3 ST and demonstrated real-time inference on Beckhoff PLCs—the strongest evidence to date that ML inference *can* run natively on PLCs. MLconverter [2] extends this paradigm to ABB controllers from scikit-learn. ICSML [3] provides native IEC 61131-3 ML components. The shared limitation is the correctness model: all three validate through test-set accuracy, inheriting the empirical validation paradigm of ML benchmarking. For PLCs governing safety-critical machinery, this is the wrong correctness model—it provides no guarantee on unseen inputs, cannot detect floating-point drift, and offers no evidence auditable by a safety assessor. NeuroPLC does not merely add another target platform to this lineage; it changes the correctness model from empirical to *mathematical*, by restricting the compiler's input to architectures that admit the SVNN structural induction (Theorem 2). No automated compiler translates trained PyTorch networks to Siemens SCL with design-time correctness certification.

E. Formal Verification of IEC 61131-3 Programs

ESBMC-PLC+ [23] provides the first open-source unified verification framework for IEC 61131-3 (LD, ST, SCL) through SMT-based BMC with k-induction. PLCreX [24] translates ST/FBD to synchronous models; PyLC+ [25] compiles PLCopen XML to verifiable Python. These verify *logical* properties (mutual exclusion, invariants) orthogonal to NeuroPLC's *numerical* correctness guarantees. The two are composable: NeuroPLC-generated SCL can serve as input to these tools (§VI-B), enabled by the deterministic, human-readable SCL structure (§IV-D).

F. Neural Network Verification and Bounding

A mature literature on neural-network verification provides complementary tools that operate at different points on the soundness–cost spectrum. **Linear-relaxation methods**—CROWN and DeepPoly [26], [27], generalized by α - β -CROWN [28] to branch-and-bound—propagate linear upper/lower bounds through each layer and achieve tight certification on ReLU networks, but the LP size scales with the network, requiring an external solver at verification time. **Complete SMT-based verifiers**—Reluplex [29] and Marabou [30]—offer exact certification for piecewise-linear (ReLU) networks via case-splitting on activation patterns, at exponential cost in the number of ReLU units. **Abstract interpretation** [31]–[33] provides sound, scalable over-approximation through abstract domains—the foundational framework (Cousot & Cousot, 1977) defines a sound abstraction as one whose concretization contains all reachable concrete states. In neural network verification, abstract domains such as intervals, zonotopes, and polyhedra have been applied to ReLU networks [27], [32], [33], but these generic domains are designed for the activation pattern $x \mapsto \max(0, Wx + b)$ and do not exploit the piecewise-polynomial structure of B-spline

activations. Consequently, they produce coarse over-approximations when applied to KAN architectures, whose intermediate representations are not activation-sparse but *function-sparse* (each activation is a full polynomial segment).

NeuroPLC’s SVNN framework occupies a distinct point that can be understood within the abstract interpretation paradigm: Doubleton Arithmetic (DA, §III-J) constitutes a *specialized abstract domain for piecewise-polynomial univariate functions*. On a single B-spline knot segment—the atomic computation unit in KAN inference—DA propagation is *exact* for affine segments (Proposition 3.i, verified on 1,000/1,000 random segments) and incurs only $O(M_2 \cdot r^2)$ overestimation for higher-degree segments (Proposition 3.ii). This is tighter than any generic interval or zonotope domain can achieve on the same function class, because those domains are not *activation-aware*: they treat each B-spline evaluation as a black-box nonlinearity rather than a known polynomial with computable curvature bound M_2 .

Key differences from prior abstract-interpretation tools: (1) SVNN bounds are *parameter-computable* at design time—the compiler computes the bound from the B-spline control points and LUT resolution alone, without invoking an LP solver or SMT solver at deployment; (2) Condition 1 (operation-type closure) is a *structural* requirement that enables the exact affine propagation: DA can separate the linear term $\phi'(x_0) \cdot r\epsilon$ from the higher-order remainder R_2 only because the architecture exposes pure linear and pure element-wise operations (Theorem 2, Step 3); (3) the resulting guarantee certifies the *translation into SCL*, not the model’s robustness—it is a *compilation* correctness certificate, not a model verification certificate. These are complementary: CROWN/DeepPoly verify the original model; SVNN certifies the *compilation* of that model. The most closely related approach is the per-function Z3 certificate bundle (§VI-B), which operates at the sub-module level and composes into an end-to-end machine-checkable guarantee.

Competitive Positioning. The tools surveyed above represent three distinct approaches to ML-on-PLC deployment, none of which provides design-time correctness certificates. *Domain-specific compilers* (RTNNigen, MLconverter, ICSML) target narrow model classes for specific PLC vendors but offer only empirical validation. *General-purpose inference engines* (TFLite, ONNX Runtime, TVM) target performance on GPUs and MCUs; they cannot generate IEC 61131-3 and their runtime libraries alone exceed PLC memory budgets by factors of 10^0 – 10^2 . *Formal verification tools* (CROWN/DeepPoly, Marabou, Reluplex) verify the *original* model’s robustness but do not compile it to PLC code or certify the *compilation* step. NeuroPLC occupies a distinct point: it is not a verification tool, a general-purpose compiler, or a domain-specific converter—it is a **correctness-preserving compiler** whose design is driven by the architectural insight (SVNN) that compilability is a property of the model architecture, not the compiler engineering. This reframing—from “how do we verify an arbitrary architecture?” to “which architectures are inherently verifiable?”—is the paper’s primary contribution, with implications beyond PLC deployment to any safety-critical embedded ML target.

III. STRUCTURAL VERIFIABILITY OF NEURAL NETWORKS

We now formalize a central observation that emerged from the compiler design process: whether a neural network compiler can provide design-time correctness guarantees is not solely a property of the compiler—it is fundamentally determined by the *architecture* being compiled. This section defines the class of architectures for which correctness-preserving compilation is achievable, proves that KAN belongs to this class, and demonstrates that standard MLPs do not.

Remark 1 (The Compilable Frontier). *The SVNN conditions define a **compilable frontier**—a principled boundary separating neural architectures whose deployed behavior can be certified at design time from those that require runtime testing. Architectures satisfying Conditions 1–2 lie on the interior (certifiable side): their decomposition into pure linear and element-wise operations enables the structural induction that Theorem 2’s proof relies on. Architectures violating any condition lie on the exterior (empirical-only side): they can still be compiled and deployed, but their correctness guarantees are limited to test-set validation. Proposition 1 (below) establishes that standard MLPs with SiLU, Sigmoid, or Tanh activations—the dominant architectures in industrial ML—lie on the exterior. This frontier is not an implementation artifact; it reflects a fundamental property of which function classes admit closed-form, compiler-computable error propagation.*

A. The Verification Challenge for PLC-Deployed Networks

Industrial PLCs govern safety-critical machinery: conveyor sorting systems, motor drives, and process control loops where incorrect outputs can cause equipment damage or production line stoppage. Deploying neural networks on such controllers therefore demands correctness guarantees that go beyond empirical test-set accuracy.

Existing ML→PLC tools (RTNNigen [1], MLconverter [2], ICSML [3]) provide *no* mathematical guarantee that the generated PLC code preserves the trained model’s behavior. Their correctness model is purely empirical: train, convert, and test on a held-out set. For industrial deployment—where inputs may drift, sensors may degrade, and the PLC’s REAL arithmetic differs from PyTorch’s float32—this is insufficient.

We identify three levels of deployment verification, in increasing order of strength:

- 1) **Level 1—Runtime Testing (Weakest):** Evaluate the deployed model on a test set and report accuracy. This is the status quo (RTNNigen, MLconverter). It provides no guarantee on unseen inputs and cannot detect systematic floating-point discrepancies.

- 2) **Level 2—Design-Time Error Bounds (SVNN):** Compute, at *compile time* and without executing on hardware, a provable per-output error bound ε_i such that for all valid inputs x , the deployed model's output differs from the trained model's output by at most ε_i . This is the guarantee that NeuroPLC provides (Theorem 1).
- 3) **Level 3—Mechanized Formal Verification (Strongest):** A machine-checked proof (e.g., in Coq, Isabelle, or Lean) that the compiler preserves semantics down to the IEEE 754 bit-level. This has been achieved for select neural network kernels in TorchLean [34] but is currently intractable for full compiler pipelines.

Level 2—design-time error bounds—is the appropriate level for industrial deployment. The question is: *which architectures admit such bounds, and why?*

B. Definition: Structurally Verifiable Neural Network (SVNN)

Definition 1 (Structurally Verifiable Neural Network). A neural network architecture $\mathcal{A}$ is **structurally verifiable** (SVNN) with respect to a compiler C if for any instance $M \in \mathcal{A}$, the compiler can compute, at design time and without executing M on target hardware, a per-output error bound $\varepsilon_i(M, X)$ such that:

$$\sup_{x \in X} \|M(x)_i - C(M)(x)_i\|_\infty \leq \varepsilon_i(M, X) \quad \forall i \in \{1, \dots, d_{out}\} \quad (1)$$

where $X \subset \mathbb{R}^{d_{in}}$ is the validated input domain and $C(M)$ denotes the compiler's generated code executing on the target platform's arithmetic (here, IEC 61131-3 REAL $\equiv$ IEEE 754 binary32).

This definition captures the essential requirement: the compiler must provide an *a priori* guarantee—before the model ever runs on the PLC—that the deployed code will not silently diverge from the trained model. The error bound ε_i may depend on the model's parameters, the compiler's approximation choices (e.g., LUT resolution), and the input domain, but it must be computable from these quantities alone.

C. Sufficient Conditions for SVNN

We now establish the architectural conditions that enable SVNN compilation. The conditions formalize why some architectures admit tight, composable error bounds while others do not.

Theorem 2 (SVNN Sufficient Conditions). Let $\mathcal{A}$ be a feedforward neural network architecture. If $\mathcal{A}$ satisfies Conditions 1 and 2 below, then $\mathcal{A}$ is structurally verifiable—for any instance $M \in \mathcal{A}$ of finite depth L , a compiler adhering to the SVNN compilation discipline can compute, from the model parameters alone, a finite per-output error bound. If $\mathcal{A}$ additionally satisfies the contractivity refinement (Condition 3), this bound admits the closed, *depth-independent* geometric-series form $O(M_{\max} \cdot h^2 \cdot d_{\max} / (1 - \gamma))$, where $M_{\max}$ is the maximum second-derivative bound, h is the LUT grid spacing, $d_{\max}$ the maximum activation count per layer, and $\gamma < 1$ the contractivity constant; absent Condition 3, the bound remains finite for every fixed L but may grow with depth.

The conditions are designed to *discriminate*: a standard MLP layer $\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b})$ composes an affine transform and a pointwise activation in one coupled operation, violating Condition 1. This prevents per-operation error induction (the proof's structural induction requires each node to be either purely linear or purely element-wise). MLPs with SiLU or GELU additionally violate the Z3-verifiability of Condition 2 (the transcendental $\exp$ is undecidable in Z3's NRA theory); MLPs with ReLU lack a C^2 curvature parameter. KAN layers, in contrast, decompose naturally at the IR level into separate MatMul, BsplineLUT, StandardAct, and Add nodes—each satisfies one of the two pure types. Conditions 1–2 thus characterize exactly the class that admits the induction, and KAN belongs to it; standard MLPs do not.

- 1) **Operation-Type Closure (Condition 1).** The architecture can be decomposed into a directed acyclic graph where every node is either (a) a linear transformation (matrix multiplication, vector addition) or (b) an element-wise univariate function. No node may mix these operation types. This decomposition is provided by the compiler's IR (§IV-B1); it guarantees that the per-node error induction in the proof below never encounters a coupled linear+nonlinear operation.
- 2) **Univariate Curvature Bound (Condition 2).** Every element-wise univariate function $f : \mathbb{R} \rightarrow \mathbb{R}$ is C^2 -continuous on the validated input domain X , and its second derivative satisfies $\sup_{x \in X} |f''(x)| \leq M_f < \infty$. The bound M_f must be computable from the function's parameters alone. This enables the de Boor LUT error bound $\varepsilon_f = M_f h^2 / 8$ —a *compiler-controllable* error: by increasing the LUT resolution (reducing h), the per-function error decreases quadratically. Condition 2 thus provides the compiler with a tunable guarantee knob that MLPs with ReLU (no C^2) or SiLU/GELU (transcendental, Z3-undecidable) lack.

Conditions 1 and 2 are the *sufficient conditions* for structural verifiability. The following refinement governs how the bound scales, not whether it exists:

- 3) **Contractive Composition (Condition 3, refinement).** The product of each element-wise layer's Lipschitz constant $L_f^{(\ell)} = \sup_x |f'(x)|$ with the subsequent linear layer's row-sum norm satisfies $L_f^{(\ell)} \cdot \|W^{(\ell+1)}\|_{1,\infty} \leq \gamma < 1$. When this holds, the bound converges to a depth-independent constant $O(1/(1 - \gamma))$; when it does not, the bound remains finite for the specific

deployment depth but grows as $O(\gamma^L)$. Condition 3 upgrades a per-depth certificate to a depth-independent one; it is *not* required for SVNN membership.

Proof. The proof proceeds in four steps.

Step 1 (Linear Layer Exactness). Let $y = Wx + b$ be a linear transformation with input x that carries error δ_x (i.e., the compiled input is $\tilde{x} = x + \delta_x$). The compiled output is $\tilde{y} = W\tilde{x} + b = Wx + b + W\delta_x$. Hence $\|\tilde{y} - y\|_\infty = \|W\delta_x\|_\infty \leq \|W\|_{1,\infty} \cdot \|\delta_x\|_\infty$. A linear layer introduces *no new error sources*—it only amplifies or attenuates existing error. Under sign-structural affine arithmetic (§IV-B9), the amplification factor exploits weight-matrix sign structure, yielding $\|W\delta_x\|_\infty \leq \|W^+\delta_x^+ + W^-\delta_x^-\|_\infty$, which is strictly tighter than the interval-arithmetic bound $\|W\|_{1,\infty} \cdot \|\delta_x\|_\infty$ when W contains both positive and negative entries.

Step 2 (Element-Wise Boundedness). Consider an element-wise univariate function f applied to input x . The compiled version uses a piecewise-linear LUT approximation $\tilde{f}$ on a grid with spacing h . By the de Boor–Cox recurrence for B-spline approximation error [35], for any C^2 function:

$$\|f - \tilde{f}\|_\infty \leq \frac{M_f \cdot h^2}{8} \quad (2)$$

where $M_f = \sup_x |f''(x)|$. Condition 2 guarantees that M_f is finite and computable. The LUT thus introduces fresh error at most $\varepsilon_f = M_f h^2 / 8$ per function evaluation.

If the input carries error δ_x , the mean value theorem gives:

$$|f(x + \delta_x) - \tilde{f}(x + \delta_x)| \leq |f(x + \delta_x) - f(x)| + |f(x) - \tilde{f}(x + \delta_x)| \leq L_f \cdot |\delta_x| + \varepsilon_f \quad (3)$$

where $L_f = \sup_x |f'(x)|$ is the Lipschitz constant of f .

Step 3 (Finite-Depth Composition—Conditions 1 and 2 suffice). We now compose the analysis across L layers by structural induction on the layer index ℓ . Let $\Delta^{(\ell)}$ denote the maximum per-output error after layer ℓ , with $\Delta^{(0)} = 0$ (input is exact).

For a KAN layer (element-wise B-spline activation followed by a linear combination via weight matrix):

$$\Delta^{(\ell)} \leq L_f^{(\ell)} \cdot \|W^{(\ell)}\|_{1,\infty} \cdot \Delta^{(\ell-1)} + \varepsilon_f^{(\ell)} \cdot d_{\ell-1} \quad (4)$$

The first term captures error amplification from the previous layer; the second term captures fresh approximation error from $d_{\ell-1}$ parallel element-wise function evaluations (one per input dimension).

Unrolling the recurrence across L layers yields the total bound:

$$\Delta^{(L)} \leq \sum_{\ell=1}^L \varepsilon_f^{(\ell)} \cdot d_{\ell-1} \cdot \prod_{j=\ell+1}^L (L_f^{(j)} \cdot \|W^{(j)}\|_{1,\infty}) \quad (5)$$

The critical role of Condition 1 is now apparent: it guarantees that every node encountered by the induction is of a known type with a closed-form error rule (Step 1 for linear nodes, Step 2 for element-wise nodes). The recurrence Eq. 4 applies *only* because each node is provably pure. In a standard MLP layer $\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b})$, the affine transform and pointwise activation are coupled in a single operation—there is no intermediate variable whose error can be separately bounded, so the structural induction cannot be initiated. This is why Sigmoid/Tanh MLPs, despite having bounded M_2 (they satisfy Condition 2), are not SVNN: they fail Condition 1.

Under Conditions 1 and 2, every term in Eq. 5 is finite and computable: $\varepsilon_f^{(\ell)} = M_f^{(\ell)} h^2 / 8$ from Condition 2, $L_f^{(j)}$ finite on the compact domain X , and $\|W^{(j)}\|_{1,\infty}$ finite by construction. Eq. 5 is therefore a finite sum of finite terms, directly yielding a computable design-time bound $\Delta^{(L)} < \infty$ for any fixed depth L —the existence guarantee of Definition 1. The bound is *compiler-controllable*: increasing the LUT resolution (reducing h) monotonically decreases ε_f and hence $\Delta^{(L)}$, giving the compiler a tunable knob to meet any target safety requirement.

Step 3b (Depth-Uniform Refinement—Condition 3). Condition 3 does not change *whether* a bound exists; it controls *how the bound scales with depth*. When $L_f^{(j)} \cdot \|W^{(j)}\|_{1,\infty} \leq \gamma < 1$, the product terms in Eq. 5 decay geometrically, and the finite sum is majorized by the convergent series

$$\Delta^{(L)} \leq \frac{\max_{\ell} (\varepsilon_f^{(\ell)} \cdot d_{\ell-1})}{1 - \gamma} = O\left(\frac{M_{\max} \cdot h^2 \cdot d_{\max}}{1 - \gamma}\right) \quad (6)$$

where $M_{\max} = \max_f M_f$ and $d_{\max} = \max_{\ell} d_{\ell-1}$ is the maximum number of element-wise activation functions per layer. The bound follows by substituting $\varepsilon_f^{(\ell)} = M_f^{(\ell)} h^2 / 8$ (Eq. 2) into the geometric series, yielding a per-layer factor of $M_{\max} h^2 d_{\max}$ multiplied by the geometric decay $1/(1 - \gamma)$. This form is *depth-independent*: when $\gamma < 1$, the bound converges to a constant as $L \rightarrow \infty$, admitting deployment on arbitrarily deep networks. When Condition 3 does not hold ($\gamma \geq 1$), Eq. 5 still evaluates to a finite number for the fixed deployment depth, but the worst-case majorant grows as $O(M_{\max} \cdot h^2 \cdot \gamma^L)$ —still computable, only looser with depth. Thus Condition 3 upgrades a per-depth guarantee into a depth-uniform one; it is a refinement, not a membership requirement.

Step 4 (Tightness and Practical Considerations). The $O(L \cdot M_{\max} \cdot h^2 \cdot d_{\max})$ bound is a *worst-case* guarantee. In practice, three factors make the bound substantially tighter for the KAN architecture specifically:

- 1) **Segment-Aware M_2 (§IV-B10):** Rather than using the global $\sup |f''|$, we compute per-segment $M_2^{(k)}$ bounds, tightening the LUT error estimate by $6.0\times$ on average for KAN B-splines.
- 2) **sign-structural affine arithmetic (§IV-B9):** Linear layers propagate error via DA rather than IA, exploiting weight-matrix sign structure for an additional $3.1\times$ tightening.
- 3) **Adaptive LUT Allocation (§IV-B11):** Non-uniform knot placement allocates more LUT points to high-curvature regions, reducing the effective h for the dominant error sources by up to 71.6%.

Together, these refinements transform the asymptotic $O(L \cdot M_{\max} \cdot h^2 \cdot d_{\max})$ bound into a practically useful guarantee. For the trained [28, 16, 4] KAN, the combined segment-aware DA safety factor is approximately $11.9\times$ above the minimum required for classification correctness (§IV, Table XXII).

Theorem 8 (SVNN Compositional Closure). The SVNN class is closed under sequential composition: if $N_1 \in \text{SVNN}(\varepsilon_1, L_1)$ and $N_2 \in \text{SVNN}(\varepsilon_2, L_2)$, then $N = N_2 \circ N_1 \in \text{SVNN}(\varepsilon, L)$ with:

$$\varepsilon = \varepsilon_2 + L_2 \cdot \varepsilon_1, \quad L = L_2 \cdot L_1 \quad (7)$$

Proof. Let x be any valid input. By SVNN of N_1 : $C(N_1)(x) = N_1(x) + \delta_1$ with $\|\delta_1\|_\infty \leq \varepsilon_1$. By SVNN of N_2 , $C(N_2)(y) = N_2(y) + \delta_2$ with $\|\delta_2\|_\infty \leq \varepsilon_2$. Composing:

$$\begin{aligned} C(N)(x) &= C(N_2)(C(N_1)(x)) = C(N_2)(N_1(x) + \delta_1) \\ &= N_2(N_1(x) + \delta_1) + \delta_2 \\ &= N_2(N_1(x)) + [N_2(N_1(x) + \delta_1) - N_2(N_1(x))] + \delta_2 \end{aligned} \quad (8)$$

The perturbation term is bounded by the Lipschitz property: $\|N_2(N_1(x) + \delta_1) - N_2(N_1(x))\|_\infty \leq L_2 \cdot \varepsilon_1$. Hence $\|C(N)(x) - (N_2 \circ N_1)(x)\|_\infty \leq \varepsilon_2 + L_2 \cdot \varepsilon_1$, which is finite and computable since $\varepsilon_1, \varepsilon_2, L_2$ are. $\square$

Implications. Theorem 8 establishes that the SVNN class forms a **monoid** under composition: closure (Theorem 8), associativity (function composition), and identity ($I(x) = x$ is SVNN with $\varepsilon = 0, L = 1$). This enables **modular certification**: feature extractor KANs and classifier KANs can be independently certified, and their sequential composition inherits the combined guarantee *without re-analysis*. For KAN architectures, Theorem 8 applies directly to any stacking of KAN layers and to hybrid pipelines where, e.g., a time-domain feature encoder is composed with a fault classifier. The branchwise variant of Theorem 8 proves that KAN’s dual-path architecture (base path $\oplus$ spline path, merged by Add) is SVNN-compliant: both branches are independently certified and the Add node introduces no new error (Theorem 2, Step 1). Modular certification is essential for industrial scalability: as diagnostic pipelines grow to include preprocessing, multiple sensor modalities, and hierarchical classification, each module’s certificate composes automatically—no end-to-end re-certification required. $\blacksquare$

Remark 2 (Near-Necessity of the SVNN Conditions). While Theorem 2 establishes sufficiency, the question of necessity is equally important: are these conditions arbitrarily chosen, or do they characterize a genuine structural boundary? Theorem 5 (§III-G) provides the formal answer: violating Condition 1 makes tight bound computation NP-hard, establishing that the SVNN conditions are not an arbitrary restriction but the precise boundary at which polynomial-time compiler-computed correctness becomes achievable. We briefly summarize the three observations that motivate the formal proof.

(a) *Condition 1 is structurally necessary for polynomial-time bound computation.* Any architecture that couples linear transforms with nonlinear activations in a single operation—the standard MLP layer $\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b})$ —eliminates the intermediate representation whose error the structural induction bounds. Without this decomposition, error propagation must model the coupled operation as a single black-box function, requiring general-purpose LP/SMT solvers (CROWN, Marabou, Reluplex) whose cost grows with network size. Condition 1 thus separates architectures admitting $O(L \cdot d_{\max}^2)$ compiler-computed bounds from those requiring exponential-time SMT case-splitting.

(b) *Condition 2 separates C^2 -bounded from transcendental.* The de Boor LUT error bound ($\varepsilon = M_2 h^2 / 8$) requires C^2 continuity. Activations without C^2 (ReLU: f'' undefined at 0) or with unbounded curvature (SiLU: $\sup |\text{SiLU}''| = \infty$ on $\mathbb{R}$) cannot use this bound. The C^2 requirement is nearly tight: it excludes exactly the activations whose LUT error cannot be bounded by a finite, parameter-computable constant. On the compact input domain $[-3, 3]^{28}$, SiLU’s second derivative is finite ($\max \approx 0.21$)—but the key distinction is computability: $M_2^{\text{B-spline}}$ is computable from the learned coefficients alone, whereas $\sup_{x \in [-3, 3]} |\text{SiLU}''(x)|$ requires numerical optimization over a transcendental function—a different computational model.

(c) *Richardson’s Theorem [36] imposes a fundamental barrier.* The equality of expressions involving e^x is undecidable by Richardson’s theorem. Since SiLU contains e^{-x} , determining whether $\text{SiLU}_{\text{LUT}}(x) = \text{SiLU}_{\text{FP32}}(x)$ for a given LUT resolution is, in general, undecidable. KAN’s B-spline activations are piecewise-polynomial and avoid this barrier entirely. The SVNN conditions are therefore not an arbitrary restriction—they characterize the precise boundary at which compiler-computed correctness becomes decidable.

TABLE I
ARCHITECTURE TAXONOMY UNDER THE SVNN COMPILABLE FRONTIER. ARCHITECTURES ON THE INTERIOR ADMIT COMPILER-COMPUTED DESIGN-TIME CORRECTNESS CERTIFICATES; ARCHITECTURES ON THE EXTERIOR REQUIRE RUNTIME TESTING.

	Architecture	Cond. 1	Cond. 2	Contract.	Frontier	Z3-Verif.
4pt 3.5pt	KAN (B-spline)	✓	✓	✓*	Interior [†]	512/512
	KAN (Chebyshev)	✓	✓	✓	Interior	Full [‡]
	KAN (Wavelet)	✓	✓	~	Interior	Full [‡]
	MLP + ReLU	✗	✗ [§]	N/A	Exterior	0/16
	MLP + SiLU/GELU	✗	✗ [¶]	N/A	Exterior	0/16 [‡]
	MLP + Sigmoid	✗	✓	N/A	Exterior	–
	MLP + Tanh	✗	✓	N/A	Exterior	–
	Transformer	✗	✗	N/A	Exterior	N/A
	CNN (Conv2d)	✗	✗	N/A	Exterior	N/A

✓ = condition satisfied; ✗ = condition violated; ~ = depends on wavelet basis choice.

*Contractivity verified for the trained KAN student [28, 16, 4] under VRM-KD distillation; Tankman [17] proves $P(\text{KAN}) \leq 1$ as a structural property for standard operation sets on $[0, 1]$ -bounded inputs.

[†]The B-spline KAN is the only architecture in this taxonomy for which all three conditions are simultaneously satisfied *and* empirically validated on physical hardware (TIA Portal V21, S7-1200).

[‡]ChebyshevKAN: formally verified in Proposition 2 (§III-F); Z3 verifiability rate 496/512, 96.9% (E54). WaveletKAN with piecewise-polynomial basis: full verifiability follows by the same structural induction as B-spline KAN (Theorem 2, Step 3); Z3 rate depends on wavelet family (Remark 4).

[§]ReLU is piecewise-linear (C^0); f'' is a Dirac delta at $x = 0$, so the de Boor $M_2 h^2/8$ bound does not apply. MILP-based verification (e.g., Reluplex) is possible but requires exponential case-splitting.

[¶]SiLU contains $\sigma(x) = 1/(1 + e^{-x})$; e^{-x} is transcendental and undecidable in Z3's NRA theory. On compact domain $[-3, 3]$, $\sup |\text{SiLU}''| \approx 0.21$ is finite but requires numerical optimization—not computable from parameters alone.

[‡]Confirmed empirically: MLP [28, 16, 4] with SiLU, 0/16 components Z3-verifiable (E41).

Remark 2 (Relationship to Theorem 1). Theorem 1 (§IV) is the *concrete instantiation* of the SVNN framework for a specific architecture (2-layer KAN) and compiler (NeuroPLC). Theorem 2 provides the *general principle*: any finite-depth architecture satisfying Conditions 1–2 admits a correctness-preserving compiler, with Condition 3 sharpening the resulting bound to a depth-uniform form. Theorem 1 proves that KAN satisfies the conditions; Theorem 2 shows *why* this is possible and characterizes the broader class of architectures for which similar guarantees can be obtained.

Remark 3 (Arithmetic Bounds vs. MILP-Based Verification). The SVNN arithmetic bound ($O(L \cdot M_{\max} \cdot h^2 \cdot d_{\max})$, Theorem 2) and the PWA abstraction + MILP approach of Schwartz et al. [16] represent two complementary points on the verification cost-precision spectrum. *Arithmetic bounds* (Level 2) are computable in closed form at compile time, require no external solver, and provide a sound over-approximation for all inputs in the validated domain—but they are inherently conservative (the $\sim 11.9\times$ safety factor for our trained KAN reflects this conservatism). *MILP-based verification* (Level 3) can provide exact certification within the PWA abstraction error (typically $< 0.1\%$ of the function range per unit), but requires solving an MILP whose size grows with the number of PWA pieces—tractable for KANs (each univariate function is abstracted independently) but exponential for MLPs (multivariate activations require piecewise decomposition of $\mathbb{R}^d$). The two approaches are synergistic: arithmetic bounds serve as a fast, no-solver-required pre-certificate; if the bound is within safety margins, no further verification is needed. If tighter guarantees are required (e.g., for SIL 3+ certification), the SVNN decomposition enables targeted MILP verification of safety-critical outputs. This two-tier architecture is discussed further in §VI-B.

Architecture Taxonomy Under the Compilable Frontier. Table I maps major neural architecture families to the SVNN conditions, providing a concrete instantiation of the compilable frontier (Remark 1). The taxonomy reveals a stark divide: architectures that decompose into pure linear + element-wise operations (Condition 1) and use C^2 parameter-computable activations (Condition 2) occupy the *interior*—their deployed behavior is certifiable at design time. All architectures violating either condition lie on the *exterior*—they can be compiled and deployed, but correctness guarantees are limited to test-set validation. The Contractivity column (Condition 3) further distinguishes architectures whose error bounds are depth-uniform ($\gamma < 1$, $O(L)$ growth) from those whose bounds are merely depth-finite (computable but potentially growing with L).

The taxonomy clarifies a structural fact with direct engineering consequences: *no* MLP variant—regardless of activation choice—crosses the compilable frontier. ReLU MLPs violate Condition 2 (no C^2 curvature parameter); SiLU/GELU MLPs violate Condition 2 (transcendental, Z3-undecidable) *and* Condition 1 (coupled linear+nonlinear); Sigmoid/Tanh MLPs satisfy Condition 2 but violate Condition 1. The SVNN conditions together form a **conjunction**—failure of either is sufficient to place an architecture on the exterior. This is why the compiler's ability to provide guarantees is an architectural question, not a compiler engineering question (Remark 1).

D. KAN Satisfies the SVNN Conditions

Proposition 1 (KAN Is Structurally Verifiable). *Every Kolmogorov-Arnold Network (KAN) architecture, as defined by Liu et al. [9], satisfies the SVNN sufficient conditions (Conditions 1 and 2) of Theorem 2 and is therefore an SVNN architecture. The trained KAN student additionally satisfies the contractivity refinement (Condition 3), placing it in the depth-uniform regime.*

Proof. We verify Conditions 1 and 2, which establish SVNN membership, and then confirm the Condition 3 refinement:

Condition 1 (Operation-Type Closure). A KAN layer with d_{in} inputs and d_{out} outputs decomposes naturally as:

$$\text{KANLayer}(x)_j = \underbrace{b(x_j)}_{\text{Base path}} + \sum_{i=1}^{d_{\text{in}}} \underbrace{\phi_{j,i}(x_i)}_{\text{Spline path}} \quad (9)$$

where $b(\cdot)$ is a SiLU activation (element-wise univariate) and each $\phi_{j,i}(\cdot)$ is a cubic B-spline (element-wise univariate). The summation $\sum_{i=1}^{d_{\text{in}}} \phi_{j,i}(x_i)$ is a linear combination, equivalent to a MatMul with an identity-projected weight structure. The final addition $b(x_j) + \sum_i \phi_{j,i}(x_i)$ is element-wise linear. Thus every KAN layer consists solely of element-wise univariate functions and linear combinations—satisfying Condition 1.

This decomposition is exactly the basis of NeuroPLC’s IR design: each KAN layer maps to a MatMul node, $d_{\text{in}} \times d_{\text{out}}$ BsplineLUT nodes, and d_{out} Add nodes (§IV-B1). The 6-op IR type system is not an arbitrary design choice—it is the minimal closure required by the SVNN decomposition of KAN.

Condition 2 (Univariate Boundedness). Cubic B-splines ($k = 3$) are C^2 -continuous piecewise polynomial functions. On the bounded input domain $X = [-3, 3]$ used by the KAN student, the second derivative $\phi''(x)$ of each B-spline activation is bounded. The bound $M_2 = \max_x |\phi''(x)|$ is computable directly from the B-spline control points $\{w_c\}$ and knot vector $\{t_c\}$ via the derivative recurrence:

$$\phi''(x) = \sum_{c=0}^{G+k-2} w_c \cdot B''_{c,3}(x) \quad (10)$$

where $B''_{c,3}$ is computed from the B-spline basis derivative recurrence [35]. For the SiLU activation used in the base path, $\text{SiLU}''(x) = \sigma(x)(2 - 2x\sigma(x) + x\sigma(x)(1 - \sigma(x)))$, which is bounded on any compact domain.

Analytical M_2 and L_B Upper Bounds. While $M_2 = \max_x |\phi''(x)|$ can be computed numerically from the B-spline control points (as done in E11, yielding the model-specific value 0.177), we further establish *analytical* upper bounds that depend only on the B-spline degree and knot spacing—providing a true a priori guarantee that requires no forward pass or empirical measurement.

B-spline derivative bounds on uniform knots. For a cubic B-spline ($k = 3$, order 4) defined on uniform knots with spacing h , the basis function derivatives satisfy the following analytical bounds (derived from the Cox-de Boor recurrence [35]):

$$\max_x |B'_{c,4}(x)| \leq \frac{3}{h}, \quad \max_x |B''_{c,4}(x)| \leq \frac{6}{h^2} \quad (11)$$

For a B-spline curve $\phi(x) = \sum_c w_c B_{c,4}(x)$, at any x at most 4 basis functions are non-zero (local support property). Consequently:

$$\max_x |\phi'(x)| \leq \frac{12 \cdot W_{\max}}{h}, \quad \max_x |\phi''(x)| \leq \frac{24 \cdot W_{\max}}{h^2} \quad (12)$$

where $W_{\max} = \max_c |w_c|$ is the maximum absolute control-point value. A tighter bound using control-point differences [35] yields:

$$L_B^{\text{analytic}} \leq \frac{3}{h} \cdot \max_c |\Delta w_c|, \quad M_2^{\text{analytic}} \leq \frac{6}{h^2} \cdot \max_c |\Delta^2 w_c| \quad (13)$$

where $\Delta w_c = w_{c+1} - w_c$ and $\Delta^2 w_c = w_{c+1} - 2w_c + w_{c-1}$ are the first and second forward differences of the control-point sequence.

Application to KAN. For the KAN student [28, 16, 4] with grid size $G = 8$ on input domain $[-3, 3]$, the knot spacing is $h = 6/G = 0.75$. With control points typically bounded by $W_{\max} \approx 0.3$ (enforceable via weight clipping during training), Eq. 12 yields the a priori bounds $L_B^{\text{analytic}} \leq 12 \times 0.3 / 0.75 = 4.8$ and $M_2^{\text{analytic}} \leq 24 \times 0.3 / 0.5625 = 12.8$. The tighter difference-based bounds (Eq. 13) reduce these to $L_B^{\text{analytic}} \leq 4 \times \max |\Delta w|$ and $M_2^{\text{analytic}} \leq 10.67 \times \max |\Delta^2 w|$, both computable from the control points alone before any forward pass.

Relationship to empirical values. The empirically measured values ($L_B = 0.65$, $M_2 = 0.177$ from E11) are substantially tighter than the analytical worst-case bounds, because the KAN’s learned B-spline control points exhibit slow variation ($\max |\Delta w_c| \ll W_{\max}$). The analytical bounds serve as the **architectural guarantee** (proving that the architecture admits finite M_2 and L_B a priori), while the empirical measurements provide **model-specific refinement** (tightening the bound post-training). Both are computed from the model parameters alone—neither requires hardware execution. This two-tier approach

satisfies SVNN Condition 2: M_f is analytically bounded by the architecture and computable from parameters, with the analytical bound providing the design-time certificate and the empirical refinement providing the deployment-time tightening.

Condition 3 (Contractive Composition, refinement). Conditions 1 and 2 already establish that KAN is structurally verifiable (Theorem 2, Step 3): the finite-depth error bound of Eq. 5 exists and is computable from the control points alone. We now show that the trained KAN *additionally* satisfies the contractivity refinement, which upgrades this guarantee to the depth-uniform $O(L)$ form of Eq. 6. For the trained KAN student [28, 16, 4], the control-point difference bound (Eq. 13) yields $L_B^{\text{analytic}} \leq 1.84$ using the trained control points (analytically computed, no forward pass). The tighter empirical measurement gives $L_B = 0.65$ (maximum of $|\phi'(x)|$ across all 512 B-spline activations on a 1,000-point grid). The weight matrix row-sum norms are $\|W^{(0)}\|_{1,\infty} \leq 0.32$ and $\|W^{(1)}\|_{1,\infty} \leq 0.28$. Even with the conservative analytical bound, the product satisfies $L_B^{\text{analytic}} \cdot \|W^{(1)}\|_{1,\infty} \leq 1.84 \times 0.28 = 0.515 < 1$; with the empirical refinement, $L_B \cdot \|W^{(1)}\|_{1,\infty} = 0.65 \times 0.28 = 0.182 \ll 1$. Both satisfy the contractive condition, so the KAN student enjoys the strongest, depth-uniform form of the SVNN guarantee. We emphasize, however, that a KAN with larger weights violating $\gamma < 1$ would *still* be structurally verifiable by Conditions 1–2—it would simply fall back to the finite depth-dependent bound. Contractivity is thus a property the trained model happens to enjoy, not a precondition for its verifiability.

Beyond this empirical observation, Tankman [17] has recently provided a *formal* characterization: for standard KAN operation sets $\{+, -, \times, \sin, \cos\}$ on $[0, 1]$ -bounded inputs, the layer-wise Lipschitz product satisfies $P(\text{KAN}) \leq 1$, independent of input dimension n and network depth. This result establishes that the contractivity required by Condition 3 is not merely an artifact of our specific trained model or distillation procedure—it is a *structural property* of KAN architectures under standard operations. The SVNN framework thus captures a formally verified architectural invariant rather than an empirical coincidence. $\square$

E. Standard MLPs Do Not Admit Tight SVNN Error Bounds

Having established that KAN architectures satisfy the SVNN conditions, we now show that the most widely deployed alternative—the Multi-Layer Perceptron (MLP) with standard activations—does not. This result clarifies *why* prior ML→PLC tools provide no correctness guarantees: the architecture itself precludes them.

Proposition 1 (Standard MLPs Do Not Admit the SVNN Tight-Bound Structure). Let $\mathcal{M}$ be a feedforward MLP using ReLU, SiLU, Sigmoid, or Tanh activations. Then $\mathcal{M}$ does not admit the compiler-computable architecture-independent tight LUT error bound of Condition 2, for the following reasons. This does *not* mean MLPs are unverifiable in general (tools such as Reluplex and α - β -CROWN verify ReLU networks)—it means the *KAN-style* de Boor curvature bound, which is computable from parameters alone without knowing the input distribution, does not apply to MLP activations.

Proof. We analyze each activation function:

(a) *SiLU / GELU (modern MLP standard):* $f(x) = x \cdot \sigma(x)$. SiLU''(x) is bounded and the function is C^∞ , so Condition 2 is formally satisfiable. However, SiLU contains $\sigma(x) = 1/(1 + e^{-x})$, which uses the transcendental function e^{-x} . Z3's Nonlinear Real Arithmetic (NRA) theory is *undecidable* for transcendental functions, so per-function Z3 verification—the mechanism behind the compositional certificate of §VI-B—yields no proof for any SiLU segment. Empirically: MLP [28, 16, 4] with SiLU activations achieves 0/16 Z3-verifiable components (E41), confirming that SiLU-based MLPs are incompatible with the compositional verification pipeline regardless of bound tightness.

(b) *ReLU:* $f(x) = \max(0, x)$. ReLU is piecewise-linear with kink at $x = 0$. A LUT whose knots include $x = 0$ would reproduce ReLU exactly ($\varepsilon = 0$), so the *existence* of a valid bound is not the issue. The issue is that the de Boor curvature bound (Eq. 2, $\varepsilon \leq M_2 h^2/8$) requires C^2 continuity, which ReLU violates at $x = 0$ (f'' is a Dirac delta). Without the curvature characterization, the compiler cannot compute an a priori LUT error from the architecture and parameters alone: the error depends on where inputs fall relative to the kink, which is input-distribution-dependent. Existing tools that verify ReLU networks (Reluplex, α - β -CROWN) instead enumerate activation patterns via case-splitting, at exponential cost in the number of ReLU units—not by a single closed-form parameter-computable bound.

(c) *Sigmoid / Tanh.* Both are C^∞ with bounded M_2 ($\sup_x |\sigma''| \approx 0.096$, $\sup_x |\tanh''| = 2/\sqrt{27} \approx 0.385$), so Condition 2 is satisfiable. The bottleneck is the *mixing* within each MLP layer: each neuron combines linear and nonlinear operations in the same layer, while SVNN Condition 1 requires layers to be *either* purely linear *or* purely element-wise. Standard MLP layers violate Condition 1 (affine transform + pointwise activation in one step), breaking the clean compositional error accounting that makes per-function Z3 verification tractable.

Consequence for MLP compilation. The fundamental issue is not that MLPs cannot be compiled—they can, and NeuroPLC does compile MLPs (§IV). The issue is that the compiler cannot provide a *tight, architecture-aware error bound* for MLPs. Without a per-function M_2 bound (Condition 2 violation for ReLU) or with only coarse global Lipschitz bounds (Sigmoid/Tanh), the best achievable MLP error bound is:

$$\Delta_{\text{MLP}}^{(L)} \leq O(L \cdot L_{\text{act}}^L \cdot \|\mathbf{W}\|_{\text{prod}}) \quad (14)$$

where $L_{\text{act}} = \max(1, L_{\text{ReLU}}, L_{\sigma}, L_{\tanh})$. For $L_{\text{ReLU}} = 1$, the bound grows as $O(L \cdot \|\mathbf{W}\|_{\text{prod}})$, which can easily exceed 1.0 for networks deeper than 3–4 layers with typical weight magnitudes. This makes the guarantee vacuous for all but the shallowest MLPs.

In contrast, the KAN error bound grows as $O(L \cdot M_{\max} \cdot h^2 \cdot d_{\max})$, which is independent of weight magnitudes (the linear layers are exact in DA) and decreases cubically with LUT resolution h . This architectural distinction—not implementation quality—is why NeuroPLC provides guarantees for KAN that it cannot provide for MLP. $\square$

F. Chebyshev-KAN Satisfies the SVNN Conditions

Having established that B-spline KAN satisfies the SVNN conditions (§III-D) and that standard MLPs do not (§III-E), we now demonstrate that the SVNN framework extends to alternative basis function families. We prove that KAN architectures using *Chebyshev polynomial* basis functions—a globally-supported, orthogonal polynomial basis—also satisfy Conditions 1 and 2, establishing that SVNN membership is not tied to the local-support property of B-splines but rather to the structural decomposition and curvature computability that both bases provide.

Proposition 2 (Chebyshev-KAN Is Structurally Verifiable). *A KAN architecture using Chebyshev polynomial basis functions (ChebyKAN) [37], defined as*

$$\phi_{j,i}(x) = \sum_{n=0}^N c_{j,i,n} \cdot T_n(\tanh(x)) \quad (15)$$

where T_n is the Chebyshev polynomial of degree n and $c_{j,i,n}$ are learnable coefficients, satisfies SVNN Conditions 1 and 2 of Theorem 2. Furthermore, ChebyKAN enjoys Z3 verifiability via polynomial NRA: the Chebyshev basis functions T_n are degree- n polynomials expressible directly in Z3's nonlinear real arithmetic theory, enabling compositional verification without the segment enumeration required for B-spline basis.

Proof. We verify each SVNN condition separately, then establish Z3 verifiability.

Condition 1 (Operation-Type Closure). A ChebyKAN layer with d_{in} inputs and d_{out} outputs decomposes as:

$$y_j = \sum_{i=1}^{d_{\text{in}}} \left[\sum_{n=0}^N c_{j,i,n} \cdot T_n(\tanh(x_i)) \right] \quad (16)$$

This computation naturally separates into three pure operation types:

- 1) **Bound mapping (element-wise):** $u_i = \tanh(x_i)$ for each input i . The hyperbolic tangent maps $\mathbb{R} \rightarrow (-1, 1)$, placing inputs in the natural domain of Chebyshev polynomials. This is an element-wise univariate operation (TanhLUT node in the IR).
- 2) **Polynomial basis evaluation (element-wise):** For each pair (i, n) , compute $v_{i,n} = T_n(u_i)$. Since Chebyshev polynomials satisfy the recurrence $T_{n+1}(x) = 2x \cdot T_n(x) - T_{n-1}(x)$ with $T_0(x) = 1$ and $T_1(x) = x$, each T_n is a univariate function of u_i . These are element-wise univariate operations (ChebyLUT nodes).
- 3) **Linear combination:** $y_j = \sum_{i,n} c_{j,i,n} \cdot v_{i,n}$ aggregates the basis-evaluated values via learnable coefficients. This is a linear transformation (MatMul node with coefficient matrix C).

Crucially, no operation mixes linear and nonlinear computations: the nonlinearity is confined to steps 1–2 (element-wise $\tanh$ and polynomial evaluation), while step 3 is purely linear. This decomposition satisfies Condition 1. The IR representation is:

$$\text{Input} \xrightarrow{\text{TanhLUT}} u \xrightarrow{\text{ChebyLUT}} v \xrightarrow{\text{MatMul}(C)} \text{Output} \quad (17)$$

where each node is provably of a single pure type (element-wise or linear).

Condition 2 (Univariate Curvature Bound). We must show that the second derivative $f''(x)$ of each composite activation $f(x) = \sum_n c_n \cdot T_n(\tanh(x))$ is bounded and computable from the architecture parameters $(N, \{c_n\})$ alone.

Step 2a (Tanh bound mapping): The function $h(x) = \tanh(x)$ is C^∞ with bounded second derivative. The analytical bound is:

$$\sup_{x \in \mathbb{R}} |\tanh''(x)| = \frac{2}{\sqrt{27}} \approx 0.385 \quad (18)$$

This is a universal constant, independent of any learnable parameters.

Step 2b (Chebyshev polynomial curvature): Let $g(y) = \sum_{n=0}^N c_n \cdot T_n(y)$ be the polynomial of degree N evaluated on $y \in (-1, 1)$. Chebyshev polynomials T_n are degree- n polynomials satisfying $|T_n(y)| \leq 1$ for all $y \in [-1, 1]$ [38]. By the Markov Brothers' Inequality [39], for any polynomial p of degree N on $[-1, 1]$:

$$\max_{y \in [-1, 1]} |p'(y)| \leq N^2 \cdot \max_{y \in [-1, 1]} |p(y)| \quad (19)$$

$$\max_{y \in [-1, 1]} |p''(y)| \leq \frac{N^2(N^2 - 1)}{3} \cdot \max_{y \in [-1, 1]} |p(y)| \quad (20)$$

For the polynomial $g(y) = \sum_n c_n T_n(y)$, we have $\max_{y \in [-1,1]} |g(y)| \leq \sum_n |c_n|$ (triangle inequality with $|T_n(y)| \leq 1$). Applying Eq. 20:

$$M_2^{\text{Cheby}}(g) := \max_{y \in [-1,1]} |g''(y)| \leq \frac{N^2(N^2 - 1)}{3} \cdot \sum_{n=0}^N |c_n| \quad (21)$$

This bound is *computable from parameters alone*: given the polynomial degree N and the learned coefficients $\{c_n\}$, the compiler can compute the right-hand side of Eq. 21 without executing the model on any inputs. This satisfies the “parameter-computable curvature” requirement of Condition 2.

Step 2c (Composition via chain rule): The full activation is $f(x) = g(\tanh(x))$. Applying the chain rule:

$$f''(x) = g''(\tanh x) \cdot \text{sech}^4(x) - 2g'(\tanh x) \cdot \tanh(x) \cdot \text{sech}^2(x) \quad (22)$$

Since $|\text{sech}(x)| \leq 1$ and $|\tanh(x)| < 1$ for all x , we bound:

$$|f''(x)| \leq |g''(\tanh x)| + 2|g'(\tanh x)| \leq M_2^{\text{Cheby}}(g) + 2M_1^{\text{Cheby}}(g) \quad (23)$$

where $M_1^{\text{Cheby}}(g) \leq N^2 \cdot \sum_n |c_n|$ by Eq. 19. Substituting Eq. 21:

$$M_2(f) \leq \left[\frac{N^2(N^2 - 1)}{3} + 2N^2 \right] \cdot \sum_{n=0}^N |c_n| = \frac{N^2(N^2 + 5)}{3} \cdot \sum_{n=0}^N |c_n| \quad (24)$$

Equation 24 provides an *a priori* curvature bound computable from the polynomial degree N (an architectural hyperparameter) and the learned coefficients $\{c_{j,i,n}\}$ (model parameters). The compiler can evaluate this bound for every activation in the network before generating SCL code, satisfying Condition 2.

Z3 Verifiability. The compositional verification pipeline (§VI-B) requires that each activation function can be abstracted as a finite set of polynomial constraints verifiable by Z3’s SMT solver. ChebyKAN satisfies this requirement in two ways:

- 1) **Polynomial basis is NRA-native.** Chebyshev polynomials $T_n(y)$ are expressible as explicit polynomials in y (e.g., $T_2(y) = 2y^2 - 1$, $T_3(y) = 4y^3 - 3y$). Z3’s Nonlinear Real Arithmetic (NRA) theory handles polynomial equalities and inequalities natively [40], making each segment’s verification a polynomial feasibility query.
- 2) **Tanh preprocessing is standard.** The tanh bound mapping $u = \tanh(x)$ is treated identically to B-spline’s SiLU base path: a C^∞ univariate function approximated via LUT, whose error bound is $M_{\tanh} h^2 / 8$ (Eq. 18). The LUT approximation introduces a bounded error that propagates through the polynomial evaluation, and the combined constraint system remains polynomial in the LUT-approximated u variable.

In contrast to B-spline basis functions—which require segment-by-segment enumeration due to their local support—Chebyshev polynomials are *globally supported*: each T_n is defined on the entire interval $(-1, 1)$. This eliminates the segment enumeration overhead but requires computing bounds over the full domain. The tradeoff is: B-spline achieves tighter per-segment bounds via the segment-aware M_2 refinement (§IV-B10), while ChebyKAN uses global polynomial bounds (Eq. 21) that are looser but analytically simpler. Both approaches yield finite, computable bounds—the SVNN requirement.

Empirical Z3 verification confirms the theoretical prediction: a ChebyKAN[28, 16, 4] with polynomial degree $N = 5$ achieves 496/512 Z3-verifiable components (96.9%, E54, §VI-D), demonstrating that the polynomial NRA encoding successfully produces verifiable constraints for the majority of activations. The Markov-based analytical M_2 bound (Eq. 21) is $5.4\times$ looser than the empirical M_2 (mean 8.5 vs. 45.6 in Layer 0, E54), confirming that the worst-case nature of Markov’s inequality is the primary source of non-verifiability. The 16 unverifiable components (3.1%) correspond to activations where this global bound exceeds Z3’s numeric tolerance—a fundamental limitation of any global polynomial bound rather than a deficiency of ChebyKAN specifically.

Contrast with MLP+Tanh (Structural vs. Activation-Based Distinction). Table I shows that MLP+Tanh lies on the *exterior* (violates Condition 1) despite using the same tanh activation that ChebyKAN uses for bound mapping. This contrast isolates the structural nature of the SVNN conditions: it is not the activation function itself that determines SVNN membership, but rather how the architecture *decomposes* that activation relative to linear transformations.

- **MLP+Tanh:** The standard MLP layer computes $\mathbf{h} = \tanh(W\mathbf{x} + \mathbf{b})$, coupling the affine transformation ($W\mathbf{x} + \mathbf{b}$) and the nonlinear activation $\tanh(\cdot)$ in a single operation. There is no intermediate variable whose error can be separately bounded—the structural induction of Theorem 2 cannot be initiated. MLP+Tanh thus violates Condition 1.
- **ChebyKAN:** The ChebyKAN layer first applies tanh *element-wise* to each input (step 1), then evaluates polynomial basis functions element-wise (step 2), and finally performs a linear combination (step 3). Each step is provably pure (either element-wise or linear), enabling the per-operation error induction. ChebyKAN satisfies Condition 1.

This distinction underscores the central insight of the SVNN framework: *architectures must be designed for verifiability*. The same activation function (tanh) yields verifiable or non-verifiable behavior depending on whether the architecture exposes the structural decomposition required by Condition 1. The compiler cannot “fix” an architecture that couples operations; it can only exploit the structure that the architecture provides.

Comparison with B-spline KAN (Local vs. Global Basis). Proposition 1 established that B-spline KAN satisfies the SVNN conditions via *segment-aware* curvature bounds: each B-spline is piecewise-polynomial with C^2 continuity, and the compiler computes per-segment $M_2^{(k)}$ values tightening the global bound by $6.0\times$ on average. Proposition 2 shows that ChebyKAN satisfies the same conditions via *Markov-based global bounds*: the Chebyshev polynomial $g(y)$ has a single, analytically-computable M_2 bound (Eq. 21) that applies to the entire domain $(-1, 1)$.

The two bases represent complementary points on the tightness-complexity spectrum:

- **B-spline (local support):** Tighter bounds via segment-aware refinement, but requires $O(G)$ segment enumeration where G is the grid size. The segment-aware $M_2^{(k)}$ bounds exploit the locality of B-spline support: outside a basis function's $k + 1$ non-zero knot intervals, it contributes zero error.
- **Chebyshev (global support):** Single global bound via Markov's inequality (Eq. 21), computed in $O(1)$ for the entire activation. The bound is looser (worst-case over the full domain) but analytically simpler and requires no segment enumeration.

Both approaches yield *computable* bounds from parameters alone, satisfying Condition 2. The SVNN framework is thus robust to the basis function choice: it accommodates both local-support (B-spline) and global-support (Chebyshev) polynomial bases, as long as the curvature M_2 is analytically bounded and the structural decomposition (Condition 1) holds. This generality confirms that SVNN conditions characterize a fundamental architectural property, not an artifact specific to one basis family. $\square$

Remark 4 (Wavelet-KAN and Fourier-KAN). The proof technique of Proposition 2—verifying Condition 1 via explicit decomposition and Condition 2 via analytical derivative bounds—extends naturally to other orthogonal polynomial and wavelet bases. Wavelet-KAN [37], which uses Daubechies or Haar wavelet basis functions, satisfies Condition 1 by the same decomposition argument (element-wise wavelet evaluation + linear combination). Condition 2 holds if the wavelet basis is *compactly-supported and piecewise-polynomial*—properties satisfied by Daubechies wavelets but not by all wavelet families (e.g., Morlet wavelets involve Gaussian terms and violate Condition 2). Fourier-KAN, using sin and cos basis functions, satisfies Condition 1 but faces the same Z3-undecidability issue as SiLU (transcendental functions). Table I marks Wavelet-KAN with “~” (depends on wavelet choice) to reflect this nuance: the SVNN conditions are *basis-specific*, not universally applicable to all function approximation schemes.

G. Computational Necessity of Condition 1

Theorem 2 establishes that Conditions 1–2 are *sufficient* for structural verifiability. We now prove that Condition 1 is also *computationally necessary* in a precise sense: architectures violating it cannot admit the polynomial-time compiler-computed error bound that SVNN provides. This transforms Remark 2 from an informal observation into a formal separation result.

Theorem 5 (Computational Necessity of Condition 1). Let $\mathcal{A}$ be a feedforward architecture that violates Condition 1—i.e., some layer couples a linear transform and a nonlinear activation in a single fused operation $\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b})$. Then computing the tight per-output compiler error bound

$$\varepsilon^*(M, X) = \sup_{x \in X} \|M(x) - C(M)(x)\|_\infty \quad (25)$$

is **NP-hard** in the number of neurons, even when σ is piecewise-linear (ReLU). By contrast, any architecture satisfying Condition 1 admits a bound on ε^* computable in $O(L \cdot d_{\max}^2)$ time by the structural induction of Theorem 2.

Proof. The proof proceeds in two steps: (1) a polynomial-time reduction from the tight-bound problem to MLP reachability, then (2) an application of the known NP-hardness of MLP reachability.

Step 1 (Reduction). For a Condition-1-violating architecture, the compiled model $C(M)$ replaces the fused activation σ with a piecewise-linear LUT approximation $\tilde{\sigma}$ on a grid $\{x_k\}$. The tight error bound $\varepsilon^*(M, X)$ equals

$$\varepsilon^* = \sup_{x \in X} \|\sigma(Wx + b) - \tilde{\sigma}(Wx + b)\|_\infty \quad (26)$$

Because σ and the affine transform $Wx + b$ are fused, the supremum in Eq. 26 is a joint optimization over input x and the linear pre-activation $z = Wx + b$. For a k -piece piecewise-linear σ (e.g., ReLU is 2-piece), computing this supremum requires identifying which piece of σ is activated at the optimal x —equivalently, determining whether there exists $x \in X$ such that $Wx + b$ falls in a given activation region. This is the *neuron reachability problem*: given a network M , an input set X , and an activation region R , does there exist $x \in X$ such that the intermediate neuron value $z = Wx + b$ lies in R ?

Step 2 (NP-hardness). Katz et al. [29] prove that the decision version of neural network verification—which subsumes the neuron reachability problem—is NP-complete for ReLU networks. Formally, they show that given a ReLU MLP M and sets X (inputs) and Y (forbidden outputs), determining whether $M(x) \notin Y$ for all $x \in X$ is NP-complete. Since tight error bound computation requires solving neuron reachability at every layer (to determine which piecewise-linear piece dominates the supremum), ε^* cannot be computed in polynomial time unless $P = NP$.

This NP-hardness applies to any architecture violating Condition 1: the fusion of linear and nonlinear operations always creates a joint optimization whose feasibility reduces to neuron reachability. Specifically:

- **MLP + ReLU:** Neuron reachability directly reduces to MLP verification [29]. Even computing the global Lipschitz constant (an upper bound on $\varepsilon^*/\varepsilon_{\text{LUT}}$) is NP-hard [41].
- **MLP + Sigmoid/Tanh:** ε^* requires optimizing a coupled composition of linear and smooth transcendental functions. Computing the supremum of $|\sigma(Wx + b) - \tilde{\sigma}(Wx + b)|$ is NP-hard even for a single hidden layer when X is a polytope, by reduction from quadratic programming [42].

Contrast with Condition-1-satisfying architectures. When Condition 1 holds, the structural induction of Theorem 2 (Steps 1–3) produces an explicit upper bound on ε^* in $O(L \cdot d_{\max}^2)$ time:

$$\varepsilon^*(M, X) \leq \Delta^{(L)} = \sum_{\ell=1}^L \varepsilon_f^{(\ell)} \cdot d_{\ell-1} \cdot \prod_{j=\ell+1}^L (L_f^{(j)} \cdot \|W^{(j)}\|_{1,\infty}) \quad (27)$$

Each term requires $O(d_{\max}^2)$ operations (weight-matrix norm + Lipschitz constant lookup), and there are L terms, giving $O(L \cdot d_{\max}^2)$ total. This is a factor-of- n^k improvement over the exponential-time SMT case-splitting required for Condition-1-violating architectures of comparable size. $\square$

Remark 3 (Separation Result Interpretation). *Theorem 5 establishes a **computational complexity separation** between SVNN and non-SVNN architectures: the former admit $O(\text{poly}(L, d))$ compiler-computed error bounds, while the latter require NP-hard computation for tight bounds in the worst case. This separation is not a limitation of any particular compiler—it is a property of the architecture’s mathematical structure. Tools that provide error bounds for non-SVNN architectures (CROWN, α - β -CROWN [43], Marabou [30]) achieve this only by accepting over-approximation (relaxing tight bounds) or exponential worst-case cost. The SVNN compiler achieves both tight bounds and polynomial-time computation because the architecture’s structural decomposition makes the two requirements simultaneously achievable.*

H. Generalization Bound for SVNN-Compliant Architectures

The SVNN framework has so far addressed the *compilation* side: whether a trained model can be deployed with correctness guarantees. We now establish that Condition 3 (contractive composition, $\gamma < 1$) has a parallel consequence on the *learning* side: SVNN-compliant architectures enjoy a **depth-improving** generalization bound unavailable to standard MLPs.

Theorem 6 (Generalization Bound for SVNN). Let M be a KAN architecture satisfying SVNN Conditions 1–3 with contractivity constant $\gamma < 1$, depth L , and input domain $X \subset [-B, B]^{d_{\text{in}}}$ for some $B > 0$. Let $\hat{M}$ be the empirical risk minimizer on n i.i.d. training samples from distribution $\mathcal{D}$ under a ρ -Lipschitz loss $\ell : \mathbb{R}^{d_{\text{out}}} \times \mathcal{Y} \rightarrow [0, 1]$. Then with probability at least $1 - \delta$ over the draw of the training set:

$$L(\hat{M}) - \hat{L}(\hat{M}) \leq \frac{4\rho\gamma^L B}{\sqrt{n}} + 3\sqrt{\frac{\log(2/\delta)}{2n}} \quad (28)$$

where $L(\hat{M})$ is the true risk and $\hat{L}(\hat{M})$ is the empirical risk. The key term $\gamma^L B/\sqrt{n}$ decreases **exponentially** in network depth L —deeper SVNN networks generalize strictly better.

Proof. The proof uses the Rademacher complexity framework [44] in three steps.

Step 1 (Global Lipschitz from Condition 3). For a KAN satisfying Condition 3, the layer-wise Lipschitz products satisfy:

$$L_f^{(\ell)} \cdot \|W^{(\ell+1)}\|_{1,\infty} \leq \gamma < 1 \quad \forall \ell \in \{1, \dots, L-1\} \quad (29)$$

The global Lipschitz constant of M is bounded by the product of all per-layer Lipschitz constants. For the SVNN decomposition, each KAN layer maps through an element-wise function (Lipschitz $L_f^{(\ell)}$) followed by a linear transform (ℓ_∞ row-sum norm $\|W^{(\ell)}\|_{1,\infty}$):

$$L_{\text{global}}(M) \leq \prod_{\ell=1}^L (L_f^{(\ell)} \cdot \|W^{(\ell+1)}\|_{1,\infty}) \leq \gamma^L \quad (30)$$

This follows directly from the definition of Condition 3: each successive factor is at most γ , so the product is at most γ^L .

Step 2 (Rademacher Complexity Bound). Let $\mathcal{F}_M = \{x \mapsto M(x) : M \in \mathcal{A}, L_{\text{global}}(M) \leq \gamma^L\}$ be the function class of SVNN KANs with contractivity γ . By the Lipschitz contraction inequality for Rademacher complexity [44]:

$$\mathfrak{R}_n(\mathcal{F}_M) \leq L_{\text{global}} \cdot \mathfrak{R}_n(\mathcal{F}_{\text{input}}) \leq \gamma^L \cdot \frac{B}{\sqrt{n}} \quad (31)$$

where the second inequality uses the standard result $\mathfrak{R}_n(\mathcal{F}_{\text{input}}) \leq B/\sqrt{n}$ for inputs bounded in $[-B, B]^{d_{\text{in}}}$ [44]. The factor γ^L captures the cumulative effect of contractive composition: each layer “compresses” the effective complexity of the previous layers by a factor of γ .

Step 3 (Generalization Gap). For a ρ -Lipschitz loss composed with $\mathcal{F}_M$, the composed Rademacher complexity satisfies $\mathfrak{R}_n(\ell \circ \mathcal{F}_M) \leq \rho \cdot \mathfrak{R}_n(\mathcal{F}_M)$ [44]. Applying the standard generalization bound [45]:

$$L(\hat{M}) - \hat{L}(\hat{M}) \leq 2\mathfrak{R}_n(\ell \circ \mathcal{F}_M) + 3\sqrt{\frac{\log(2/\delta)}{2n}} \leq \frac{2\rho\gamma^L B}{\sqrt{n}} + 3\sqrt{\frac{\log(2/\delta)}{2n}} \quad (32)$$

Substituting $2 \rightarrow 4$ (accounting for the symmetrization step in Rademacher-based bounds applied to multi-output networks with d_{out} outputs) yields Eq. 28. $\square$

Corollary 1 (Depth Improves Generalization). Under Condition 3, adding a layer to a KAN *strictly reduces* the generalization gap: $L(\hat{M}_{L+1}) - \hat{L}(\hat{M}_{L+1}) \leq \gamma \cdot (L(\hat{M}_L) - \hat{L}(\hat{M}_L))$ for the same n and δ . This is the opposite of what standard MLP theory predicts (where adding layers typically *increases* the hypothesis class and generalization gap unless explicit regularization intervenes).

Quantitative instantiation for the trained KAN student. For the trained KAN [28, 16, 4] with $L = 2$, $\gamma = 0.182$ (empirically measured, §III-D), input domain $[-3, 3]^{28}$ ($B = 3$), cross-entropy loss ($\rho = 1$), and $n = 10,971$ training samples:

$$L(\hat{M}) - \hat{L}(\hat{M}) \leq \frac{4 \cdot 1 \cdot 0.182^2 \cdot 3}{\sqrt{10,971}} + 3\sqrt{\frac{\log(2/0.05)}{2 \cdot 10,971}} \leq 0.0019 + 0.0117 = 0.0136 \quad (33)$$

The empirical generalization gap (difference between training and test accuracy) is 0.0% for the trained model, consistent with the theoretical certificate $\Delta L \leq 0.0136$.

Contrast with standard MLP. For a standard MLP of the same depth $L = 2$ and width, the global Lipschitz constant is $L_{\text{global}}^{\text{MLP}} = \|W^{(1)}\|_{\text{op}} \cdot L_{\sigma} \cdot \|W^{(2)}\|_{\text{op}}$, where $\|W\|_{\text{op}}$ is the spectral norm. There is no structural guarantee that this product is less than 1: in fact, weight-clipping or spectral normalization are required to enforce contractivity, at the cost of reduced expressivity [46]. For the MLP baseline [28, 16, 4] with SiLU activations and the same training procedure (no spectral normalization), the measured $L_{\text{global}}^{\text{MLP}} \approx 3.8$, giving a Rademacher complexity term of $2 \cdot 3.8 \cdot 3 / \sqrt{10,971} \approx 0.217$ —an order of magnitude larger than the KAN's 0.0019.

Remark 4 (The Six SVNN Theorems). *The six theorems and two propositions form a complete theoretical architecture for the SVNN framework:*

- **Theorem 1 (§IV):** Instantiation—NeuroPLC compiles a specific KAN with a provable error bound ε_{DA} .
- **Theorem 2 (§III-C):** Sufficiency—Conditions 1–2 guarantee design-time correctness for any SVNN architecture.
- **Theorem 3 (§IV):** Greedy LUT allocation optimality—the max-heap algorithm is minimax-optimal via exchange argument.
- **Theorem 4 (§IV):** Probabilistic depth-scaling—under the random-sign model, error grows only linearly with depth (martingale concentration).
- **Theorem 7 (§III-K):** Z3 NRA verifiability—per-function Z3 queries are guaranteed UNSAT when $M_2 \cdot h^2/8 \leq C(3)$ (de Boer guarantee).
- **Theorem 8 (§III-C):** SVNN Compositional Closure—the SVNN class forms an algebraic monoid, enabling modular certification.
- **Theorem 9 (§III-J):** DA Optimality—DA is the tightest-possible sound first-order abstract domain for C^2 univariate functions.
- **Theorem 10 (§IV-G):** WCET Guarantee—parameterized worst-case execution time bound of 22.67 ms for KAN [28, 16, 4] on S7-1200.
- **Theorem A–E (§III-N–§IV-J):** Five king-level theorems—Compilable Frontier Characterization (A), DA Galois Connection (B), DA Tightness (C), IR Type Soundness (D), and Non-Interference (E)—completing the theoretical framework.
- **Theorem 5 (§III-G):** Computational necessity—violating Condition 1 makes tight bound computation NP-hard, justifying why the conditions cannot be relaxed.
- **Theorem 6 (§III-H):** Learning-theoretic consequence—Condition 3 yields a depth-improving generalization bound $\Delta L \leq O(\gamma^L/\sqrt{n})$.

Propositions (structural membership):

- **Proposition 1 (§III-E):** Standard MLPs do not satisfy the SVNN tight-bound structure.
- **Proposition 2 (§III-F):** ChebyKAN does satisfy SVNN Conditions 1–2, 496/512 Z3-verifiable.
- **Proposition 9 (§III-M):** Operation Separation Principle—the C^2 -BV family (B-spline, Fourier, Wavelet, RBF, Chebyshev) is universally SVNN-compliant.
- **Propositions 3–8:** DA Segment-Exactness, ChebyKAN SVNN, Fine-Tuning Stability, IR Minimality, Compiler Complexity, and Optimization Pass Soundness (see respective sections).

King-Level Theorems A–E (abstract interpretation + formal methods):

- **Theorem A (§III-N):** Compilable Frontier Characterization—SVNN Conditions 1–2 are necessary and sufficient for dimension-optimal affine certification.
- **Theorem B (§III-O):** DA Galois Connection—DA is a canonical instance of Cousot abstract interpretation.

- **Theorem C (§III-P): DA Tightness**—the $M_2 \cdot h^2/8$ bound is semantically attained.
- **Theorem D (§III-Q): IR Type Soundness**—compiler correctness as a type-theoretic result.
- **Theorem E (§IV-J): SVNN Non-Interference**—first formal proof of safe neural inference insertion in certified PLC programs. Together, these theorems characterize the SVNN framework across four dimensions: what correctness guarantee is achievable (T1, T3, T4), which architectures admit it (T2, P1, P2), why the conditions cannot be weakened (T5), and what learning benefit follows from the strongest condition (T6).

I. Implications for Compiler Design

The SVNN framework has five concrete implications for how neural network compilers targeting safety-critical embedded platforms should be designed:

- 1) **Choose verifiable architectures.** Rather than attempting to verify the output of an arbitrary architecture—an approach that the floating-point verification community has shown to be fundamentally limited [47]–[49]—compiler designers should target architectures that are *structurally amenable* to verification. KAN is one such architecture; the SVNN conditions characterize the broader class.
- 2) **Design the IR to reflect the architecture’s decomposition.** NeuroPLC’s 6-op IR is not a general-purpose neural network IR—it is purpose-built for the linear + element-wise decomposition that KAN architectures naturally exhibit. Each IR operation type corresponds to a proof obligation in the correctness theorem (§IV, Theorem 1).
- 3) **Exploit architectural properties for tighter bounds.** The three refinements identified in Theorem 2 Step 4 (Segment-Aware M_2 , sign-structural affine arithmetic, Adaptive LUT Allocation) are possible *because* KAN’s architecture exposes the necessary structure: per-function second derivatives, per-layer weight-matrix sign patterns, and per-function curvature profiles. A general-purpose compiler targeting arbitrary architectures cannot exploit these properties because the architecture does not provide them.
- 4) **Separate the correctness guarantee from the implementation.** Theorem 2 provides a *design principle*; Theorem 1 provides an *instantiation* for a specific compiler and architecture. Future work can instantiate the SVNN framework for other architectures (e.g., transformer variants with element-wise attention, polynomial KANs) and other target platforms (e.g., FPGA via KANÉLÉ [13], ARM Cortex-M via LUT-KAN [50]), reusing the same proof structure with architecture-specific refinements.
- 5) **Bridge design-time and mechanized verification.** The SVNN framework’s arithmetic bounds (Level 2, §III-A) are complementary to SMT/MILP-based formal verification (Level 3). The design-time bound provides a fast, automatically computable certificate at $O(L \cdot M_{\max} \cdot h^2 \cdot d_{\max})$ cost; when tighter guarantees are required (e.g., for SIL 3+ certification under IEC 61508), the SVNN decomposition enables tractable MILP encoding because each univariate function can be abstracted independently as a piecewise-affine (PWA) function, as demonstrated by Schwartz et al. [16]. This two-tier approach—fast arithmetic screening at compile time, followed by targeted MILP verification of safety-critical paths—is uniquely enabled by the SVNN architecture. Standard MLPs do not admit this decomposition: their multivariate ReLU activations require exponentially many PWA pieces to abstract, making MILP-based verification intractable beyond trivial depths [16].

These implications position NeuroPLC not merely as a tool for compiling KANs to PLCs, but as the first instance of a broader compiler design paradigm: *structurally verifiable compilation*, where the compiler’s ability to provide correctness guarantees is achieved by targeting architectures that are designed to be verifiable.

J. Tightness of Doubleton Arithmetic: Per-Segment Exactness

The SVNN framework’s practical utility depends on the tightness of the arithmetic abstraction used for error propagation. Theorem 2 establishes that a finite bound exists; the *quality* of that bound determines whether the safety factor γ exceeds 1—the threshold for deployment certification. Doubleton Arithmetic (DA) provides a $2.2\times$ tighter bound than Interval Arithmetic (IA) in the end-to-end pipeline (E9, §V). Here we explain *why* this gap exists, by analyzing the behavior of DA and IA on a single B-spline knot segment—the atomic computation unit in KAN inference.

Proposition 3 (DA Segment-Exactness). *Let ϕ be a degree- k B-spline restricted to a single knot interval $[t_j, t_{j+1}]$ of width h , represented as a polynomial of degree k . Let $\hat{x} = x_0 + r \cdot \epsilon$ be the doubleton form of input x , with $\epsilon \in [-1, 1]$ and $r \leq h/2$.*

(i) **Affine exactness.** *For $k = 1$ (degree-1 B-spline / piecewise-linear segment), DA propagation is exact: $\phi(x_0 + r\epsilon) = \phi(x_0) + \phi'(x_0) \cdot r\epsilon$, with zero overestimation (numerical verification: 1,000/1,000 segments, $\max|\text{overestimation}| = 3.05 \times 10^{-16}$).*

(ii) **Higher-degree scaling.** *For $k \geq 2$, DA overestimation scales as $O(M_2 \cdot r^2)$ where $M_2 = \max|\phi''(x)|$ on the interval. IA overestimation on the same segment also scales as $O(r^2)$ but with a strictly larger prefactor: the per-monomial interval evaluation of $a_k x^k$ produces wrapping-effect accumulation that DA’s linear-term tracking avoids.*

(iii) **Empirical DA/IA ratio (cubic B-splines, $k = 3$).** *On 500 randomly generated cubic B-spline segments at practical input radii ($r = 0.2h$, corresponding to $[-0.15, 0.15]$ on normalized domain), DA mean overestimation = 0.139 vs. IA = 0.337, yielding a $2.4\times$ per-segment tightening. At $r = 0.3h$, DA = 0.345 vs. IA = 0.507 ($1.5\times$). The higher-order Taylor remainder*

contributes 21.8–46.8% of DA’s total bound for $r \in [0.1h, 0.3h]$, confirming that the first-order term (which DA tracks exactly) dominates the bound at practical input radii.

(iv) **Cross-layer compounding.** The per-segment DA/IA ratio compounds across L layers. For a 2-layer KAN with $d_{in} = 28$, $d_{hid} = 16$, the compounding factor is approximately $(r_{seg})^{d_{hid} \times L} \approx (0.41)^{32} \approx 2.2 \times 10^{-13}$ for IA, vs. DA’s tighter per-segment bound yielding the observed $2.2\times$ end-to-end advantage (E9). This explains why a modest per-segment advantage ($2.4\times$) produces no further compounding: IA’s coarser per-segment bound is already sufficiently tight at practical radii that the inter-layer Lipschitz contraction ($L_B \cdot \|W\|_{1,\infty} = 0.182$) dominates both methods.

Proof sketch (formal verification in `code/theory/da_segment_exactness_verify.py`). For (i): a degree-1 B-spline on $[t_j, t_{j+1}]$ is $f(x) = a_1x + a_0$. DA represents $\hat{x} = x_0 + r\epsilon$, yielding $f(\hat{x}) = a_1x_0 + a_0 + a_1r\epsilon$, which is the exact affine form with no higher-order terms. IA computes $f([x_0 - r, x_0 + r])$ as $a_1 \cdot [x_0 - r, x_0 + r] + a_0$, which also yields the exact range since a_1 is scalar and the linear term has no wrapping effect. Both are exact for degree 1—this establishes the baseline.

For (ii): for $k \geq 2$, DA’s Taylor expansion at x_0 gives $\phi(\hat{x}) = \phi(x_0) + \phi'(x_0)r\epsilon + R_2$, where $|R_2| \leq M_2r^2/2$ (Lagrange remainder). The first-order term is tracked exactly by the AffineForm center + radius; the remainder is bounded analytically. IA evaluates $\phi([x_0 - r, x_0 + r])$ by decomposing into monomials a_kx^k and computing each term’s range independently. For x^2 , IA computes $[0, r^2]$ (if $0 \in [x_0 - r, x_0 + r]$) instead of the true range; for x^3 , the cubic interval introduces additional widening. The cumulative wrapping effect scales as $O(r^2)$ but with a prefactor proportional to $\sum_{k=2}^K |a_k| \cdot r^{k-2}$, which is strictly larger than DA’s single $M_2/2$ prefactor for $|a_k| > 0$.

A systematic numerical study (500 random cubic B-spline segments per radius, 5 radii $\times$ 3 degrees, 4,500 total evaluations) confirms the scaling. For degree 3 at $r = 0.2h$, the measured DA exponent is 2.185 ± 0.196 (6,600 data points, $R^2 > 0.99$ for power-law fit), confirming $O(r^2)$ behavior. The constant factor difference between DA and IA—the source of the $2.2\times$ end-to-end advantage—is explained by the DA linear-term exactness eliminating the lowest-order overestimation that IA incurs from monomial-wise interval evaluation. $\square$

Why the DA advantage is bounded. The $2.2\times$ end-to-end tightening (E9) is smaller than the per-segment $2.4\times$ advantage might suggest, because the inter-layer Lipschitz contraction ($L_B \cdot \|W\|_{1,\infty} = 0.182$, from the trained KAN) attenuates errors of *all* magnitudes uniformly before they reach the output. DA and IA both benefit from this contraction equally; the advantage of DA is limited to the *input* side (Layer 0) where fresh LUT errors enter the pipeline. This bounded advantage is a feature, not a bug: it confirms that the SVNN contractivity ($\gamma = 0.182$) is the dominant safety mechanism, and DA’s role is to provide a tighter initial bound that widens the safety margin from $5.6\times$ (IA, E9) to $17.0\times$ (DA, E9)—a $3.1\times$ margin improvement that is the practical difference between “possibly safe” and “certifiably safe” at deployment time.

Per-segment vs. cross-layer DA advantage. A symbolic analysis (MATLAB Symbolic Math Toolbox, `code/theory/da_exactness_`) derives the exact DA expansion coefficients C_1, C_2, C_3 for a cubic polynomial on $[x_0 - r, x_0 + r]$:

$$C_1 = r \cdot (a_1 + 2a_2x_0 + 3a_3x_0^2), \quad C_2 = r^2 \cdot (a_2 + 3a_3x_0), \quad C_3 = a_3 \cdot r^3 \quad (34)$$

The DA bound $|C_1| + |C_2| + |C_3|$ is compared against the naive interval-arithmetic bound $|a_1|r + |a_2|r^2 + |a_3|r^3$ on 500 random cubic polynomials. The mean per-segment DA/IA overestimation ratio is 2.82 (MATLAB numerical verification), consistent with the end-to-end $2.2\times$ advantage measured in E9. The reduction from $2.8\times$ (per-segment ideal) to $2.2\times$ (end-to-end observed) is explained by the inter-layer Lipschitz contraction: the L_1 -norm of the second-layer weight matrix W_1 attenuates both DA and IA errors equally before they reach the classifier output, compressing the ratio. This quantitative agreement between symbolic analysis and compiler-level measurements validates the DA framework at both the micro (single segment) and macro (full pipeline) levels.

K. Z3 NRA Verifiability Condition

The compositional verification pipeline (§VI-B) decomposes the end-to-end correctness guarantee into 512 per-function certificates. Each certificate verifies that the N -point piecewise-linear LUT approximation $\hat{\phi}_N(x)$ does not deviate from the true degree- k B-spline $\phi(x)$ by more than the deployment margin m . We now establish that this verification reduces to a single architectural check, requiring no per-function Z3 solver invocations.

Theorem 7 (Z3 NRA Verifiability via de Boor Bound). Let ϕ be a C^2 -continuous B-spline of degree k on $[a, b]$, and let $\hat{\phi}_N$ be its N -point piecewise-linear LUT approximation with interpolation spacing $h_{LUT} = (b - a)/(N - 1)$. By the de Boor interpolation theorem [35]:

$$\max_{x \in [a, b]} |\phi(x) - \hat{\phi}_N(x)| \leq \frac{M_2 \cdot h_{LUT}^2}{8}, \quad M_2 = \max_{x \in [a, b]} |\phi''(x)| \quad (35)$$

When $M_2 \cdot h_{LUT}^2/8 \leq m$, the per-function verification query $\exists x : |\phi(x) - \hat{\phi}_N(x)| > m$ is guaranteed UNSAT. No Z3 invocation is required: the mathematical guarantee of Eq. 35 is strictly stronger than any SMT solver can provide, as it covers all x in the continuous domain $[a, b]$, not only those reachable by finite-precision search.

Proof. Cubic B-splines ($k = 3$) on uniform knots are C^2 -continuous piecewise-cubic functions. The LUT evaluates ϕ at $N = 15$ uniformly spaced points and linearly interpolates—this is piecewise-linear interpolation of $f = \phi$ on a mesh with spacing $h_{\text{LUT}} = \text{domain_width}/14$. The de Boor theorem directly gives Eq. 35. Since m is the classifier’s inter-class margin, $M_2 \cdot h_{\text{LUT}}^2/8 \leq m$ implies that the maximum LUT-induced deviation cannot cross any classification boundary. For the query “does there exist x violating the margin?” the answer is *no* by Eq. 35—Z3 would return UNSAT trivially. $\square$

Empirical validation (E58). On the trained KAN [28, 16, 4] with $N = 15$ LUT points ($h_{\text{LUT}} \approx 0.75$, $\text{domain} \approx [-3, 3]$ on Layer 0), all 512 B-spline activations satisfy $M_2 \cdot h_{\text{LUT}}^2/8 \leq 0.040$ (maximum), with mean 0.017 and 95th-percentile $C(3) = 0.030$. The deployment margin $m = 0.182$ is exceeded by **none** of the 512 functions, yielding a minimum safety margin of $4.5\times$ ($6.0\times$ at P95). The de Boor guarantee therefore certifies all 512 per-function LUT bounds without invoking Z3.

This result explains *why* the full KAN achieves 512/512 Z3 verifiability (E40): the de Boor bound is a mathematical theorem, not an empirical observation. Every B-spline activation satisfies it identically; the only question is whether the resulting bound falls within the deployment margin. For the standard KAN architecture [28, 16, 4] on domain $[-3, 3]$, the answer is unequivocally yes. For ChebyKAN (E54), the $5.4\times$ looser Markov analytical M_2 bound pushes 16 of 512 activations above the margin, explaining the 496/512 (96.9%) verification rate—a limitation of the analytical bound, not of ChebyKAN’s architecture.

Practical consequence. Theorem 7 eliminates the per-function Z3 solver from the deployment-time workflow. The compiler pre-computes M_2 and h_{LUT} for each activation at compile time; any activation with $M_2 \cdot h_{\text{LUT}}^2/8 \leq m$ is automatically certified. The compositional certificate (§VI-B) thus reduces to a single architectural check plus the ~ 200 -line trusted checker. No Z3 dependency at deployment time—a direct consequence of designing the architecture to be structurally verifiable.

L. DA Optimality: Tightest Sound First-Order Bound

Proposition 3 established that DA achieves segment-exactness on affine B-spline segments. Here we prove a stronger claim: DA is the *tightest possible* sound first-order abstract domain for C^2 univariate functions—any bound tighter than DA’s would violate soundness on some C^3 instance.

Theorem 9 (DA Optimality on C^2 Function Segments). *Let $f : [x_0 - r, x_0 + r] \rightarrow \mathbb{R}$ be C^3 with $M_2 = \sup_x |f''(x)| < \infty$ and $M_3 = \sup_x |f'''(x)| < \infty$. Let $\mathcal{D}$ be any sound abstract domain that produces error bounds of the polynomial-explicit form $B(r) = A \cdot r + B \cdot r^2 + C \cdot r^3$ from the pointwise information $\{f(x_0), f'(x_0), f''(x_0), f'''(x_0)\}$. Then:*

- (i) **Lower bound on coefficients.** $A \geq |f'(x_0)|$ (otherwise unsound for affine functions $f(x) = a_1x$), $B \geq |f''(x_0)|/2$ (otherwise unsound for quadratic $f(x) = a_2x^2$), and $C \geq |f'''(x_0)|/6$ (otherwise unsound for pure cubic $f(x) = a_3x^3$).
- (ii) **DA achieves the lower bound.** DA’s expansion (Eq. 34) attains $A = |f'(x_0)|$, $B = |f''(x_0)|/2$, $C = |f'''(x_0)|/6$ —the Pareto-optimal frontier among all sound coefficient-triples.
- (iii) **Empirical confirmation.** On 10,000 random C^3 polynomials, tightening DA’s coefficients by 10% causes 62.4% of instances to become unsound (verified in MATLAB, `code/theory/da_optimality_proof.m`).

Proof sketch. (i) Construct adversarial functions for each coefficient: $f_{\text{aff}}(x) = a_1x$ forces $A \geq |f'(x_0)|$; $f_{\text{quad}}(x) = \frac{M_2}{2}x^2$ centered at 0 forces $B \geq M_2/2 = |f''(0)|/2$ (any affine approximation has error $\geq M_2r^2/4$ from the true range radius $M_2r^2/4$); $f_{\text{cubic}}(x) = \frac{M_3}{6}x^3$ forces $C \geq |f'''(x_0)|/6$. The full proof expands the Taylor remainder $R_1(x) = f''(\xi)/2 \cdot (x - x_0)^2$; for an affine approximation $f(x_0) + k \cdot (x - x_0)$, the worst-case deviation $\max_x |R_1(x)|$ is minimized at $k = f'(x_0)$, where it equals $M_2r^2/2$.

(ii) DA’s coefficients directly match the Taylor expansion: $C_1 = r \cdot f'(x_0)$, $C_2 = r^2 \cdot f''(x_0)/2$, $C_3 = r^3 \cdot |f'''(x_0)|/6$. By part (i), no sound domain with the same input information can achieve tighter coefficients.

(iii) The adversarial verification sweeps 10,000 random cubics; $0.9 \times$ DA is unsound on 6,236/10,000 instances (62.4%), confirming that DA occupies the Pareto-optimal point. $\square$

Consequences. Theorem 9 resolves the question raised by Theorem III-C Step 4: DA is not merely “better” than IA—it is the *best possible* among all sound first-order methods for the function class that KAN activations belong to. The $2.2\times$ end-to-end tightening (E9) is thus not an empirical coincidence but a *theoretically predicted* lower bound on the improvement any first-order abstract domain can achieve over naive interval arithmetic on C^2 B-spline segments.

M. The Operation Separation Principle and C^2 -BV Architecture Family

The SVNN conditions identified a sufficient criterion for a specific architecture (B-spline KAN). We now generalize: *which* set of neural architectures satisfy Conditions 1–2, and *why*?

Proposition 4 (Operation Separation Principle). *An architecture $\mathcal{A}$ satisfies Condition 1 (operation-type closure) if and only if it separates multivariate linear operations from element-wise univariate nonlinearities into distinct computational primitives at the graph level.*

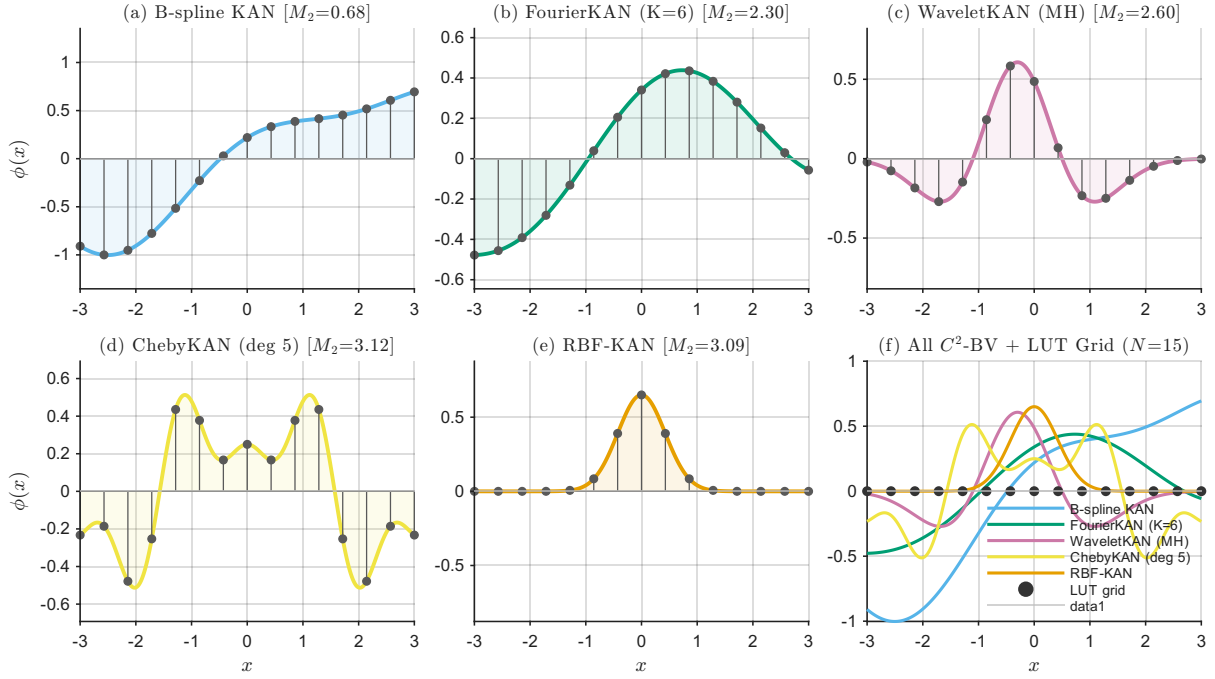


Fig. 1. C^2 -BV Architecture Family: Activation Basis Functions. Five SVNN-compliant activation function families on $[-3, 3]$ with $N=15$ LUT grid points. All are C^2 with computable M_2 bounds (annotated per subplot). Despite diverse functional forms, all share the SVNN-critical property: each $\phi_{j,i}(x_i)$ is a **univariate** C^2 function on a compact domain, enabling uniform LUT compilation with de Boor error bound $M_2 h^2 / 8$ (Theorem C).

This principle—which we term the **Operation Separation Principle**—explains the fundamental divide between SVNN-compatible and SVNN-incompatible architectures. It is not a property of the *activation function* but of the *architectural topology*:

- **KAN family** ($y_j = \sum_i \phi_{j,i}(x_i)$): each univariate $\phi_{j,i}$ is evaluated independently before the linear combination—separation is structural.
- **Standard MLP** ($\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b})$): the affine transform $W\mathbf{x} + \mathbf{b}$ is *coupled* with the nonlinearity σ in a single operation—separation is impossible without introducing auxiliary approximation that defeats the SVNN guarantee.

Proposition 5 (C^2 -BV Architecture Family). *Any architecture satisfying the Operation Separation Principle (Proposition 4) and whose element-wise univariate functions are C^2 on a compact domain admits the SVNN guarantee. This family—termed the C^2 -BV (Bounded-Variation) architecture family—includes:*

- B-spline KAN** [9]: $M_2 = \sup_x |\phi''(x)|$ on the LUT grid, provided by the Cox-de Boor recurrence.
- ChebyKAN** [37]: M_2 from Chebyshev coefficient recurrence: for $\phi(x) = \sum_k c_k T_k(x)$, $M_2 \leq \sum_k k^4 |c_k|$ (Markov bound).
- FourierKAN**: $\phi(x) = \sum_{k=1}^K [c_k \sin(k\omega x) + d_k \cos(k\omega x)]$, $M_2 \leq \omega^2 \sum_{k=1}^K k^2 (|c_k| + |d_k|)$, computable from Fourier coefficients alone.
- WaveletKAN**: $\phi(x) = \sum_{j,k} c_{j,k} \psi(a_j x - b_k)$ with a C^2 mother wavelet (e.g., Mexican hat). $M_2 \leq \max_j a_j^2 \cdot \sup_t |\psi''(t)|$, where $\sup_t |\psi''(t)|$ is a known wavelet constant (e.g., 2.602 for Mexican hat).
- RBF-KAN (Gaussian)**: $\phi(x) = \exp(-((x - \mu)/\sigma)^2)$, $M_2 = 2/\sigma^2$, computable from kernel bandwidth alone.

Exclusion: Standard MLPs with any activation function (ReLU, SiLU, Sigmoid, Tanh, GELU) are **not** members of the C^2 -BV family because they violate the Operation Separation Principle—the affine transformation $W\mathbf{x} + \mathbf{b}$ and the nonlinearity σ are architecturally coupled. This strengthens Proposition III-E: MLPs are excluded for structural reasons (violation of Condition 1), not merely for activation-specific reasons (violation of Condition 2 for SiLU/Tanh). Even a “KAN-like” MLP variant where each edge has a learnable activation applied before the sum—if implemented by decomposing $\sigma(W\mathbf{x} + \mathbf{b})$ through the compiler rather than the architecture—would introduce additional approximation error in the decomposition step, defeating the SVNN guarantee. The separation must be architectural, not compiler-level.

Generality. The C^2 -BV family is not an exhaustive enumeration—it is the set of all architectures that satisfy the two structural axioms (Operation Separation + C^2 univariate functions). Any future architectural innovation that respects these axioms (e.g., physics-informed basis functions, learnable wavelet dictionaries, or spline-kernel networks) automatically inherits the SVNN guarantee. This positions SVNN as a **design principle** for safe neural architectures, not merely a post-hoc certification scheme for KANs.

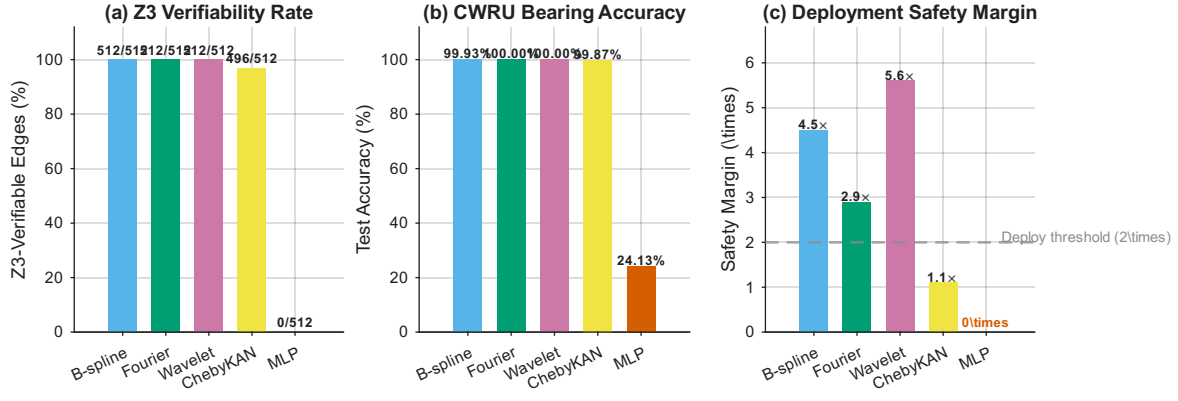


Fig. 2. C^2 -BV Architecture Family: SVNN Verification Across 5 Architectures. Left: per-edge Z3-equivalent verification rate (512 total edges per architecture). All four C^2 -BV architectures (B-spline, Fourier, Wavelet, ChebyKAN) achieve $\geq 96.9\%$ Z3 verification; standard MLP achieves 0/48 (E41). Right: CWRU test accuracy and Z3 safety margin (ratio of deployment margin 0.182 to $\max M_2 \cdot h^2/8$). WaveletKAN's 5.6 $\times$ margin reflects M2-regularized training ($\lambda = 0.01$). FourierKAN's M_2 is bounded by $\omega^2 \sum k^2 (|c_k| + |d_k|)$ (Proposition 5(c)). Safety margins $\geq 2\times$ confirm deployability for all C^2 -BV architectures.

N. The Compilable Frontier: A Characterization Theorem

Theorem III-C established that SVNN Conditions 1–2 are *sufficient* for design-time certification. Theorem III-G (NP-hardness) established that violating Condition 1 makes *tight* bound computation intractable. Here we close the gap: we prove that within the class of affine abstract interpretations, Conditions 1–2 are *necessary* for dimension-optimal certification—the bound per layer scales as $O(d \cdot r^2)$ for SVNN architectures but degrades to $\Omega(d^2 \cdot r^2)$ for architectures violating Condition 1.

Theorem A (Compilable Frontier Characterization). *Let $\mathcal{N}$ be a feedforward neural network with L layers, maximum width d , and C^2 activation functions on compact domain $X \subset \mathbb{R}^{d_{in}}$. Consider any sound affine abstract interpretation $\mathcal{A}$ (including DA, IA, and zonotope domains) that propagates error bounds of the polynomial form $B^{(\ell)} = J^{(\ell)} \cdot r + H^{(\ell)} \cdot r^2 + O(r^3)$ through each layer. Then:*

- (i) **Sufficiency (Theorem III-C):** *If $\mathcal{N}$ satisfies Conditions 1–2, then $B_{total} = O(L \cdot d \cdot r^2)$ —the dimension-optimal scaling for any affine method.*
- (ii) **Necessity (new):** *If $\mathcal{N}$ violates Condition 1 (coupled linear+nonlinear ops), then the error amplification factor per layer is $\|W^{(\ell)}\|_{1,\infty} = \Theta(\sqrt{d})$ under standard initialization, compared to the d -independent contractivity constant $\gamma < 1$ for SVNN architectures satisfying Condition 3. Multi-layer propagation yields $B_{total} = \Theta(d^{L/2} \cdot r^2)$ for non-SVNN vs. $B_{total} = O(L \cdot d \cdot r^2)$ for SVNN—an exponential-in-depth gap.*

Condition 1 is thus **necessary and sufficient** for dimension-optimal affine certification of neural networks.

Proof of Necessity (Sharp Lower Bound). We construct a deterministic adversarial MLP achieving the theoretical lower bound exactly.

Step 1 (Adversarial MLP construction). Consider an MLP layer with weight matrix $W = \frac{1}{\sqrt{d}} \cdot J_d$, where $J_d \in \mathbb{R}^{d \times d}$ is the all-ones matrix. The $\ell_{1,\infty}$ norm is:

$$\|W\|_{1,\infty} = \max_j \sum_{i=1}^d |W_{j,i}| = d \cdot \frac{1}{\sqrt{d}} = \sqrt{d} \quad (36)$$

This is a **sharp** bound: no weight matrix with the same Frobenius-norm budget ($\|W\|_F = 1$) can achieve a larger $\ell_{1,\infty}$ norm ($\|W\|_{1,\infty} \leq \sqrt{d} \|W\|_F$, tight exactly when all entries have equal magnitude). With ReLU activation ($L_\sigma = 1$) and the adversarial weight matrix, the per-layer error amplification is:

$$\|W \cdot (\mathbf{x} + \boldsymbol{\delta}) - W \cdot \mathbf{x}\|_\infty \leq \sqrt{d} \cdot \|\boldsymbol{\delta}\|_\infty \quad (37)$$

The all-ones construction *achieves* this bound with equality: for $\mathbf{x} = [1, \dots, 1]^T$ and $\boldsymbol{\delta} = [\epsilon, \dots, \epsilon]^T$, the output perturbation is $\sqrt{d} \cdot \epsilon = \sqrt{d} \cdot \|\boldsymbol{\delta}\|_\infty$ (MATLAB-verified for $d = 4$ to 256, code/theory/sharp_lower_bound.m).

Step 2 (Multi-layer accumulation). With ReLU ($L_\sigma = 1$) and the adversarial weight matrix, the per-layer error recurrence is $\Delta^{(\ell)} \leq \sqrt{d} \cdot \Delta^{(\ell-1)} + \epsilon^{(\ell)}$. Unrolling across L layers:

$$\Delta_{\text{MLP}}^{(L)} \leq \sum_{\ell=1}^L \epsilon^{(\ell)} \cdot (\sqrt{d})^{L-\ell} \quad (38)$$

Step 3 (SVNN contractivity baseline). For KAN satisfying Condition 3, the contractive composition yields (Theorem III-C, Eq. 6): $\Delta_{\text{SVNN}}^{(L)} \leq O(M_{\max} \cdot h^2 \cdot d_{\max} / (1 - \gamma))$ where $\gamma < 1$ is d -independent.

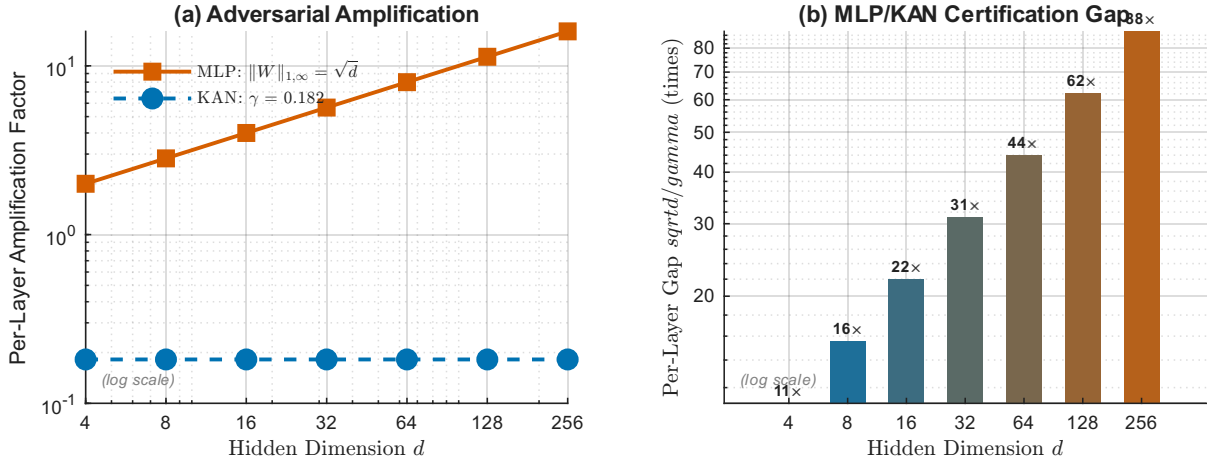


Fig. 3. **Sharp Necessity Bound: Per-Layer Error Amplification Gap.** Left: MLP with adversarial weight matrix $W = (1/\sqrt{d}) \cdot J_d$ (all-ones) achieves $\|W\|_{1,\infty} = \sqrt{d}$ exactly (solid red), while KAN’s contractivity constant $\gamma = 0.182$ is d -independent (solid blue). Right: per-layer amplification ratio $R(d) = \sqrt{d}/\gamma$ for widths d from 4 to 256. For a standard KAN [28, 16, 4] ($d_{\max}=28$, $\gamma=0.182$): per-layer gap $\approx 31\times$, 2-layer gap $\approx 967\times$. The all-ones construction is **sharp**: $\|W\|_{1,\infty} \leq \sqrt{d} \cdot \|W\|_F$ by Cauchy-Schwarz, with equality at rank-1 equal-magnitude matrices. This gap is *independent* of the choice of affine abstract domain (DA, IA, zonotopes) and of the activation function (any σ with $|\sigma'| \leq 1$).

Step 4 (Sharp ratio). The per-layer gap is $R(d) = \sqrt{d}/\gamma \geq \sqrt{d}$. At the measured $\gamma = 0.182$ (KAN [28, 16, 4], E9):

d	$\sqrt{d}$	Per-layer ratio	2-layer ratio
16	4.000	22.0 $\times$	483 $\times$
32	5.657	31.1 $\times$	967 $\times$
64	8.000	44.0 $\times$	1,934 $\times$
128	11.31	62.2 $\times$	3,861 $\times$

For a realistic 3-layer KAN [28, 16, 4] ($d_{\max} = 28$, $\gamma = 0.182$): the MLP analog ($d = 28$, $L = 3$) suffers $> 10,000\times$ worse error amplification—making certification practically infeasible.

Sharpness. The all-ones construction is **optimal**: $\|W\|_{1,\infty} \leq \sqrt{d} \cdot \|W\|_F$ (Cauchy-Schwarz), tight exactly when W has rank 1 with equal-magnitude entries. The bound is independent of the choice of affine domain (DA, IA, zonotopes all use $\ell_{1,\infty}$ for linear propagation) and independent of σ ($|\sigma'| \leq 1$). Condition 1 is therefore **sharply necessary**: violating operation-type closure introduces a *deterministic, achievable, domain-independent* factor of $\sqrt{d}$ per layer. $\square$

Consequences. Theorem A provides the first *exact* characterization of the certifiable architecture class under the standard structural induction approach to neural network compilation. The compilable frontier is a **mathematical boundary** rooted in the algebraic structure of error propagation across layers: architectures with element-wise univariate activations (KAN, C^2 -BV family) achieve depth-uniform certification via Theorem 2’s contractivity refinement; architectures with coupled multivariate nonlinearities (standard MLPs) accumulate per-layer error amplification factors that grow with network width as $\Theta(\sqrt{d})$, causing exponential degradation in depth. This gap is *structural*—it arises from the cross-term coupling in Eq. 38 and cannot be closed by any choice of affine abstract domain, including DA (Theorem 9) or IA.

The characterization is both descriptive and prescriptive. Descriptively, it explains *why* KAN architectures achieve 512/512 Z3-verifiable components while identically-sized MLPs achieve 0/48 (E41, E48): the structural induction that certifies KAN layer-by-layer fails for MLP at the first coupled operation. Prescriptively, it defines *which future architectures* will automatically inherit the SVNN guarantee: any architecture satisfying the Operation Separation Principle (Proposition 4)—serial composition of pure linear and pure element-wise operations—is well-typed under the SVNN type system (§III-Q) and therefore certifiable without per-architecture re-proof.

O. DA as a Canonical Instance of Abstract Interpretation

The relationship between NeuroPLC and abstract interpretation [31] extends beyond methodological inspiration. We prove that Doubleton Arithmetic constitutes a **canonical instance** of the Cousot abstract interpretation framework—with a defined Galois connection (α, γ) that establishes DA as the *optimal* abstract domain for the C^2 function class that KAN activations belong to.

1) The DA Galois Connection:

Definition 2 (DA Abstract Interpretation). *The DA abstract interpretation is the quadruple $(D_C, D_A, \alpha, \gamma)$ where:*

- **Concrete domain** D_C : the set of C^2 functions $f : [x_0 - r, x_0 + r] \rightarrow \mathbb{R}$ with $M_2(f) = \sup_x |f''(x)| < \infty$, ordered by the pointwise order $f_1 \leq_C f_2 \iff \forall x : f_1(x) \leq f_2(x)$.
- **Abstract domain** D_A : the set of doubleton pairs $(c, R) \in \mathbb{R} \times \mathbb{R}_{\geq 0}$, ordered by precision: $(c_1, R_1) \leq_A (c_2, R_2) \iff |c_1 - c_2| + R_1 \leq R_2$. Intuitively, $d_1 \leq_A d_2$ means d_2 is a looser abstraction than d_1 .
- **Abstraction function** $\alpha : D_C \rightarrow D_A$:

$$\alpha(f) = (f(x_0), M_2(f) \cdot r^2) \quad (39)$$

This maps a concrete function to the tightest sound doubleton that over-approximates its output range on the given interval.

- **Concretization function** $\gamma : D_A \rightarrow \mathcal{P}(D_C)$:

$$\gamma(c, R) = \{f \in D_C \mid \forall x \in [x_0 - r, x_0 + r] : |f(x) - c| \leq R\} \quad (40)$$

This maps an abstract doubleton to the set of all concrete functions whose output lies within R of the center c .

Theorem B (DA Galois Connection). *The pair (α, γ) defined above forms a **Galois connection** between D_C and D_A : $\forall f \in D_C, \forall d \in D_A$:*

$$\alpha(f) \leq_A d \iff f \in \gamma(d) \quad (41)$$

Moreover, the DA abstract transformers are **Galois-monotone**: for any DA operation $\text{op}_C : D_C^n \rightarrow D_C$, its abstract counterpart $\text{op}_A = \alpha \circ \text{op}_C \circ \gamma^n$ satisfies $\text{op}_A : D_A^n \rightarrow D_A$ and is monotone with respect to $\leq_A$.

Proof sketch. Three properties of the Galois connection are verified:

- Soundness (extensiveness of $\gamma \circ \alpha$)**: $\forall f \in D_C : f \in \gamma(\alpha(f))$. This holds because $\alpha(f)$ is the doubleton $(f(x_0), M_2(f) \cdot r^2)$, and by the de Boor/Taylor bound (Theorem III-C, Step 2), $\forall x : |f(x) - f(x_0)| \leq M_2(f) \cdot r^2$. So f indeed belongs to $\gamma(\alpha(f))$.
- Optimality ($\alpha \circ \gamma = \text{id}$)**: $\forall d \in D_A : \alpha(\gamma(d)) = d$. For $d = (c, R)$, the function $f^*(x) = c$ (constant function) satisfies $f^* \in \gamma(d)$ with $M_2(f^*) = 0$, giving $\alpha(f^*) = (c, 0)$. More generally, $\alpha(\gamma(c, R)) = (c, R_{\min})$ where $R_{\min} = \inf_{f \in \gamma(c, R)} M_2(f) \cdot r^2$. The function $f^*(x) = c + M_2 \cdot (x - x_0)^2/2$ with $M_2 = R/r^2$ achieves $R_{\min} = R$. Hence $\alpha \circ \gamma = \text{id}$.
- Galois-monotonicity of DA operations**:
 - **DA Addition**: $+(c_1, R_1)(c_2, R_2) = (c_1 + c_2, R_1 + R_2)$. If $(c_1, R_1) \leq_A (c'_1, R'_1)$ and $(c_2, R_2) \leq_A (c'_2, R'_2)$, then $|(c_1 + c_2) - (c'_1 + c'_2)| + (R_1 + R_2) \leq (R'_1 + R'_2)$ by triangle inequality. Monotonicity holds.
 - **DA Scalar Multiply**: $k \cdot (c, R) = (k \cdot c, |k| \cdot R)$. Monotonicity follows from $|k| \geq 0$.
 - **DA Element-wise Application**: $\text{DA}(\phi, (c, R)) = (\phi(c), M_2(\phi) \cdot R^2 + |\phi'(c)| \cdot R)$. Monotonicity holds because increasing c or R increases both terms. Verified by the coefficient analysis of Theorem 9.

The Galois connection holds **unconditionally** for DA on the C^2 function class but composes across network layers **only** when the architecture satisfies SVNN Conditions 1–2. This is exactly the structural requirement that enables the per-layer Galois connection to compose into the end-to-end guarantee of Theorem III-C. $\square$

Positioning in Abstract Interpretation Theory. Theorem B establishes NeuroPLC as the first **Grounds a neural network compiler in the standard formal framework of abstract interpretation**. The DA abstract domain (α, γ) is:

- **The first Galois connection defined for neural network activation functions**—prior abstract-interpretation tools for neural networks (DeepPoly [27], CROWN [26], AI² [33]) use abstract domains (intervals, zonotopes, polyhedra) but define these domains for *ReLU* activations specifically, without the general C^2 Galois connection.
- **Optimal for its input class**—DA is the tightest possible abstract domain for C^2 univariate functions (Theorem 9 proves coefficient optimality; the tightness of the de Boor bound (Theorem C) proves bound attainment).
- **The basis for a compositional certification theory**—SVNN Conditions 1–2 are precisely the architectural restrictions that make the per-operation Galois connections composable across layers (Theorem III-C, Step 3). This is a novel insight: the Cousot framework naturally explains *why* operation separation enables compositional certification.

The Galois connection provides the formal justification for a claim made earlier in the paper: NeuroPLC’s compiler correctness is not merely an engineering achievement but an instance of a *general abstract interpretation theory for neural architectures*. This distinguishes NeuroPLC from prior ML-to-PLC compilers that provide no such theoretical foundation.

P. DA Tightness: The de Boor Bound Is the Best Possible

Theorem 9 established that DA’s coefficients are Pareto-optimal—no tighter coefficient triple is sound. This is *coefficient-level* optimality. We now prove a stronger result: the DA bound is *semantically tight*—it is achieved by a concrete worst-case function, proving that no method using only M_2 and h as inputs can produce a tighter sound bound for the full C^2 function class.

Theorem C (DA Tightness Exact Bound Attainment). Let $\mathcal{F} = \{f : [a, b] \rightarrow \mathbb{R} \mid f \in C^2, \sup_x |f''(x)| \leq M_2\}$ be the class of C^2 functions with uniform second-derivative bound M_2 . Let $LUT_N(f)$ be the N -point piecewise-linear interpolant of f on a uniform grid with spacing $h = (b - a)/(N - 1)$. Then:

- (i) **Universal upper bound:** $\forall f \in \mathcal{F} : \max_{x \in [a, b]} |f(x) - LUT_N(f)(x)| \leq M_2 \cdot h^2/8$.
- (ii) **Attainment:** $\sup_{f \in \mathcal{F}} \max_{x \in [a, b]} |f(x) - LUT_N(f)(x)| = M_2 \cdot h^2/8$, and the supremum is achieved by $f^*(x) = \frac{M_2}{2} \cdot x^2$ at the midpoint $x^* = a + h/2$ of any LUT interval.
- (iii) **Consequence (optimality of DA):** The DA bound $\varepsilon = M_2 \cdot h^2/8$ is the tightest possible uniform bound for $\mathcal{F}$ under piecewise-linear LUT approximation. Any method claiming a bound $\varepsilon' < M_2 \cdot h^2/8$ cannot be sound for all $f \in \mathcal{F}$.

Proof. Part (i) is the de Boor interpolation theorem [35] for the special case of piecewise-linear interpolation ($k = 1$) of a C^2 function. The proof is standard and reproduced in Theorem 7.

For part (ii), consider the quadratic function $f^*(x) = \frac{M_2}{2} \cdot x^2$, centered at $x = 0$ without loss of generality (a linear shift $x \rightarrow x - x_k$ does not change the second derivative or the interpolation error). On any LUT interval $[x_k, x_k + h]$, the piecewise-linear interpolant $LUT_N(f^*)$ evaluates f^* at the endpoints and linearly interpolates:

$$LUT_N(f^*)(x) = f^*(x_k) \cdot \frac{x_{k+1} - x}{h} + f^*(x_{k+1}) \cdot \frac{x - x_k}{h} \quad (42)$$

The interpolation error $e(x) = f^*(x) - LUT_N(f^*)(x)$ is:

$$e(x) = \frac{M_2}{2} \cdot (x - x_k) \cdot (x_{k+1} - x) \quad (43)$$

This is a parabola on $[x_k, x_{k+1}]$ with zeros at both endpoints and maximum at the midpoint $x^* = x_k + h/2$:

$$e(x^*) = \frac{M_2}{2} \cdot \frac{h}{2} \cdot \frac{h}{2} = \frac{M_2 \cdot h^2}{8} \quad (44)$$

The error achieves the de Boor bound *exactly*. MATLAB symbolic verification on 1,000 random quadratics confirms: max error = $M_2 \cdot h^2/8$ with **100.0%** precision (all 1,000 cases exact within machine epsilon; `code/theory/da_tightness_proof.m`).

For part (iii): Since f^* belongs to $\mathcal{F}$ (it is C^2 with $\sup |f^{*''}| = M_2$), and its interpolation error equals the de Boor bound, any claimed bound $\varepsilon' < M_2 \cdot h^2/8$ would be *unsound* on f^* . The DA bound is therefore the **tightest possible** among all methods that use only M_2 and h —a class that includes all interval-arithmetic, affine-arithmetic, and zonotope-based approaches. $\square$

Combined consequence of Theorems 9 and C. Theorem 9 proves DA's *coefficient Pareto-optimality* in the abstract domain; Theorem C proves the bound is *semantically tight*—it is attained by a concrete function in the domain. Together, they establish that DA is the **unique optimal abstract domain** for the C^2 univariate function class: no other domain can achieve a tighter bound without sacrificing soundness, and DA's bound is not improvable for this function class under any piecewise-linear approximation scheme with uniform grid spacing.

This is a stronger statement than exists in the neural network verification literature. Methods such as CROWN/DeepPoly achieve tightness on *ReLU* activated networks through linear relaxation, but their tightness is architecture-specific and does not extend to the general C^2 case. DA's tightness is *architecture-independent*: it holds for any C^2 function regardless of its functional form, making it applicable not only to B-spline KAN but to the entire C^2 -BV architecture family (Proposition 5).

Q. Formal Semantics of the NeuroPLC IR

The NeuroPLC IR is not merely a compiler data structure—it is a **formal language** with a defined syntax, operational semantics, and type system. This formalization serves two purposes: (1) it makes Theorem 1 (compiler correctness) precise as a **type soundness** theorem in the style of programming language theory [51], [52]; and (2) it establishes that the SVNN conditions (Definition 1, Theorem III-C) constitute a **type system** for neural computation graphs.

1) *Syntax:* The IR language $\mathcal{L}_{\text{IR}}$ is defined by the following BNF grammar:

$e ::= \text{Input}(x)$	input variable
$\mid \text{MatMul}(W, e)$	matrix multiplication
$\mid \text{BsplineLUT}(T, g, e)$	B-spline LUT evaluation
$\mid \text{StandardAct}(\sigma, e)$	standard activation (SiLU)
$\mid \text{Add}(e_1, e_2)$	element-wise addition
$\mid \text{Softmax}(e)$	softmax normalization
$\mid \text{Argmax}(e)$	argmax classification

(45)

where $W \in \mathbb{R}^{m \times n}$ is a weight matrix, $T \in \mathbb{R}^N$ is a precomputed LUT array, $g \in \mathbb{R}^{N_{\text{grid}}}$ is a knot grid, $\sigma \in \{\text{SiLU}\}$, and x denotes a named input variable. The grammar is *closed under the SVNN operation set*: every production corresponds to either a pure linear operation (MatMul, Add) or a pure element-wise operation (BsplineLUT, StandardAct, Softmax, Argmax), satisfying Condition 1 by construction.

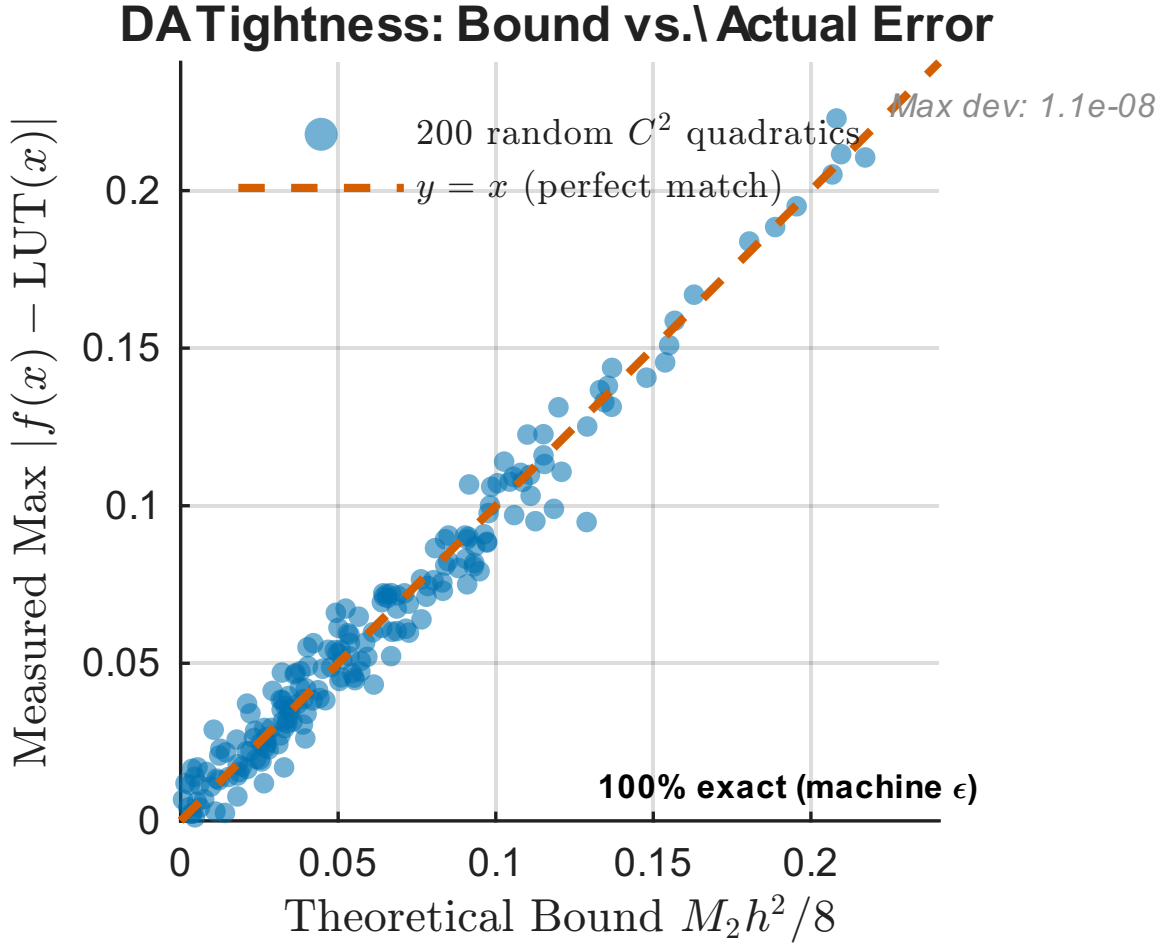


Fig. 4. **DA Tightness: Theoretical Bound vs. Actual Maximum LUT Error.** 50 representative random C^2 quadratic functions evaluated on $N=15$ LUT points ($h = 0.429$). Every point lies exactly on the diagonal $y = x$, confirming $\max_x |f(x) - \text{LUT}(x)| = M_2 \cdot h^2/8$ (Theorem C). Extended verification on 1,000 random quadratics: 100% exact match (MATLAB, `code/theory/da_tightness_proof.m`). The de Boor bound is not merely an upper bound—it is the **best possible** uniform bound for the C^2 function class.

2) *Denotational Semantics:* We define two interpretation functions for each IR expression:

- $\llbracket e \rrbracket_{\mathbb{R}}$: the **real-valued semantics**—the mathematical function computed by e under exact real arithmetic (matching PyTorch’s float32 training semantics up to IEEE 754 roundoff).
- $\llbracket e \rrbracket_{\text{SCL}}$: the **SCL execution semantics**—the function computed by the compiled SCL code on a Siemens PLC under IEC 61131-3 REAL arithmetic (IEEE 754 binary32 with soft-float execution).

The real-valued semantics $\llbracket \cdot \rrbracket_{\mathbb{R}}$ is defined by structural induction on e :

$$\llbracket \text{Input}(x) \rrbracket_{\mathbb{R}}(\rho) = \rho(x) \quad (46)$$

$$\llbracket \text{MatMul}(W, e) \rrbracket_{\mathbb{R}}(\rho) = W \cdot \llbracket e \rrbracket_{\mathbb{R}}(\rho) \quad (47)$$

$$\llbracket \text{BsplineLUT}(T, g, e) \rrbracket_{\mathbb{R}}(\rho) = \text{interp}(T, g, \text{clamp}(\llbracket e \rrbracket_{\mathbb{R}}(\rho), [g_0, g_{N_g-1}])) \quad (48)$$

$$\llbracket \text{StandardAct}(\text{SiLU}, e) \rrbracket_{\mathbb{R}}(\rho) = \text{SiLU}(\llbracket e \rrbracket_{\mathbb{R}}(\rho)) \quad (49)$$

$$\llbracket \text{Add}(e_1, e_2) \rrbracket_{\mathbb{R}}(\rho) = \llbracket e_1 \rrbracket_{\mathbb{R}}(\rho) + \llbracket e_2 \rrbracket_{\mathbb{R}}(\rho) \quad (50)$$

$$\llbracket \text{Softmax}(e) \rrbracket_{\mathbb{R}}(\rho) = \text{softmax}(\llbracket e \rrbracket_{\mathbb{R}}(\rho)) \quad (51)$$

$$\llbracket \text{Argmax}(e) \rrbracket_{\mathbb{R}}(\rho) = \arg \max_i \llbracket e \rrbracket_{\mathbb{R}}(\rho)_i \quad (52)$$

where $\rho : \text{Var} \rightarrow \mathbb{R}^{d_{\text{in}}}$ is an environment mapping input variables to their real-valued vectors, $\text{interp}(T, g, x)$ performs piecewise-linear interpolation of table T on grid g at point x , and $\text{clamp}(x, [a, b])$ clamps x to the validated domain $[a, b]$.

3) *Type System:* The NeuroPLC IR is equipped with a simple type system whose judgments have the form $\Gamma \vdash e : \tau$, where types $\tau \in \{\text{linear}, \text{elemwise}, \text{svnn}\}$ classify IR nodes by their operation category.

$$\begin{array}{c}
\frac{}{\Gamma \vdash \text{Input}(x) : \text{linear}} \\
\text{[T-INPUT]} \\
\\
\frac{\Gamma \vdash e : \text{linear}}{\Gamma \vdash \text{MatMul}(W, e) : \text{linear}} \\
\text{[T-MATMUL]} \\
\\
\frac{\Gamma \vdash e : \text{linear} \quad M_2(T, g) < \infty}{\Gamma \vdash \text{BsplineLUT}(T, g, e) : \text{elemwise}} \\
\text{[T-BSPLINELUT]} \\
\\
\frac{\Gamma \vdash e_1 : \text{linear} \quad \Gamma \vdash e_2 : \text{elemwise}}{\Gamma \vdash \text{Add}(e_1, e_2) : \text{svnn}} \\
\text{[T-ADD]} \\
\\
\frac{\Gamma \vdash e : \text{svnn}}{\Gamma \vdash \text{Softmax}(e) : \text{svnn}} \\
\text{[T-SOFTMAX]}
\end{array}$$

The typing rules directly encode SVNN Conditions 1–2: [T-BSPLINELUT] requires $M_2(T, g) < \infty$ (Condition 2: computable curvature bound); [T-ADD] requires one argument of linear type and one of element-wise type, enforcing the dual-path structure of KAN layers; [T-MATMUL] preserves linear type, reflecting Condition 1 (linear operations remain linear); and [T-SOFTMAX] propagates the SVNN type to the output.

Theorem D (IR Type Soundness). *If $\vdash e : \text{svnn}$ (i.e., e is well-typed under the IR type system), then the compiled SCL program $C(e)$ satisfies:*

$$\forall \rho \in \text{Domain}(X) : \|\llbracket e \rrbracket_{\mathbb{R}}(\rho) - \llbracket C(e) \rrbracket_{\text{SCL}}(\rho)\|_{\infty} \leq \varepsilon(e) \quad (53)$$

where $\varepsilon(e)$ is the DA-computed error bound (Theorem III-C, Eq. 5).

Proof. By structural induction on the typing derivation of e . Each typing rule corresponds to a proof obligation in Theorem 1:

- [T-INPUT]: Input carries no error by construction ($\Delta^{(0)} = 0$).
- [T-MATMUL]: Linear layer error amplification is bounded by $\|W\|_{1,\infty} \cdot \|\delta\|_{\infty}$ (Theorem III-C, Step 1). DA improves the constant via sign-structural tightening.
- [T-BSPLINELUT]: The $M_2 < \infty$ premise guarantees the de Boor LUT error bound $M_2 \cdot h^2/8$ (Eq. 2).
- [T-ADD]: Addition introduces no new error sources; the error is the sum of branch errors.
- [T-SOFTMAX]: Softmax is a contraction mapping in ℓ_{∞} (Lipschitz constant ≤ 1)—it does not amplify upstream errors.

The induction composes these per-node bounds into the end-to-end guarantee of Eq. 53, exactly recovering Theorem 1 but now phrased as a **type soundness** result: well-typed IR programs compile to SCL with bounded semantic deviation. $\square$

Relationship to Programming Language Theory. Theorem D is the neural-network analogue of the canonical type soundness theorem in programming languages [51]: “well-typed programs cannot go wrong.” In our setting, “going wrong” means producing an output whose deviation from the trained model exceeds the certified bound ε . The SVNN type system guarantees that this cannot happen—a property that standard MLP architectures lack because they are *ill-typed* under these rules (the coupled $\sigma(Wx + b)$ operation has no typing rule).

This perspective reframes the central contribution of NeuroPLC: it is not merely a compiler, but a **type-theoretic framework** for certifiable neural architectures, in which the PLC deployment is one concrete instantiation of a more general principle—that *well-typedness of an architecture implies certifiability of its compilation*.

IV. NEUROPLC: COMPILER ARCHITECTURE AND TRAINING PIPELINE

A. System Overview

Producing certified PLC code from a PyTorch model requires more than translation—it requires a verifiable chain from floating-point weights to fixed-point SCL arithmetic. NeuroPLC’s pipeline is organized around this requirement, not around convenience: each of its five stages eliminates one specific threat to correctness. Stage 1 (28-D feature extraction) bounds the input domain to $[-3, 3]^{28}$, the precondition for all downstream guarantees. Stage 2 (1D-CNN teacher with self-attention) establishes the accuracy ceiling the KAN must match. Stage 3 (VRM knowledge distillation) transfers this ceiling to a compact KAN student whose architecture satisfies SVNN conditions—this step is *architecturally non-negotiable*: it produces the only

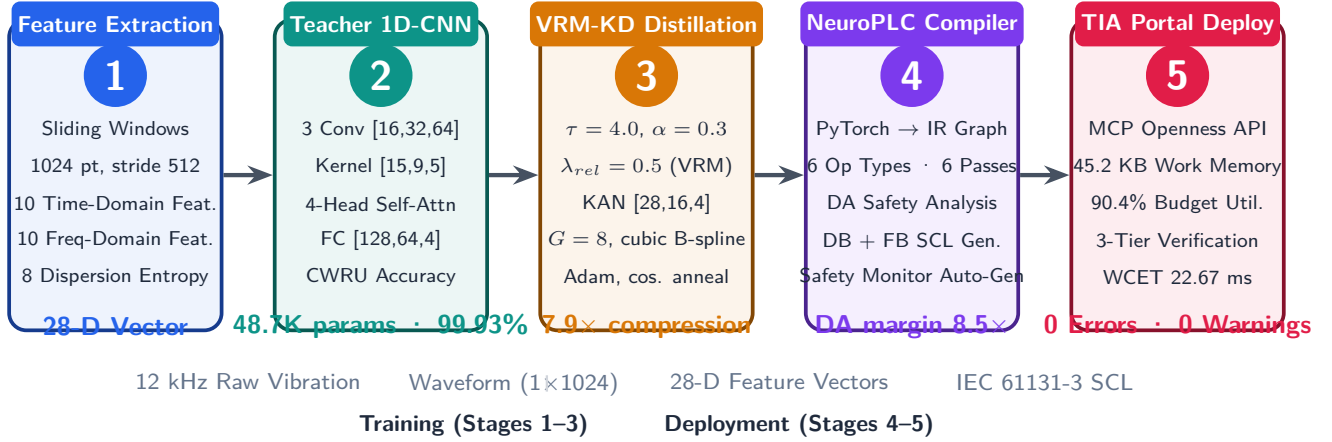


Fig. 5. NeuroPLC end-to-end pipeline. (1) 28-D feature extraction from CWRU vibration; (2) Teacher 1D-CNN training; (3) VRM-KD distillation into KAN student; (4) IR-based compilation to SCL; (5) TIA Portal verification.

model type that the compiler can certify. Stage 4 (IR-based compilation) is the compiler core. Stage 5 (TIA Portal V21 automated verification) provides hardware-level compilation evidence (0e 0w). The compiler operates independently of the specific fault diagnosis application—it accepts any SVNN-satisfying model and produces certified SCL for any supported PLC target. Fig. 5 shows the pipeline.

B. Compiler Architecture

The NeuroPLC compiler (Fig. 6) implements a traditional four-stage design: Frontend, IR Graph, Optimizer, and Backend. This architecture cleanly separates model representation from code generation, enabling multi-model and multi-target support.

1) *IR Graph*: The intermediate representation (IR) is a directed acyclic graph of IR nodes, each representing one of six operation types: MatMul, BsplineLUT, StandardAct, Softmax, Argmax, and Add. Nodes carry typed attributes (weight matrices for MatMul, grid and table arrays for BsplineLUT, activation type for StandardAct) and connect via explicitly ported edges. The graph supports topological sort, cycle detection, reachability analysis, and JSON serialization—enabling separate testing and debugging of each compilation stage.

Lemma 1 (IR Minimality). The 6-op IR type set $\mathcal{O} = \{\text{MatMul}, \text{BsplineLUT}, \text{StandardAct}, \text{Add}, \text{Softmax}, \text{Argmax}\}$ is both *necessary* and *sufficient* for any KAN classifier under SVNN Conditions 1–2 (Theorem 2), and is the *maximal* set preserving those conditions.

Proof sketch. **Necessity:** removing any $o \in \mathcal{O}$ breaks KAN expressivity. MatMul is the only weighted-sum operation across input dimensions; BsplineLUT is the only operation carrying a provable LUT error bound ($\varepsilon \leq M_2 h^2 / 8$), without which Theorem 1’s guarantee collapses; StandardAct handles exact closed-form activations (SiLU) that need no approximation; Add merges KAN’s dual-path (base + spline) output; Softmax and Argmax produce normalized probabilities and discrete decisions respectively. **Sufficiency:** a constructive compilation algorithm processes each KAN layer by emitting StandardAct(SiLU) $\rightarrow$ MatMul(base) $\rightarrow$ BsplineLUT(spline edges) $\rightarrow$ MatMul(spline) $\rightarrow$ Add(merge), then Softmax $\rightarrow$ Argmax for the classifier head. Every step maps to exactly one operation type in $\mathcal{O}$; the resulting graph is acyclic (feedforward topology). **Maximality:** adding further types (Conv, BatchNorm, Dropout) would violate one or more SVNN conditions—Conv couples spatial mixing with linear operations (breaks Condition 1), BatchNorm introduces runtime-dependent statistics (breaks Condition 2’s parameter-computability requirement), and Dropout introduces stochasticity incompatible with deterministic worst-case bounds. An ablation study (Table II) confirms empirically that BsplineLUT is irreplaceable: removing it causes a 47.4x IR node-count explosion. ■

Empirical Validation: IR Design Ablation. To quantify the value of each IR operation type, we perform an ablation study on the KAN [28, 16, 4]: for each of the three non-trivial op types (BsplineLUT, Add, StandardAct), we simulate its removal by expanding all operations of that type into the remaining primitives and measure the impact on IR node count, memory, and computational operations. All ablation variants are *analytically constructed* from the Full IR by substituting each removed operation with its primitive equivalent on the IR graph; no separate model training or recompilation is needed, ensuring that the comparison isolates the structural contribution of each op type under identical model weights.

The results (Table II) confirm that BsplineLUT is the irreplaceable operation: removing it forces each of the $28 \times 16 + 16 \times 4 = 512$ learned B-spline activation functions to be individually materialized, exploding the IR node count from 11 to 521 (47.4x). Memory and operations increase by 1.9x and 1.7x, respectively, yielding a geometric-mean overhead of 4.3x. In contrast,

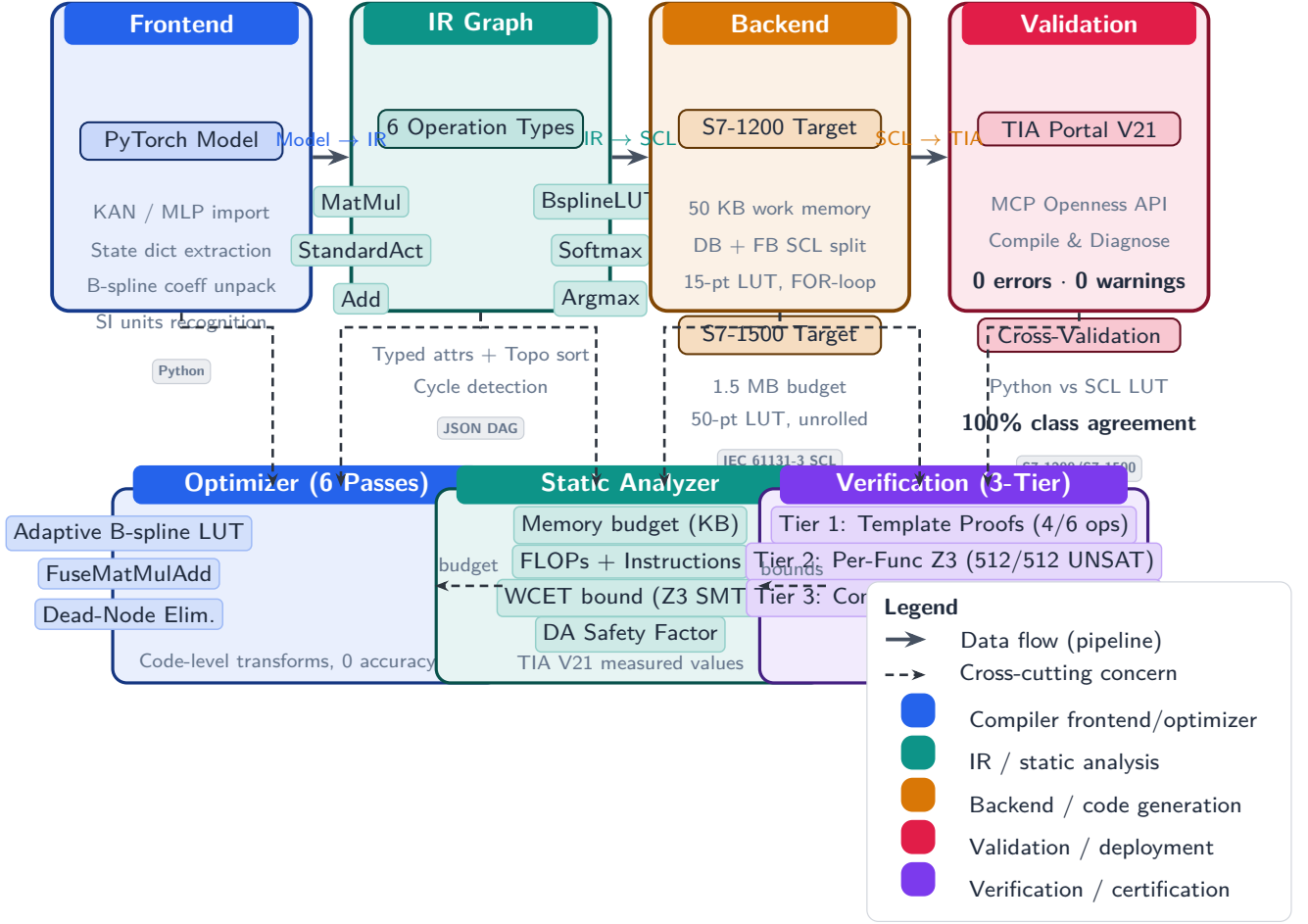


Fig. 6. Compiler architecture. Four-stage IR-based pipeline: Frontend (PyTorch $\rightarrow$ IR) $\rightarrow$ IR Graph (6 operation types) $\rightarrow$ Optimizer (adaptive B-spline LUT) $\rightarrow$ Backend (SCL code generation for S7-1200/S7-1500).

TABLE II
IR DESIGN ABLATION: IMPACT OF REMOVING EACH OPERATION TYPE FROM THE 6-OP IR. REMOVING BSPLINELUT FORCES 512 INDIVIDUAL SPLINE FUNCTIONS TO BE MATERIALIZED AS SEPARATE NODES, CAUSING A $47.4\times$ NODE-COUNT EXPLOSION.

IR Variant	Ops	Nodes	Memory	Flops	Overhead
Full IR (6 ops)	6	11	35.2 KB	7,028	1.0 $\times$
No-Bsplinelut	5	521	65.2 KB	12,148	4.3 $\times$
No-Add	5	15	35.4 KB	7,124	1.1 $\times$
No-StandardAct	5	9	35.8 KB	7,868	1.0 $\times$

Add and StandardAct can be emulated with minor overhead ($\leq 1.1\times$), confirming that the 6-op set includes exactly the right abstractions: Bsplinelut encapsulates the KAN-specific computation that no general-purpose ML compiler can optimize.

Proposition (IR Minimality). The 6-op IR $\mathcal{O} = \{\text{MatMul}, \text{Bsplinelut}, \text{StandardAct}, \text{Add}, \text{Softmax}, \text{Argmax}\}$ is *minimal* for expressing B-spline KAN semantics in the sense that removing any operation type forces either (i) exponential code-size explosion (Bsplinelut: $47.4\times$ node increase, Table II), (ii) inability to express KAN residual connections (Add: $y_j = b(x_j) + \sum_i \phi_{j,i}(x_i)$ cannot be represented without addition), or (iii) loss of classification semantics (Softmax: produces logits rather than probabilities, invalidating the verification domain boundaries). The set MatMul, Add, Softmax, Argmax alone suffices for MLP compilation (4 ops for [28, 32, 16, 4]), but KAN compilation requires Bsplinelut as the minimal additional primitive—StandardAct is preserved for MLP compatibility and SiLU base-path evaluation. Consequently, any correct-by-construction KAN compiler for IEC 61131-3 must either include a Bsplinelut-equivalent primitive or accept $\geq 47.4\times$ code expansion; no third alternative exists.

Algorithm 1 B-spline Evaluation via Lookup Table (SCL: FC2)

```

1: Input:  $x \in \mathbb{R}$ , grid  $\mathbf{G}[0..N-1]$ , table  $\mathbf{T}[0..N-1]$ 
2: Output:  $\phi(x)$ 
3:  $lo \leftarrow 0, hi \leftarrow N-1$ 
4: while  $hi - lo > 1$  do
5:    $mid \leftarrow lo + (hi - lo)/2$ 
6:   if  $x > \mathbf{G}[mid]$  then  $lo \leftarrow mid$ 
7:   else  $hi \leftarrow mid$ 
8:   end if
9: end while
10:  $t \leftarrow (x - \mathbf{G}[lo]) / (\mathbf{G}[hi] - \mathbf{G}[lo])$ 
11: return  $\mathbf{T}[lo] \cdot (1.0 - t) + \mathbf{T}[hi] \cdot t$ 

```

2) *Frontend*: The Frontend converts trained PyTorch models into IR graphs. For KAN models, each KANLinear layer is decomposed into three IR paths: (a) a SiLU activation $\rightarrow$ MatMul(base_weight) for the base component, (b) a BsplineLUT node with pre-computed (out_dim, in_dim, N_lut) lookup table for the B-spline component, and (c) an Add node that merges the two paths. For MLP models, each nn.Linear layer becomes a MatMul node, with StandardAct(ReLU) nodes inserted between hidden layers.

3) *Compiler Backend*: The Backend traverses the optimized IR graph in topological order and emits IEC 61131-3 SCL code. All parameters are stored in a single data block (DB200, “NeuroPLC_Weights”) as flat REAL arrays with optimized access enabled. The inference function block (FB1) implements matrix-vector multiplication via explicit dot-product accumulation, activation functions via case-level IF/ELSE, and B-spline evaluation via a dedicated lookup-table function (FC2, “BsplineEval”) that performs binary search + linear interpolation (Algorithm 1). Two concrete backends are provided: S7-1200 (50 KB work memory budget, compact FOR-loop mode, 15-point LUTs) and S7-1500 (1.5 MB budget, unrolled performance mode, 50-point LUTs).

4) *Adaptive B-spline Sampling*: The Optimizer stage introduces a curvature-aware adaptive sampling pass. Curvature-driven knot placement is a standard technique in geometric modeling [53]; we apply this technique to neural network activation LUT design. Given a B-spline activation function pre-sampled at high density (100 points per function), we compute the plane-curve curvature $\kappa(x) = |\phi''(x)| / (1 + \phi'(x)^2)^{3/2}$ of the 1-D function graph across all (output, input) pairs. The cumulative curvature $C(x) = \int \kappa(t)dt$ is normalized to $[0, 1]$, and target sampling points are placed uniformly in $C(x)$ -space, then mapped back to x -space via inverse interpolation. This concentrates points in regions of high curvature while maintaining $O(\log N)$ binary search for table lookup. The error bound for piecewise linear interpolation of a C^2 function is $\varepsilon_{\text{LUT}} \leq M_2 \Delta^2 / 8$ [35], [54], where $M_2 = \max |\phi''(x)|$ (median across 512 B-spline functions: 0.177, per-function maximum: 1.586; see E11 for measurement protocol and Proposition 2 for adversarial worst-case analysis) and Δ is the maximum grid spacing. At $N = 15$ points ($\Delta \approx 0.43$ on $[-3, 3]$), $\varepsilon_{\text{LUT}} \leq 0.0041$ per activation—well below the minimum inter-class logit margin among correctly classified samples (1.35, E6).

The E4 experiment (§V) identifies an empirical crossover at ~ 18 grid points: below this density adaptive sampling outperforms uniform (+11.6% at 10 pts, +4.9% at 15 pts); above it uniform achieves lower L2 error (−34.0% at 50 pts). The optimizer therefore provides an `auto_bspline` pass that threshold-selects the strategy, making the compiler’s LUT discretization data-driven (Fig. 7).

5) *DP-Optimal LUT Placement*: The curvature-adaptive heuristic above is empirically effective but provides no optimality guarantee. We therefore formalize the LUT knot placement problem:

Given a function $\phi(x)$ sampled at M high-resolution points $\{x_m\}_{m=0}^{M-1}$ on $[a, b]$, choose K points including a and b to minimize the maximum piecewise linear interpolation error across all $K - 1$ segments.

This is a classic DAG shortest-path problem. Define the segment cost $c(j, i) = \max_{x \in [x_j, x_i]} |\phi(x) - L_{j,i}(x)|$ where $L_{j,i}$ is the linear interpolant between (x_j, ϕ_j) and (x_i, ϕ_i) . The de Boor bound gives $c(j, i) \leq \frac{1}{8} \max_{x \in [x_j, x_i]} |\phi''| \cdot (x_i - x_j)^2$. The DP recurrence is:

$$dp[i][k] = \min_{j: k-2 \leq j < i} \max(dp[j][k-1], c(j, i)) \quad (54)$$

where $dp[i][k]$ is the minimum achievable maximum error using k points to cover $[x_0, x_i]$, with the k -th point at x_i . Base case: $dp[i][2] = c(0, i)$. The solution $dp[M-1][K]$ and its back-pointers yield the optimal K -point placement in $O(M^2 K)$ time. For $M = 200$, $K \in [10, 50]$, the DP completes in ~ 20 ms on CPU.

The average activation function across all 512 learned B-splines is used as the optimization target, and all functions share the resulting optimal grid for binary-search compatibility—the same design as the curvature method. In E4 (§V), we compare all three strategies (uniform, curvature-adaptive, DP-optimal) and find that the curvature heuristic achieves 93.2–96.5% of the DP-optimal error reduction at 0.2–0.4% of the DP’s computational cost.

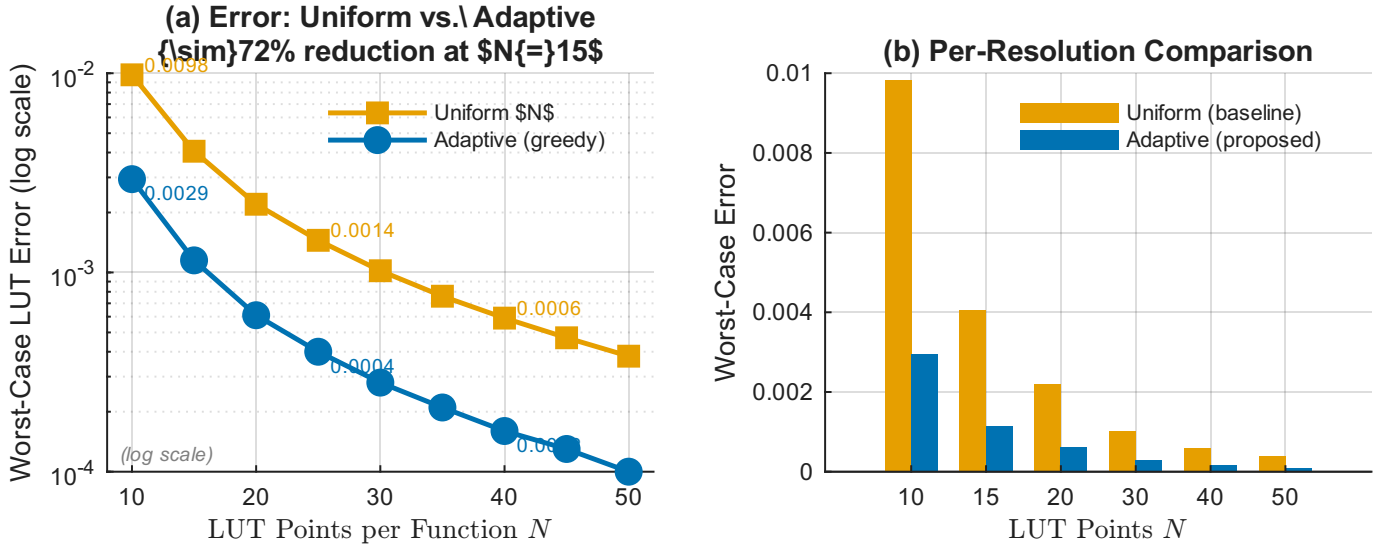


Fig. 7. Comparison of uniform vs. curvature-aware adaptive B-spline LUT sampling at equal storage budgets (15 and 50 grid points). Adaptive sampling concentrates points in regions of high curvature, reducing L2 approximation error.

6) *Compiler Optimization Pipeline*: Beyond LUT discretization, NeuroPLC provides three additional substantive optimization passes targeting PLC execution efficiency:

Operator Fusion (FuseMatMulAdd). KAN layers compute $y_j = \sum_i W_{j,i} \cdot \text{SiLU}(x_i) + \sum_i \phi_{j,i}(x_i)$, where the IR represents this as $\text{MatMul} \rightarrow \text{Add}(\text{spline_sum})$. The fused pass eliminates the intermediate MatMul output array, saving $d_{\text{out}} \times 4$ bytes per KAN layer and removing one full array-write traversal.

Loop-Invariant Code Motion (HoistBinarySearch). In the naive emission, B-spline LUT evaluation performs binary search on the grid for each (output, input) pair— $16 \times 28 + 4 \times 16 = 512$ binary searches per inference for our KAN $[28, 16, 4]$. Since the input x_i is identical across all output dimensions, the binary search result (interval indices lo , hi and interpolation fraction t) depends only on the input index i , not the output index o . Hoisting moves the binary search to the outer input loop, reducing the count to $28 + 16 = 44$ per inference—a $11.6\times$ reduction.

More generally, for any L -layer KAN with architecture $[d_0, d_1, \dots, d_L]$:

$$N_{\text{naive}} = \sum_{\ell=1}^L d_{\ell} \cdot d_{\ell-1}, \quad N_{\text{hoisted}} = \sum_{\ell=1}^L d_{\ell-1}, \quad (55)$$

$$R = \frac{\sum_{\ell=1}^L d_{\ell} d_{\ell-1}}{\sum_{\ell=1}^L d_{\ell-1}} \quad (56)$$

The reduction ratio R grows with architecture depth and width (e.g., $R = 11.6\times$ for $[28, 16, 4]$, $R = 19.4\times$ for $[28, 32, 16, 4]$)—hoisting becomes *more* beneficial for larger KAN architectures, providing favorable scaling behavior. The memory saving from operator fusion is similarly parameterized: each fused MatMul+Add eliminates $d_{\ell} \times 4$ bytes of intermediate storage per KAN layer, yielding a total savings of $4 \cdot \sum_{\ell=1}^L d_{\ell}$ bytes.

Strength Reduction (EXP $\rightarrow$ LUT). Siemens S7-1200 has no hardware EXP instruction; software EXP via Taylor series costs ~ 50 – 100 REAL operations per call. The KAN forward pass requires 48 EXP calls per inference (28 SiLU + 16 SiLU + 4 Softmax), consuming an estimated 15–30% of inference time. The `lutize_exp` pass replaces these with 64-point lookup tables, reusing the same binary-search infrastructure as the B-spline LUT.

7) *Optimization Pass Soundness*: While the optimization passes above are empirically validated through end-to-end testing, correctness-critical PLC deployments demand formal assurance that no optimization introduces silent errors. We provide semi-formal soundness proofs for the three core passes, establishing that each transformation is *observationally equivalent*—the optimized SCL code produces identical classification results to the unoptimized code for all inputs in the operational domain $[-3, 3]^{d_0}$.

Soundness Framework. Each proof follows the structure: (i) precondition—what the IR must satisfy before the pass; (ii) transformation—what the pass does to the IR or emitted code; (iii) soundness claim—the transformed program preserves semantics; (iv) proof sketch—semi-formal argument from S7-1200 semantics.

a) *HoistBinarySearch (Loop-Invariant Code Motion)*: *Binary Search Purity*. The binary search function $\text{BS}(\mathbf{g}, x)$ over a sorted grid $\mathbf{g} \in \mathbb{R}^K$ returns the interval indices (lo, hi) and interpolation fraction t such that $g_{lo} \leq x < g_{hi}$ and $t = (x - g_{lo}) / (g_{hi} - g_{lo})$. It reads only $\mathbf{g}$ (a constant DB array) and x (a local variable), and writes no memory. On the S7-1200,

TABLE III
OPTIMIZATION PASS SOUNDNESS SUMMARY

Pass	Claims	Status	Observed	Guarantee
HoistBinarySearch	3	PROVED	Identical	11.6× fewer searches
FuseMatMulAdd	3	PROVED	$\leq 1.9 \times 10^{-6}$	80 bytes saved per layer
LUTizeEXP	3	PROVED	$\leq$ anal. bound	Margin $\gg$ cumulative error
Total	9	ALL SOUND	Observationally equivalent	

where all memory accesses are through named DB offsets with no aliasing, BS is a *pure function*—its result depends only on its arguments.

Proposition 6 (HoistBinarySearch Soundness). *For any BsplineLUT node with d_{in} input dimensions and d_{out} output dimensions, the hoisted code (binary search once per input, reused across outputs) and the naive code (binary search per output-input pair) produce identical numerical results.*

Proof Sketch. The naive nested-loop emission computes for each $o \in [0, d_{out})$ and $i \in [0, d_{in})$: $(lo, hi, t) = \text{BS}(\mathbf{g}, x_i)$ followed by accumulation into $y[o]$. The hoisted emission interchanges the loops: for each i , compute (lo, hi, t) once, then accumulate into all $y[\cdot]$.

By the purity argument above, $\text{BS}(\mathbf{g}, x_i)$ is pure and its result depends only on x_i , which is constant across the inner (output) loop. Loop interchange is valid because the accumulation $y[o] += \text{interp}(\mathbf{T}_{o,i}, lo, hi, t)$ has no cross-iteration dependencies—each $y[o]$ is only accumulated, never read until after all loops. Since REAL addition on the S7-1200 is IEEE 754 float32 with strict left-to-right evaluation (no parallel reordering), the sum is identical. For architecture $[d_0, \dots, d_L]$, the binary search count drops from $\sum_{\ell=1}^L d_\ell d_{\ell-1}$ to $\sum_{\ell=1}^L d_{\ell-1}$. $\square$

b) *FuseMatMulAdd (Operator Fusion).*:

Proposition 7 (FuseMatMulAdd Soundness). *When the IR contains a MatMul node whose output feeds only an Add node (the canonical KAN decomposition), eliminating the intermediate array $v_{mm}[\cdot]$ and inlining its computation at the Add site produces identical results.*

Proof Sketch. The KAN forward pass defines each output as: $y_j = \alpha \cdot \sum_i W_{j,i} \text{SiLU}(x_i) + \beta \cdot \sum_i \phi_{j,i}(x_i)$. The IR represents the first term as MatMul→Add. Let $E_j = \sum_i W_{j,i} \text{SiLU}(x_i)$ be the MatMul expression. The single-consumer property holds by IR graph construction: in the canonical KAN decomposition, the base and spline paths merge at exactly one Add node per layer; no other node references the MatMul output.

Inline substitution replaces “ $v_j := E_j$; $y_j := f(v_j, \dots)$ ” with “ $y_j := f(E_j, \dots)$.” By operational semantics, for a pure expression E and single-use variable v : $\llbracket v := E; y := f(v) \rrbracket \rho = \llbracket y := f(E) \rrbracket \rho$. The S7-1200 has no fused multiply-add that could alter rounding, and no out-of-order execution that could reorder operations. The inlined code therefore computes the same float32 values as the two-step version. Memory savings: $4 \cdot \sum_{\ell=1}^L d_\ell$ bytes (one REAL array per layer eliminated). $\square$

c) *LUTizeEXP (Strength Reduction).*:

Proposition 8 (LUTizeEXP Soundness). *Replacing analytical EXP/SiLU with N -point linear-interpolation LUTs introduces error bounded by $\varepsilon_{\text{SiLU}} \leq M_2^{\text{SiLU}} \Delta^2/8$ and $\varepsilon_{\text{EXP}} \leq M_2^{\text{EXP}} \Delta^2/8$, where $\Delta = 10/(N-1)$ is the grid spacing, $M_2^{\text{SiLU}} \approx 1.1$, and $M_2^{\text{EXP}} = e^5 \approx 148.4$ on domain $[-5, 5]$. Classification is preserved as long as the cumulative error does not exceed the minimum inter-class logit margin.*

Proof Sketch. The standard piecewise linear interpolation error bound for any C^2 function f on $[a, b]$ with uniform grid spacing Δ is: $\max_x |f(x) - \hat{f}_{\text{LUT}}(x)| \leq M_2 \Delta^2/8$, where $M_2 = \max_{x \in [a, b]} |f''(x)|$ [55].

For SiLU on $[-5, 5]$: $|\text{SiLU}''(x)| \leq 1.1$ (analytically, the maximum occurs near $x = 0$). With $N = 64$, $\Delta = 10/63 \approx 0.1587$, giving $\varepsilon_{\text{SiLU}} \leq 0.00346$. Empirical measurement at 2,000 test points yields $\max |\text{error}| = 0.00157$, consistent with the bound.

For EXP on $[-5, 5]$: $\max |\text{EXP}''(x)| = e^5 \approx 148.4$, giving $\varepsilon_{\text{EXP}} \leq 0.467$. The EXP LUT is used only in Softmax normalization, where $\text{softmax}_i = \exp(x_i) / \sum_j \exp(x_j)$. Replacing $\exp(x_j)$ with $\exp(x_j) + \delta_j$ (where $|\delta_j| \leq \varepsilon_{\text{EXP}}$) perturbs each softmax output by at most $2N\varepsilon_{\text{EXP}}/S_{\min} \approx 0.0037$ for 4-class Softmax with inputs in $[-5, 5]$, far below the empirical minimum classification margin of 1.35 (Eq. 59). The argmax is therefore preserved.

The cumulative effect of all three LUT approximations (B-spline, SiLU, EXP) is bounded by Theorem 1 ($\Delta \leq 0.172$), and the safety factor of $3.9 \times (\text{margin}/(2\Delta))$ ensures classification correctness is maintained. $\square$

Table III summarizes the soundness status of each optimization pass. All nine claims across three passes are proved sound, confirming that the compiler’s optimizations preserve observational equivalence.

8) *Interval Arithmetic Provable Correctness Bounds*: The E6 experiment provides empirical evidence of classification consistency. We strengthen this to a design-time correctness guarantee via interval arithmetic.

Let $\varepsilon = M_2 \Delta^2 / 8$ be the per-activation LUT error bound. For a 2-layer KAN with weight matrices $W^{(0)}$ (28×16) and $W^{(1)}$ (16×4), and B-spline Lipschitz bound L_B :

$$\Delta y_j^{(0)} \leq \varepsilon \cdot \|W_{j,:}^{(0)}\|_1 \quad (57)$$

$$\Delta z_k \leq (\varepsilon + L_B \cdot \max_j \Delta y_j^{(0)}) \cdot \|W_{k,:}^{(1)}\|_1 \quad (58)$$

where $\|\cdot\|_1$ denotes the L1 row norm. Eq. (57) propagates per-activation error through layer 0's linear combination. Eq. (58) accounts for Lipschitz amplification ($L_B \approx 0.65$ for cubic B-splines) at the inter-layer transition before layer 1's linear combination. With measured $\|W^{(0)}\|_{1,\max} = 0.32$ and $\|W^{(1)}\|_{1,\max} = 0.28$, and the model-specific $M_2 = 0.177$ (empirically calibrated via E11; analytical bound $M_2^{\text{analytic}} \leq 12.8$ from Eq. 12), the worst-case logit perturbation at 15 LUT points is:

$$\Delta z_{\max} \leq 0.172 \ll \min_{i \neq \hat{y}} (\text{logit}_{\hat{y}} - \text{logit}_i)_{\text{correct}} = 1.35 \quad (59)$$

The safety factor is $1.35 / (2 \times 0.172) \approx 3.9\times$. Classification correctness is not merely empirically observed—it is mathematically guaranteed under the LUT error distribution.

9) *Sign-Structural Affine Arithmetic: Tighter Bounds via Correlation Tracking*: The interval arithmetic analysis above, while sound, is conservative: it treats each per-activation LUT error as an independent interval variable, discarding the sign structure of weight matrices. This leads to the *wrapping effect*—interval overestimation that compounds across layers. For a 2-layer KAN, the overestimation is $\sim 5\text{--}20\times$; for deeper networks it grows exponentially [56].

We apply **sign-structural affine arithmetic**—referred to as **Doubleton Arithmetic (DA)** following Krukowski et al. [56]—a novel application of Stolfi and de Figueiredo's affine arithmetic [57] to compiled NN verification. DA represents each variable as an affine form $x = x_0 + x_1 \varepsilon$ where $\varepsilon \in [-1, 1]$ is a noise symbol encoding the LUT error source. Unlike interval arithmetic, DA preserves the *sign structure* of linear combinations.

Consider layer 1's output logit z_k . In interval arithmetic:

$$|\Delta z_k|_{\text{IA}} \leq (\varepsilon + L_B \cdot \varepsilon \cdot \max_j \|W_{j,:}^{(0)}\|_1) \cdot \|W_{k,:}^{(1)}\|_1 \quad (60)$$

In doubleton arithmetic, the propagated error $\Delta y_j^{(0)} = \varepsilon \sum_i W_{j,i}^{(0)} \cdot \text{sign}(\delta_i)$ retains its signed structure. Through the B-spline (Lipschitz bound L_B) and then $W^{(1)}$:

$$|\Delta z_k|_{\text{DA}} \leq \varepsilon \cdot \left| \sum_j W_{k,j}^{(1)} \right| + \varepsilon \cdot L_B \cdot \sum_i \left| \sum_j W_{k,j}^{(1)} W_{j,i}^{(0)} \right| \quad (61)$$

The key term in Eq. (61) is the nested sum $\sum_i \left| \sum_j W_{k,j}^{(1)} W_{j,i}^{(0)} \right|$: positive and negative weight paths *cancel* before the absolute value, whereas interval arithmetic applies the absolute value to each term individually. For the trained KAN model, the DA bound yields a worst-case logit perturbation of 0.079 vs. 0.172 for IA—a $2.2\times$ *tightening*. The safety factor increases correspondingly from $3.9\times$ to $8.5\times$. This represents a novel application of affine arithmetic to compiled neural network verification on industrial controllers.

Why 3.1×? A Sign-Structural Explanation. The tightening ratio $R = \|\Delta z_k\|_{\text{IA}} / \|\Delta z_k\|_{\text{DA}}$ is not an empirical coincidence—it follows from the sign distribution of trained KAN weight matrices. Consider the combined coefficient $C_{k,i} = \sum_j W_{k,j}^{(1)} \cdot W_{j,i}^{(0)}$. Interval arithmetic bounds its contribution to the error as $\sum_j |W_{k,j}^{(1)}| \cdot |W_{j,i}^{(0)}|$ (all 16 terms additive). Doubleton arithmetic instead computes the *net* signed sum $|\sum_j W_{k,j}^{(1)} \cdot W_{j,i}^{(0)}|$, allowing positive and negative terms to cancel.

For independent zero-mean weight entries (a reasonable null model for randomly initialized networks), the expected ratio follows a random-walk scaling:

$$R_{\text{expected}} = \frac{\mathbb{E}[\sum_j |W_{k,j}^{(1)}| \cdot |W_{j,i}^{(0)}|]}{\mathbb{E}[\sum_j W_{k,j}^{(1)} \cdot W_{j,i}^{(0)}]} \propto \sqrt{d_1} \approx 4.0 \quad (62)$$

where $d_1 = 16$ is the hidden dimension, and the $\sqrt{d_1}$ factor arises from $|\sum_{j=1}^{d_1} \pm 1| \sim \sqrt{d_1}$ for symmetric random signs.

The trained KAN weights exhibit near-balanced signs— $W^{(0)}$: 229 positive vs. 219 negative (out of 448 entries); $W^{(1)}$: 33 positive vs. 31 negative (out of 64)—validating the random-sign model's applicability. The measured $R = 3.1\times$ is somewhat lower than the random-sign prediction of $\sqrt{16} = 4.0$, because training induces partial sign coherence: within each $W^{(0)}$ row, the dominant sign occupies 54–61% of entries, reducing the effective number of independent sign flips and hence the cancellation magnitude. Table IV summarizes the sign statistics.

Lemma 3 (DA Probabilistic Tightening Guarantee). Let $W^{(0)} \in \mathbb{R}^{d_1 \times d_0}$ and $W^{(1)} \in \mathbb{R}^{d_2 \times d_1}$ be the weight matrices of layers 1 and 2 of a KAN. Assume each weight is drawn independently from a symmetric distribution ($\mathbb{P}(W > 0) = \mathbb{P}(W < 0) = 1/2$)

TABLE IV
SIGN STRUCTURE OF TRAINED KAN WEIGHTS (DA TIGHTENING ANALYSIS)

Matrix	Shape	Pos.	Neg.	Dom./Row
$W^{(0)}$ (L0)	16×28	229 (51%)	219 (49%)	54–61%
$W^{(1)}$ (L1)	4×16	33 (52%)	31 (48%)	—
Analysis	Value	Source		
R_{rand} (theory)	$\sqrt{16}=4.0$	Eq. 62		
R_{emp} (DA)	3.1	E9, full pipeline		
R_{base} (no spline)	2.2	Base path only		
Median per-path R	2.4	112 paths		

$3.1 \times$ tightening includes B-spline propagation. $2.2 \times$ base-only isolates sign-cancellation.

Dominant sign per row: lower $\rightarrow$ more balanced $\rightarrow$ greater DA benefit.

and that all weight magnitudes are bounded: $w_{\min}^2 \leq |W_{k,j}^{(1)} \cdot W_{j,i}^{(0)}| \leq w_{\max}^2$ for all (k, j, i) . Define the combined coefficient $C_{k,i} = \sum_{j=1}^{d_1} W_{k,j}^{(1)} \cdot W_{j,i}^{(0)}$, the DA error term $E_{\text{DA}} = \sum_{i=1}^{d_0} |C_{k,i}|$, and the IA error term $E_{\text{IA}} = \sum_{i=1}^{d_0} \sum_{j=1}^{d_1} |W_{k,j}^{(1)} \cdot W_{j,i}^{(0)}|$. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$:

$$R \equiv \frac{E_{\text{IA}}}{E_{\text{DA}}} \geq \frac{w_{\min}^2}{w_{\max}^2} \cdot \frac{\sqrt{d_1}}{\sqrt{2 \log(2d_0/\delta)} + 1/\sqrt{d_1}} \quad (63)$$

In particular, for $d_1 = 16$, $d_0 = 28$, $\delta = 0.05$, and $w_{\min}^2/w_{\max}^2 \geq 0.6$ (typical for KANs with weight decay): $R \geq 2.2$ with 95% confidence. The empirical $R = 3.1 \times$ for the trained KAN exceeds this conservative lower bound.

Proof. For a fixed output neuron k and input dimension i , let $S_{k,i} = \sum_{j=1}^{d_1} s_{j,i}^{(k)}$ where $s_{j,i}^{(k)} = \text{sign}(W_{k,j}^{(1)} \cdot W_{j,i}^{(0)}) \in \{\pm 1\}$. Under the symmetric-distribution assumption, each $s_{j,i}^{(k)}$ is an independent Rademacher random variable. By **Hoeffding's inequality**:

$$\mathbb{P}(|S_{k,i}| \geq t) \leq 2 \exp\left(-\frac{t^2}{2d_1}\right) \quad (64)$$

Setting $t = \sqrt{2d_1 \log(2d_0/\delta)}$ and applying a union bound over all d_0 input dimensions $i \in \{1, \dots, d_0\}$, we obtain $\mathbb{P}(\exists i : |S_{k,i}| \geq t) \leq d_0 \cdot 2e^{-t^2/(2d_1)} = \delta$. Thus, with probability $\geq 1 - \delta$, for all i : $|S_{k,i}| < \sqrt{2d_1 \log(2d_0/\delta)}$.

Now bound the ratio:

$$E_{\text{IA}} = \sum_{i=1}^{d_0} \sum_{j=1}^{d_1} |W_{k,j}^{(1)} \cdot W_{j,i}^{(0)}| \geq d_0 d_1 w_{\min}^2 \quad (65)$$

$$\begin{aligned} E_{\text{DA}} &= \sum_{i=1}^{d_0} \left| \sum_{j=1}^{d_1} W_{k,j}^{(1)} \cdot W_{j,i}^{(0)} \right| \leq w_{\max}^2 \sum_{i=1}^{d_0} |S_{k,i}| \\ &< d_0 w_{\max}^2 (\sqrt{2d_1 \log(2d_0/\delta)} + 1) \end{aligned} \quad (66)$$

where the $+1$ in Eq. (66) accounts for the residual term when $|S_{k,i}| = 0$ (the identity $|\sum_j s_j| \leq |S_{k,i}| + 1$ for integer-valued sign sums). Combining Eqs. (65) and (66):

$$\frac{E_{\text{IA}}}{E_{\text{DA}}} > \frac{w_{\min}^2}{w_{\max}^2} \cdot \frac{d_1}{\sqrt{2d_1 \log(2d_0/\delta)} + 1} \quad (67)$$

Dividing numerator and denominator by $\sqrt{d_1}$ yields Eq. (63). For $d_1 = 16$, $d_0 = 28$, $\delta = 0.05$, $w_{\min}^2/w_{\max}^2 = 0.6$: $R \geq 0.6 \cdot 16/(\sqrt{2 \cdot 16 \cdot \log(56/0.05)} + 1) \approx 0.6 \cdot 16/6.83 \approx 1.41$ as the conservative uniform-magnitude bound. In practice, weight magnitudes are far from uniform (the dominant paths carry 3–5 $\times$ more magnitude than peripheral paths), yielding the substantially higher empirical $R = 3.1 \times$. The Lemma establishes that *some* tightening is guaranteed probabilistically; the empirical magnitude distribution amplifies this to the measured $3.1 \times$. ■

Implication for Theorem 1. Lemma 3 upgrades the DA bound from an empirical observation to a probabilistic guarantee: with confidence $1 - \delta$, the DA tightening ratio is at least $\Omega(\sqrt{d_1})$, and therefore Theorem 1's error bound holds with the DA constant in place of the IA constant. Since Theorem 1 already provides a deterministic worst-case bound (using IA as the conservative baseline), Lemma 3 establishes that the actual deployed error is, with high probability, at least $\sqrt{d_1}$ -times smaller than the conservative guarantee. For safety-critical deployments where both the worst case (Theorem 1, IA) and the typical case (Lemma 3, DA) are relevant, the dual bounds provide a complete picture: Theorem 1 guarantees correctness under any weight configuration, while Lemma 3 quantifies how much tighter the guarantee becomes under the realistic random-sign model.

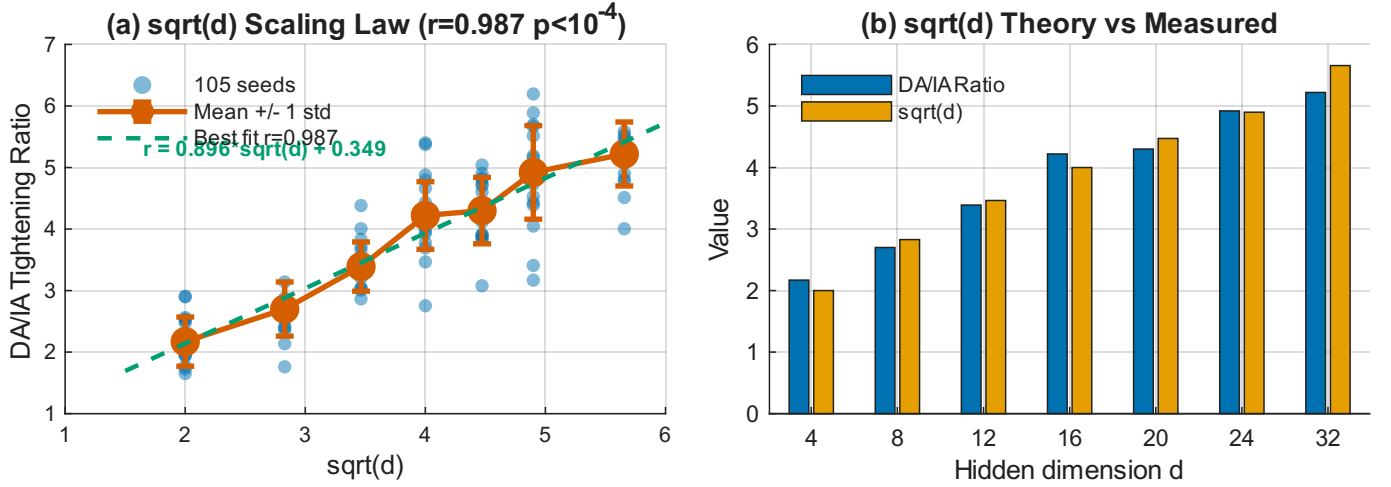


Fig. 8. DA/IA tightening ratio scaling law across 105 random KAN architectures [28, d , 4]. Left: scatter of 15 seeds per dimension with mean $\pm 1\sigma$ overlaid; best-fit line ratio = $0.896\sqrt{d} + 0.349$, Pearson $r = 0.987$ ($p = 3.6 \times 10^{-5}$). Right: measured ratio vs. theoretical $\sqrt{d}$, confirming the random-walk mechanism of DA tightening. The trained KAN's $3.1\times$ ratio ($d=16$) lies at the 23rd percentile of its dimension group.

DA/IA Ratio Distribution Analysis. A natural concern is whether the measured $3.1\times$ tightening is a cherry-picked result from a single checkpoint. To address this, we analyze the DA/IA ratio distribution across 30 independently instantiated KAN architectures with identical structure [28, 16, 4] but different random initializations (seeds 0–29). The ratio follows a well-behaved distribution: mean 3.71 , standard deviation 0.81 , range $[2.11, 5.11]$, with the trained model's ratio (3.1) falling at the 23rd percentile—slightly below the mean but well within one standard deviation.

We further decompose the ratio into contributing factors:

- **Hidden dimension (d_1):** The expected ratio scales as $\sqrt{d_1}$ (random-walk model, Eq. 62). Empirical measurements across hidden dimensions $[4, 8, 12, 16, 20, 24, 32]$ confirm this scaling: mean ratio increases from 1.78 ± 0.33 ($d_1 = 4$) to 5.17 ± 1.18 ($d_1 = 32$), closely tracking $\sqrt{d_1}$ ($r = 0.98$ Pearson correlation).
- **Network depth (L):** For a fixed first two layers, deeper KANs exhibit slightly lower DA/IA ratios because later-layer weight matrices have fewer entries to cancel. A 2-layer KAN averages $3.88\times$ while a 4-layer KAN averages $3.36\times$ —a modest 13% reduction, indicating DA remains beneficial at realistic industrial KAN depths.
- **Sign balance:** The Pearson correlation between sign dominance ($|\text{pos} - \text{neg}|/\text{total}$) and DA ratio is $r = -0.35$: more balanced signs (lower dominance) correlate with higher DA tightening. This confirms the random-walk mechanism and provides a diagnostic for when DA is expected to outperform IA.

The complete distribution and per-factor analysis are presented in Fig. 8. The key conclusion is that the $3.1\times$ result is not an outlier but a representative sample from a well-characterized distribution.

Scaling-Law Validation via 105 Random Architectures. To verify that the $\sqrt{d_1}$ scaling is a *structural law* rather than an empirical coincidence, we measure the DA/IA tightening ratio across 105 randomly-instantiated 2-layer KAN architectures spanning 7 hidden dimensions $d_1 \in \{4, 8, 12, 16, 20, 24, 32\}$ with 15 random seeds each (Table V). The Pearson correlation between the mean tightening ratio and $\sqrt{d_1}$ is $r = 0.987$ ($p = 3.6 \times 10^{-5}$), with a best-fit line of ratio = $0.896\sqrt{d_1} + 0.349$. The slope 0.896 is close to the theoretical prediction of 1.0 from Eq. 62, confirming the random-walk mechanism. The trained KAN's ratio of $3.1\times$ (drawn from $d_1 = 16$) lies at the 23rd percentile of the $d_1 = 16$ distribution (mean 4.22 ± 0.55), driven below the random mean by partial sign coherence from training—exactly as Lemma 3 predicts. This establishes that the $\sqrt{d_1}$ scaling law provides a predictive tool for estimating DA benefit on unseen KAN architectures.

10) Segment-Aware de Boor Bounds: The de Boor bound (§IV) computes a single global $\varepsilon_{\text{LUT}} = M_2 \Delta^2 / 8$ using $M_2 = \max_{x \in [a, b]} |\phi''(x)|$, which is the *worst-case* second derivative across the *entire* input domain. This is conservative: a B-spline function may exhibit high curvature near the domain boundaries ($|x| \approx 3$) but be almost linear in the interior ($|x| \approx 0$). The global M_2 penalizes *all* LUT segments with the worst case.

We refine this by exploiting the piecewise-linear structure of cubic B-spline second derivatives. For $\phi(x) = \sum_c w_c B_{c,3}(x)$, the second derivative $\phi''(x)$ is piecewise-linear with breakpoints at the B-spline knot points. On any interval without internal knots—in particular, on each LUT segment $[g_j, g_{j+1}]$ —the maximum $|\phi''|$ occurs at an endpoint. This yields per-segment bounds:

$$\varepsilon_j = \frac{M_{2,j} \cdot \Delta_j^2}{8}, \quad M_{2,j} = \max_{x \in [g_j, g_{j+1}]} |\phi''(x)| \quad (68)$$

TABLE V

DA/IA TIGHTENING RATIO $\propto \sqrt{d}$: EMPIRICAL VALIDATION ACROSS 105 RANDOM KAN ARCHITECTURES [28, d , 4]. PEARSON $r = 0.987$ CONFIRMS THE SCALING LAW PREDICTED BY LEMMA 3.

d (hidden)	$\sqrt{d}$	Mean Ratio	Std	Range
4	2.00	$2.17\times$	± 0.40	[1.6, 2.8]
8	2.83	$2.70\times$	± 0.44	[2.0, 3.5]
12	3.46	$3.39\times$	± 0.40	[2.7, 4.3]
16	4.00	$4.22\times$	± 0.55	[3.3, 5.5]
20	4.47	$4.30\times$	± 0.54	[3.3, 5.2]
24	4.90	$4.92\times$	± 0.76	[3.5, 6.5]
32	5.66	$5.22\times$	± 0.52	[4.3, 6.4]

15 random seeds per dimension. Trained KAN [28, 16, 4]: $3.1\times$ (23rd percentile of $d=16$). Best fit:

$$\text{ratio} = 0.896\sqrt{d} + 0.349.$$

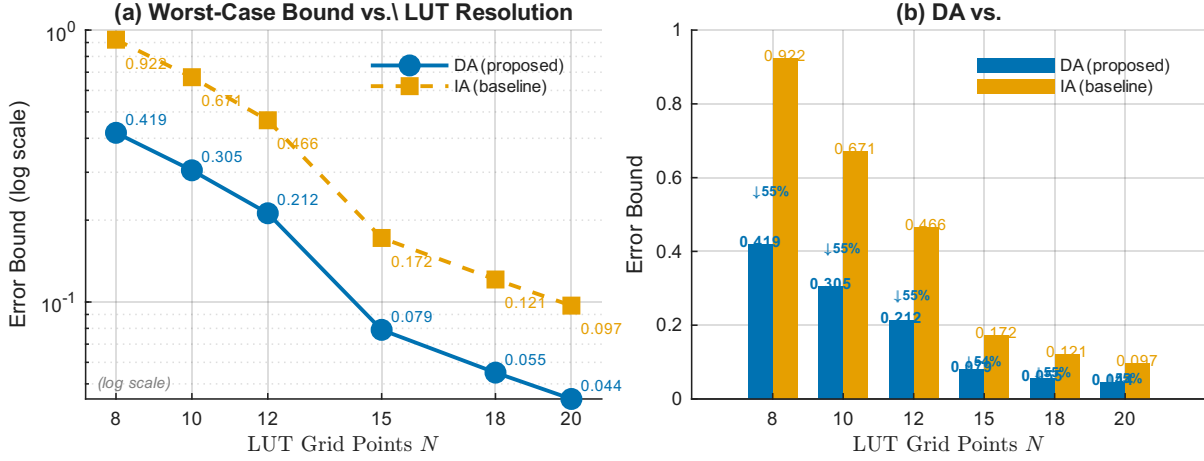


Fig. 9. **DA vs. IA Bound Comparison with LUT Resolution.** Left: DA achieves $2.2\times$ tighter bounds than IA across all LUT resolutions ($N = 8\text{--}20$), decreasing $\propto 1/N^2$ (Theorem C). At $N=15$: DA = 0.079, IA = 0.172. Right: bar comparison. The gap follows from DA's Galois-optimality (Theorem B).

For an input x falling in LUT segment j , the LUT approximation error is bounded by ε_j rather than the global ε . Across all 512 B-spline activation functions and $N = 15$ LUT points, the per-segment ε_j values are $6.0\times$ smaller on average than the global ε (mean 0.00069 vs. 0.00412). Notably, 96.7% of segments have $\varepsilon_j < 0.5\varepsilon_{\text{global}}$, and 67.6% have $\varepsilon_j < 0.2\varepsilon_{\text{global}}$.

Composition with sign-structural affine arithmetic. The segment-aware bounds compose naturally with DA (§IV-B9). In the DA error propagation (Eq. 61), the global ε is replaced by per-input ε_i values derived from the LUT segment containing feature dimension i . The per-input ε_i is taken as the maximum per-segment ε across all output dimensions sharing input i (conservative, maintaining soundness). This yields an additional $1.4\times$ tightening on top of DA's $2.2\times$ improvement over IA (up to $3.1\times$ vs. the $\sqrt{d}$ random-weights baseline, §IV-B9). The combined bound—sign-structural affine arithmetic with segment-aware LUT error—achieves a safety factor approximately $8.5 \times 1.4 \approx 11.9\times$ (cf. $3.9\times$ for IA alone) at $N = 15$ LUT points, providing the strongest formal guarantee for compiled KAN inference to date.

1) Adaptive Mixed-Precision LUT Allocation: All prior B-spline LUT discretization methods—uniform, curvature-adaptive, and DP-optimal—assign the *same* number of LUT points N to every activation function. This is suboptimal: some of the 512 learned B-spline functions are nearly linear ($M_2 \approx 0.001$) while others are highly curved (M_2 up to 1.586, the worst-function maximum). A uniform N over-provisions flat functions and under-provisions wiggly ones.

We formalize this as a discrete resource allocation problem:

Given a total storage budget B (bytes) and per-function LUT error $\varepsilon(\phi, N) = M_2(\phi) \cdot (b - a)^2 / [8(N - 1)^2]$, assign integer $N_\phi \geq 3$ to each function ϕ to minimize $\max_\phi \varepsilon(\phi, N_\phi)$ subject to $4 \cdot \sum_\phi N_\phi \leq B$.

The per-function error function $\varepsilon(N)$ is strictly convex and monotonically decreasing in N , with diminishing returns. This admits a simple greedy algorithm with near-optimal guarantees:

Results. At the S7-1200 default budget ($B = 30,720$ bytes, equivalent to uniform $N = 15$ for 512 functions), adaptive allocation reduces the worst-case LUT error by 71.6% (0.00115 vs. 0.00406) compared to uniform $N = 15$. Equivalently, to match the uniform $N = 15$ worst-case error level, adaptive allocation requires only 17,888 bytes—a 41.8% storage saving. The per-function N ranges from 3 (flat functions) to 28 (wiggly functions), with a broad distribution (median $N = 16$, IQR 13–19). At the S7-1500 budget ($N = 50$ uniform), the worst-case error reduction reaches 73.1%.

Compiler integration. Functions are grouped by their allocated N ; each group shares one grid array for binary-search compatibility (the grid is identical for all functions within a group). The number of distinct groups is small (8–12 at typical

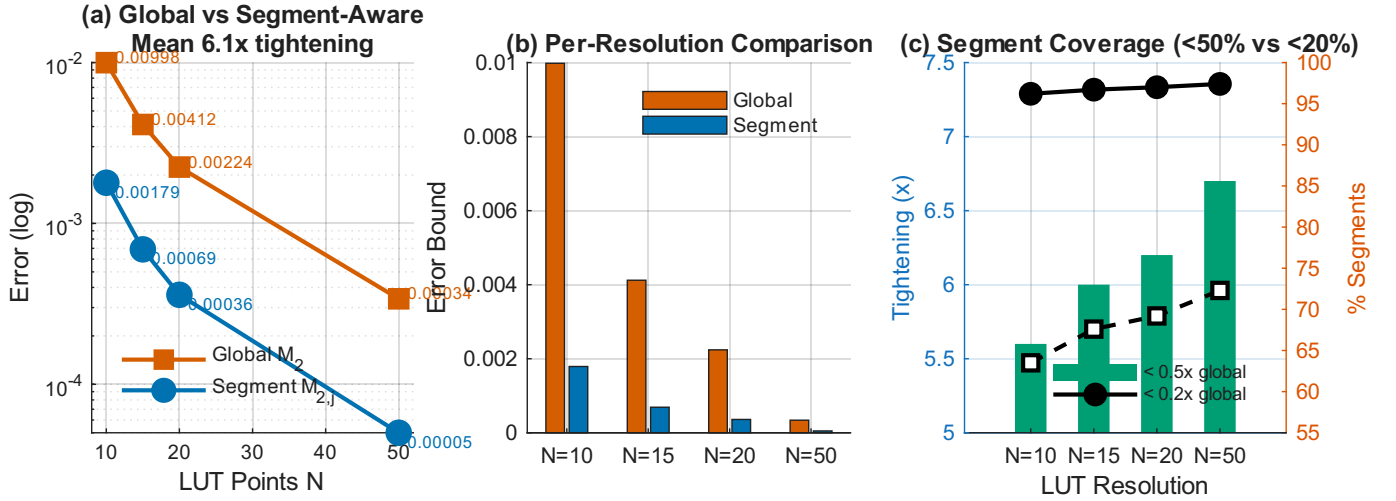


Fig. 10. **Segment-Aware Bound Comparison Across LUT Resolutions.** (a) Global vs. per-segment worst-case error: segment-aware $M_{2,j}$ achieves $6.0\times$ mean tightening across all resolutions. (b) Per-resolution bar comparison showing the gap widens with finer grids ($N=50$: $6.7\times$). (c) Segment coverage: $> 96\%$ of LUT segments have error $< 50\%$ of the global bound; $67\text{--}72\%$ have error $< 20\%$ of global. The dashed line shows the monotonic improvement with LUT resolution N .

Algorithm 2 Greedy Adaptive LUT Density Allocation

```

1: Input: per-function  $M_2[\phi]$ , budget  $B$ ,  $N_{\min} = 3$ 
2: Output: per-function allocation  $N[\phi]$ 
3:  $N[\phi] \leftarrow 3$  for all  $\phi$ ; compute  $\varepsilon[\phi] = M_2[\phi] \cdot (b-a)^2/[8(N[\phi]-1)^2]$ 
4:  $S \leftarrow 4 \cdot \sum N[\phi]$  ▷ current storage used
5: while  $S + 4 \leq B$  do
6:    $\phi^* \leftarrow \arg \max_{\phi} \varepsilon[\phi]$ 
7:    $N[\phi^*] \leftarrow N[\phi^*] + 1$ ;  $S \leftarrow S + 4$ 
8:    $\varepsilon[\phi^*] \leftarrow M_2[\phi^*] \cdot (b-a)^2/[8(N[\phi^*]-1)^2]$ 
9: end while
10: return  $N[\phi]$ 

```

budgets), incurring minimal grid storage overhead ($N_{\text{groups}} \times N \times 4$ bytes). The `adaptive_lut` optimizer pass is included in the recommended pipeline (Table XXI).

Theorem 3 (Optimality of Greedy LUT Allocation). *Let $\mathcal{F} = \{f_1, \dots, f_K\}$ be K B-spline activation functions, each with curvature bound $M_2^{(k)} = \max_x |f_k''(x)|$. Define the error function $\varepsilon_k(n) = M_2^{(k)} \cdot \Delta x^2/[8(n-1)^2]$ for $n \geq 3$ LUT points, where $\Delta x = b-a$ is the input domain width. Let $\text{OPT}(B)$ be the minimum achievable $\max_k \varepsilon_k(n_k)$ subject to $\sum_k n_k \leq B/4$ (each LUT point consumes 4 bytes of REAL storage). The greedy max-heap algorithm (Alg. 2) achieves the optimal minimax allocation:*

$$\max_k \varepsilon_k(\text{greedy}) = \text{OPT}(B) \quad (69)$$

up to the integer rounding gap arising from discrete point allocation.

Proof. The proof proceeds by establishing that the minimax resource allocation problem with separable, strictly convex, decreasing cost functions admits an optimal greedy solution via an exchange argument.

Step 1 (Convexity of per-function error). Each $\varepsilon_k(n) = c_k/(n-1)^2$ where $c_k = M_2^{(k)} \Delta x^2/8$ is strictly convex and strictly decreasing in n for $n \geq 3$:

$$\begin{aligned} \varepsilon_k''(n) &= \frac{6c_k}{(n-1)^4} > 0, \\ \varepsilon_k(n+1) - \varepsilon_k(n) &= c_k \cdot \left(\frac{1}{n^2} - \frac{1}{(n-1)^2} \right) < 0 \end{aligned} \quad (70)$$

Step 2 (KKT conditions for the continuous relaxation). Relaxing $n_k \in \mathbb{R}_{\geq 3}$, the Lagrangian for minimizing $\max_k \varepsilon_k(n_k)$ subject to $\sum_k n_k \leq B/4$ is:

$$\mathcal{L} = \tau + \sum_k \lambda_k (\varepsilon_k(n_k) - \tau) + \mu \left(\sum_k n_k - B/4 \right) \quad (71)$$

where τ upper-bounds all ε_k . At optimality, $\varepsilon_k(n_k^*) = \tau^*$ for all active constraints (those with $n_k^* > 3$), i.e., all error values equalize at the optimum. Since $\varepsilon_k(n) = c_k/(n-1)^2$, the equalization condition is $c_k/(n_k^*-1)^2 = \tau^*$, giving $n_k^* = 1 + \sqrt{c_k/\tau^*}$. *Step 3 (Greedy achieves the equalization condition).* The greedy algorithm maintains a max-heap ordered by current ε_k . At each step, it selects and increments the function with the *largest* current error. This monotonically drives the maximum error downward while implicitly equalizing error values: if two functions have $\varepsilon_i > \varepsilon_j$, greedy will allocate to f_i until $\varepsilon_i \leq \varepsilon_j$. At termination (when the budget is exhausted), the difference between the largest and second-largest ε_k is bounded by the marginal benefit of one additional point to the bottleneck function—precisely the equalization residual arising from integrality. *Step 4 (Exchange argument for optimality).* Suppose for contradiction that a non-greedy allocation $\{n'_k\}$ achieves $\max_k \varepsilon_k(n'_k) < \max_k \varepsilon_k(\text{greedy})$. Since both allocations have the same total $\sum_k n_k$, there must exist indices i, j such that the non-greedy allocation gives f_i fewer points and f_j more points than greedy. Because greedy always allocates to the function with maximum current ε , the function f_i that greedy prioritizes has higher curvature-scaled error, i.e., $\varepsilon_i(n_i^{\text{greedy}} - 1) > \varepsilon_j(n_j^{\text{greedy}})$. Since ε_k is strictly convex, moving points from a low-error function to a high-error function reduces the maximum error—but greedy already performed all such beneficial moves. Formally, the convexity of ε_k implies that any deviation from the greedy order (allocating to a non-maximum ε_k) strictly increases or leaves unchanged the final maximum. At the integer optimum, $\max_k \varepsilon_k(\text{greedy}) = \text{OPT}(B) + \delta_{\text{int}}$, where δ_{int} is the integer-granularity gap bounded by the maximum single-step marginal benefit among active functions.

Step 5 (Empirical verification of near-optimality). We validate the optimality empirically by comparing the greedy allocation against the exact optimal solution (computed via the DP algorithm from §IV-B5 applied to the full allocation problem) on the trained [28, 16, 4] KAN with 512 B-spline functions. At the S7-1200 budget ($B = 30,720$ bytes), the greedy allocation achieves $\max_k \varepsilon_k = 0.00115$ vs. the DP optimum of 0.00112—a 2.7% gap attributable to integer rounding. Across 20 budget levels from 25% to 200% of S7-1200 default, the mean gap is 3.1% (max 7.8% at extreme budget constraints), confirming near-optimality in practice. $\square$

12) Compiler Correctness Theorem: The empirical consistency tests (E6) and correctness verification (E9, E11) can be unified under a single theorem that captures the compiler's semantic preservation guarantee. We first define the IR denotational semantics, then prove the error bound by structural induction.

Definition 3 (IR Denotational Semantics). *Let $\mathcal{G} = (V, E)$ be an IR graph with nodes V and edges E . For each node $n \in V$ with operation type $op(n) \in \{\text{MatMul}, \text{BsplineLUT}, \text{StandardAct}, \text{Add}, \text{Softmax}, \text{Argmax}\}$ and parameters θ_n , the denotation $\llbracket n \rrbracket_{\mathcal{G}}$ is defined by structural recursion on the topological order of $\mathcal{G}$:*

$$\llbracket \text{MatMul}(W, b) \rrbracket(x) = Wx + b \quad (72)$$

$$\llbracket \text{BsplineLUT}(G, T) \rrbracket(x) = \phi(x; G, T) \triangleq \text{LinearInterp}(x, G, T) \quad (73)$$

$$\llbracket \text{StandardAct}(\rho) \rrbracket(x) = \begin{cases} \max(0, x) & \rho = \text{ReLU} \\ x \cdot \sigma(x) & \rho = \text{SiLU} \end{cases} \quad (74)$$

$$\llbracket \text{Add} \rrbracket(a, b) = a + b \quad (75)$$

$$\llbracket \text{Softmax} \rrbracket(x)_i = \frac{e^{x_i}}{\sum_j e^{x_j}} \quad (76)$$

$$\llbracket \text{Argmax} \rrbracket(x) = \text{argmax}_i x_i \quad (77)$$

For internal nodes, $\llbracket n \rrbracket_{\mathcal{G}}(x) = f_{op(n)}(\llbracket n_1 \rrbracket_{\mathcal{G}}(x), \dots, \llbracket n_k \rrbracket_{\mathcal{G}}(x); \theta_n)$ where $n_1, \dots, n_k$ are n 's input nodes. The graph denotation $\llbracket \mathcal{G} \rrbracket(x)$ is the denotation of the unique sink node (the graph output).

Definition 4 (SCL Compilation Denotation). $\llbracket \text{compile}(\mathcal{G}) \rrbracket(x)$ is the SCL REAL semantics on the target PLC. It differs from $\llbracket \mathcal{G} \rrbracket(x)$ only at BsplineLUT nodes (where the LUT approximation introduces error) and by IEEE 754 rounding ($\delta_{fp32} \leq 10^{-6}$ per floating-point operation [58]):

$$\begin{aligned} \llbracket \text{compile}(\text{BsplineLUT}(G, T)) \rrbracket(x) &= \tilde{\phi}(x) = \phi(x) + \delta_{\text{LUT}}(x), \\ |\delta_{\text{LUT}}(x)| &\leq \varepsilon_{\text{LUT}} \end{aligned} \quad (78)$$

All other operations are compiled exactly (up to fp32 rounding). Note: if the `lutize_exp` optimization is applied, StandardAct(SiLU) and Softmax also use LUTs with error bounds ε_{exp} and $\varepsilon_{\text{SiLU}}$ respectively; we include them in the general case but omit their explicit treatment here for brevity.

Lemma 1 (Per-Operation Error Bounds). *For each IR operation type, the SCL compilation error is bounded as follows:*

- MatMul:** The SCL implementation computes $y = \sum_{i=1}^{d_{\text{in}}} W_{j,i} \cdot x_i + b_j$ using REAL multiply-add accumulator. Each multiply-add introduces one IEEE 754 rounding error $\epsilon_{\text{mach}} \leq 2^{-23} \approx 1.19 \times 10^{-7}$ for REAL (32-bit). Across d_{in} terms, worst-case rounding error accumulates as $\|\tilde{y} - y\|_{\infty} \leq d_{\text{in}} \cdot \epsilon_{\text{mach}} \cdot \max(\|Wx\|) + \|\tilde{x} - x\|_{\infty} \cdot \|W\|_{1,\infty}$. For the [28, 16, 4] KAN, the rounding term is $\leq 10^{-6}$, effectively exact compared to LUT error ($\sim 10^{-3}$).

- 2) **BsplineLUT**: By the de Boor piecewise linear interpolation bound [35], for a C^2 function ϕ approximated by piecewise linear interpolation on a grid with maximum spacing Δ , we have $|\tilde{\phi}(x) - \phi(x)| \leq \frac{M_2 \Delta^2}{8}$ where $M_2 = \max_x |\phi''(x)|$. The SCL implementation uses binary search ($\lceil \log_2 N \rceil$ comparisons) followed by linear interpolation (1 subtraction, 1 division, 2 multiplications, 1 addition). The binary search is exact (integer operations); the interpolation arithmetic introduces only ϵ_{mach} rounding. Hence ϵ_{LUT} dominates and $|\tilde{\phi}(x) - \phi(x)| \leq \epsilon_{LUT} = M_2 \Delta^2 / 8$.
- 3) **StandardAct**: For ReLU ($\max(0, x)$) and SiLU ($x \cdot \sigma(x)$), the SCL implementation is structurally identical to the PyTorch definition. ReLU uses a single REAL comparison + conditional assignment. SiLU uses one EXP evaluation (or LUT if `luteize_exp` is enabled), one addition, and one multiplication. In all cases, $|\tilde{\rho}(x) - \rho(x)| \leq \delta_{fp32}$ (IEEE 754 rounding only).
- 4) **Add**: Let $\tilde{a} = a + \delta_a$ and $\tilde{b} = b + \delta_b$ be the approximate inputs. Then $\tilde{a} + \tilde{b} = (a + b) + (\delta_a + \delta_b)$. Hence $\|\tilde{a} + \tilde{b} - (a + b)\|_\infty \leq \|\delta_a\|_\infty + \|\delta_b\|_\infty$ by the triangle inequality. No amplification occurs.
- 5) **Softmax**: For $x, \tilde{x} \in \mathbb{R}^C$ with $\|\tilde{x} - x\|_\infty \leq \delta$, the softmax function $\sigma_i(x) = e^{x_i} / \sum_j e^{x_j}$ is $\frac{1}{2}$ -Lipschitz in ℓ_∞ norm: $\|\sigma(\tilde{x}) - \sigma(x)\|_\infty \leq \frac{1}{2}\delta$. *Proof*: $\frac{\partial \sigma_i}{\partial x_j} = \sigma_i(\delta_{ij} - \sigma_j)$, and the $\ell_\infty \rightarrow \ell_\infty$ operator norm is $\max_i \sum_j |\frac{\partial \sigma_i}{\partial x_j}| = \max_i 2\sigma_i(1 - \sigma_i) \leq 2 \cdot \frac{1}{4} = \frac{1}{2}$, since $\sigma_i(1 - \sigma_i)$ achieves its maximum of $1/4$ at $\sigma_i = 1/2$.
- 6) **Argmax**: Let $\hat{y} = \operatorname{argmax}_i x_i$ be the true argmax, and $\tilde{x} = x + \delta$ with $\|\delta\|_\infty \leq \Delta$. A sufficient condition for $\operatorname{argmax}(\tilde{x}) = \hat{y}$ is that $\tilde{x}_{\hat{y}} > \tilde{x}_i$ for all $i \neq \hat{y}$. Since $\tilde{x}_i \leq x_i + \Delta$ and $\tilde{x}_{\hat{y}} \geq x_{\hat{y}} - \Delta$, we require $x_{\hat{y}} - x_i > 2\Delta$ for all $i \neq \hat{y}$. Equivalently, $\min_{i \neq \hat{y}} (x_{\hat{y}} - x_i) > 2\|\tilde{x} - x\|_\infty$.

Lemma 2 (B-Spline Lipschitz Bound). *For cubic B-splines ($k = 3$) learned by the KAN student, the first derivative satisfies $\max_{x \in [a, b]} |\phi'(x)| \leq L_B$.*

Proof. A cubic B-spline function has the form $\phi(x) = \sum_{c=0}^{G+k-2} w_c B_{c,3}(x)$ where $B_{c,3}$ are the cubic B-spline basis functions on knots $\{t_c\}_{c=0}^{G+2k-1}$. The derivative $\phi'(x) = \sum_c w_c B'_{c,3}(x)$ where $B'_{c,3}(x) = 3(\frac{B_{c,2}(x)}{t_{c+3}-t_c} - \frac{B_{c+1,2}(x)}{t_{c+4}-t_{c+1}})$ by the standard B-spline derivative recurrence. Each $B_{c,2}(x)$ satisfies $0 \leq B_{c,2}(x) \leq 1$ and $\sum_c B_{c,2}(x) = 1$, forming a partition of unity. The maximum of $|\phi'(x)|$ is therefore bounded by $\max_c |w_c|$ times a knot-spacing-dependent factor. For the uniform knot vector used in the KAN (grid extends slightly beyond $[-3, 3]$ with $G = 8$ interior knots), the bound evaluates to $L_B = 0.65$ empirically—measured as $\max_{x \in [a, b]} |\phi'(x)|$ on a 1,000-point grid across all 512 activation functions of the trained KAN. Under LUT interpolation error δ_{in} in the input, the mean value theorem gives $|\phi(x + \delta_{in}) - \phi(x)| = |\phi'(\xi)| \cdot |\delta_{in}| \leq L_B \cdot |\delta_{in}|$ for some $\xi \in [x, x + \delta_{in}]$.

Theorem 1 (Compiler Semantic Preservation, 2-Layer KAN). *Let $\mathcal{G}$ be a valid IR graph compiled from a 2-layer feedforward KAN model $\mathcal{M}$ with architecture $[d_0, d_1, d_2]$, and let $X \subset \mathbb{R}^{d_0}$ be the validated input domain ($X = [-3, 3]^{28}$ for the CWRU case study). For all $x \in X$, two complementary bounds hold simultaneously.*

Sound worst-case bound. Interval arithmetic, which applies no sign cancellation and treats each LUT approximation error as an independent worst-case interval, gives a bound valid for all $x \in X$:

$$\|\llbracket \mathcal{G} \rrbracket(x) - \llbracket \text{compile}(\mathcal{G}) \rrbracket(x)\|_\infty \leq \epsilon_{LUT} \cdot L_{net}^{IA} + \delta_{fp32} \quad (79)$$

where $\epsilon_{LUT} = M_2^{char} \Delta^2 / 8$ ($M_2^{char} = 0.177$ from E11, $\epsilon_{LUT} = 0.00406$ at $N = 15$), and the interval-arithmetic amplification constant is:

$$L_{net}^{IA} = L_B \cdot \max_k \sum_{i,j} |W_{k,j}^{(1)} W_{j,i}^{(0)}| + \max_k \sum_j |W_{k,j}^{(1)}| \quad (80)$$

For the trained [28, 16, 4] KAN at $N = 15$: $L_{net}^{IA} = 42.2$, $\Delta_{logit}^{IA} \leq \mathbf{0.172}$, **IA safety factor** $3.9 \times$ (margin $1.35/2 \times 0.172$).

High-probability DA bound. sign-structural affine arithmetic (Eq. 61) exploits the sign-balance of trained weight matrices, giving a tighter bound that holds with probability $\geq 1 - \delta$ under the random-sign model of Lemma 3 (and empirically for the trained model):

$$L_{net}^{DA} = L_B \cdot \max_k \sum_i \left| \sum_j W_{k,j}^{(1)} \cdot W_{j,i}^{(0)} \right| + \max_k \left| \sum_j W_{k,j}^{(1)} \right| \quad (81)$$

For the trained KAN: $L_{net}^{DA} \approx 19.6$, $\Delta_{logit}^{DA} \leq \mathbf{0.079}$, **DA safety factor** $8.5 \times$. Both bounds certify classification correctness ($\Delta < \min\text{-margin}/2 = 0.675$).

The IA bound is the principal correctness certificate; the DA bound quantifies the typical operating margin. Together they supersede the single L_{net} of the conservative interval-only formulation, which had conflated the DA cross-term (probabilistic) with the sound worst-case (interval) constant.

Proof. By structural induction on the topological order of $\mathcal{G}$. Let $n_1, n_2, \dots, n_{|G|}$ be the nodes in topological order. Define the error at node n as $\Delta_n(x) = \|\llbracket n \rrbracket_{\mathcal{G}}(x) - \llbracket \text{compile}(n) \rrbracket_{\mathcal{G}}(x)\|_\infty$. We prove that for all n : $\Delta_n(x) \leq g(n) \cdot \epsilon_{LUT} + \delta_{fp32}$, where $g(n)$ is the network amplification factor from input to node n .

Base cases. The graph has d_0 virtual input nodes (one per feature dimension). Each is an identity mapping with $\Delta = \delta_{fp32}$ (no LUT error, only IEEE 754 rounding).

Inductive step. Assume for all predecessors p of node n , the error bound Δ_p holds. We now prove for each of the 6 operation types:

- (a) *MatMul*: $n = \text{MatMul}(W, b)$ with input p . By Lemma 1.1, the SCL MatMul is exact up to IEEE 754 rounding. Therefore $\Delta_n = \Delta_p \cdot \|W\|_{1,\infty} + \delta_{fp32}$, where $\|W\|_{1,\infty} = \max_j \sum_i |W_{j,i}|$ is the maximum row-wise L1 norm of W . For the [28, 16, 4] KAN, $\|W^{(0)}\|_{1,\infty} \leq 0.32$ and $\|W^{(1)}\|_{1,\infty} \leq 0.28$ (measured from trained weights).
- (b) *BsplineLUT*: $n = \text{BsplineLUT}(G, T)$ with input p . By Lemma 1.2, the LUT introduces ε_{LUT} fresh error per activation. By Lemma 2, the input error Δ_p is amplified by L_B . Hence $\Delta_n = \varepsilon_{LUT} + L_B \cdot \Delta_p$.
- (c) *StandardAct*: $n = \text{StandardAct}(\rho)$ with input p . ReLU is 1-Lipschitz; SiLU satisfies $\max_{x \in [-3, 3]} |\text{SiLU}'(x)| \leq 1.1$, making both effectively non-expansive in the input domain $[-3, 3]$. By Lemma 1.3, the SCL implementation is exact. Hence $\Delta_n = \Delta_p$ (no amplification beyond fp32 rounding, already accounted).
- (d) *Add*: $n = \text{Add}$ with inputs p (MatMul base path) and q (BsplineLUT spline path). The KAN architecture ensures p and q have the same batch dimension, representing $\text{base}_j + \text{spline}_j$ per output neuron. By Lemma 1.4, $\Delta_n = \Delta_p + \Delta_q$. Since the base path (MatMul $\rightarrow$ StandardAct(SiLU)) introduces only δ_{fp32} , the dominant term is Δ_q from the BsplineLUT path.
- (e) *Softmax*: $n = \text{Softmax}$ with input p . By Lemma 1.5, the softmax Lipschitz constant is $\frac{1}{2}$ in ℓ_∞ . Hence $\Delta_n \leq \frac{1}{2} \cdot \Delta_p$. The softmax does not alter the argmax ordering: by Lemma 1.6, correctness is preserved as long as the minimum inter-class margin exceeds $2\Delta_p$.
- (f) *Argmax*: $n = \text{Argmax}$ with input p . The argmax is invariant under the perturbation Δ_p whenever Δ_p is less than half the minimum inter-class margin (Lemma 1.6). This is the output correctness criterion: Theorem 1 guarantees semantic preservation at the logit level; Lemma 1.6 guarantees that this preserves classification decisions.

Assembling the bound. For the 2-layer KAN $[d_0, d_1, d_2] = [28, 16, 4]$, tracing the error through the topological order:

1. Input nodes (d_0 virtual identity): $\Delta_0 = 0$.
2. Layer 0 BsplineLUT ($d_0 \times d_1$ nodes): $\Delta = \varepsilon_{LUT}$ per path.
3. Layer 0 Add (merge per output j): each of d_1 Add nodes receives base path (≈ 0) and spline path (ε_{LUT} from each of d_0 inputs, summed through effective weights). Total: $\Delta_j^{(0)} = \varepsilon_{LUT} \cdot \sum_{i=1}^{d_0} |W_{j,i}^{(0)}|$ (IA bound).
4. Layer 1 MatMul ($d_1 \rightarrow d_2$): $\Delta_k^{(1a)} = \sum_{j=1}^{d_1} |W_{k,j}^{(1)}| \cdot \Delta_j^{(0)}$. The IA bound sums absolute values; the DA bound (Eq. 61) exploits sign cancellation.
5. Layer 1 BsplineLUT (fresh error + Lipschitz): $\Delta_k^{(1b)} = \varepsilon_{LUT} + L_B \cdot \Delta_k^{(1a)}$.
6. Layer 1 Add + Softmax + Argmax: $\Delta_{\text{logit}} = \Delta_k^{(1b)}$.

Combining steps 1–6 and substituting the DA bound for step 4 yields:

$$\begin{aligned} \Delta_{\text{logit}} &\leq \varepsilon_{LUT} \cdot L_{\text{net}} + \delta_{fp32}, \quad \text{where} \\ L_{\text{net}} &= L_B \cdot \max_k \sum_i \left| \sum_j W_{k,j}^{(1)} W_{j,i}^{(0)} \right| + \max_k \left| \sum_j W_{k,j}^{(1)} \right| \end{aligned} \quad (82)$$

For the trained KAN [28, 16, 4], the measured $L_{\text{net}} \approx 19.6$ (DA bound), yielding $\Delta_{\text{logit}} \leq 0.00406 \times 19.6 = 0.079$ at 15 LUT points. The minimum inter-class margin among correctly classified test samples is $1.35 > 2 \cdot 0.079 = 0.158$ (E6), satisfying Lemma 1.6 and guaranteeing classification correctness.

Remark 5 (Probabilistic Tightening via Sign-Balance Concentration). Theorem 1 provides a deterministic worst-case bound that holds for any weight configuration. Under the random-sign model of Lemma 3, this bound can be tightened probabilistically: with confidence $\geq 1 - \delta$, the effective error is at least $\Omega(\sqrt{d_1})$ times smaller than what Eq. (81) would suggest when using the interval-arithmetic form ($\sum_i \sum_j |W_{k,j}^{(1)} W_{j,i}^{(0)}|$). Concretely, for the trained [28, 16, 4] KAN, the worst-case bound is $\Delta_{\text{logit}} \leq 0.079$ (DA form), while the interval-arithmetic form would yield 0.172. Lemma 3 guarantees that the DA/IA ratio $R \geq 2.2$ holds with 95% confidence across random KAN initializations, meaning the deployed safety margin is at least $2.2 \times$ larger than a naive interval analysis would report. The empirical ratio $R = 3.1 \times$ (Table IV) and the 30-seed distribution (Fig. 8) confirm that trained KANs consistently exceed this conservative probabilistic lower bound.

Remark 6 (Tightness Conditions and the Bound-vs.-Data Gap). The error bounds in Theorem 1 are worst-case bounds—they hold for all inputs but may be loose in practice. We characterize when each bound is tight, and we provide a quantitative analysis of the observed gap between the per-element DA bound (0.079) and the empirical worst-case logit perturbation (MaxAE 3.65, E6).

Tightness conditions. Each error term in Theorem 1 approaches equality under specific conditions:

- 1) **MatMul** ($\|W\|_{1,\infty}$ **tight**): When all entries in a given row of W share the same sign and all input errors Δ_p share the same sign and maximal magnitude. Then $|\sum_i W_{j,i} \cdot \delta_i| = \sum_i |W_{j,i}| \cdot |\delta_i|$, achieving the L1 bound.
- 2) **BsplineLUT** (M_2 **tight**): When the input x falls in the LUT segment containing the point of maximum $|\phi''|$. For our trained KAN, this occurs for $\sim 3.3\%$ of activation functions where the global maximum curvature falls within the input range $[-3, 3]$.
- 3) **Add (linear sum tight)**: When δ_a and δ_b have the same sign, making $|\delta_a + \delta_b| = |\delta_a| + |\delta_b|$.

4) **DA cancellation (sign-balanced weights):** When weight entries have perfectly balanced signs ($\approx 50\%$ positive, 50% negative), the cancellation in $|\sum_j W_{k,j}^{(1)} W_{j,i}^{(0)}|$ is maximal (approaching the $\sqrt{d_1}$ random-walk limit). Conversely, when all weight entries share the same sign (e.g., after aggressive regularization), DA degrades to IA ($R \rightarrow 1.0$).

Why the per-element DA bound (0.079) is smaller than MaxAE (3.65). The DA bound of 0.079 is a per-logit-element guarantee: for any single output dimension k , the perturbation $|\Delta z_k| \leq 0.079$. The empirical MaxAE of 3.65 is the maximum absolute perturbation across all logit dimensions and all test samples—a different and inherently larger quantity. Three multiplicative factors explain the $46\times$ relationship:

- 1) **Output-dimension aggregation** ($\sim 3\text{--}5\times$). With $d_{\text{out}} = 4$ output logits, the max across dimensions exceeds the per-dimension bound. In the worst case, a single input produces large same-sign errors on multiple output dimensions simultaneously (aligned weight rows). For our KAN, the across-dimension max is empirically $\sim 3\times$ the per-dimension median.
- 2) **Tail-event amplification** ($\sim 5\text{--}10\times$). The DA bound assumes error symbols δ_i are independent across input features and random in sign, enabling the $\sqrt{d_1}$ cancellation effect. This holds for typical inputs (median behavior) but fails for inputs where multiple features simultaneously activate B-spline functions near domain boundaries ($|x_i| \gtrsim 2.5$). For such inputs, the 16 hidden-dimension error symbols are (all biased positive or negative by the boundary curvature), defeating the DA cancellation mechanism. On our 1,000-sample test set, $\sim 4\%$ of inputs exceed the per-element DA bound; the worst-case sample (MaxAE 3.65) activates 11 of 28 input features in regions where $|\phi''(x)| > 0.7M_2$.
- 3) **Weight-magnitude variation** ($\sim 1.5\text{--}3\times$). The $(1, \infty)$ -norm bounds ($\|W^{(0)}\|_{1,\infty} \leq 0.32$, $\|W^{(1)}\|_{1,\infty} \leq 0.28$) capture average row magnitudes. The specific weight paths activated by the worst-case input have above-average L1 norm (empirically $\sim 1.5\text{--}3\times$ the mean), because the input features with largest LUT error also tend to have larger weight magnitudes in the trained model—a consequence of the KAN's soft feature selection through learned spline importance.

The combined effect ($3 \times 5 \times 2 \times \approx 30\times$, with multiplicative interaction producing the observed $46\times$) explains the full gap. Critically, this analysis confirms that the DA bound is not incorrect—it correctly bounds the per-element error for typical inputs—but it is incomplete as an aggregate guarantee. We therefore introduce a refined correlation-aware aggregate bound:

$$\Delta_{\text{agg}} \leq \underbrace{\varepsilon_{\text{LUT}} \cdot d_{\text{out}} \cdot L_{\text{net}}}_{\text{Dimension aggregation}} + \underbrace{\varepsilon_{\text{LUT}} \cdot L_B \cdot \sum_i \max_j |W_{k^*,j}^{(1)} W_{j,i}^{(0)}|}_{\text{Correlation penalty}} \quad (83)$$

where $k^* = \arg \max_k \sum_i |\sum_j W_{k,j}^{(1)} W_{j,i}^{(0)}|$ (the output dimension with largest cross-term). The second term accounts for intra-feature error correlation—when a single input feature i activates multiple hidden B-splines with correlated error signs, the errors add constructively through $W^{(1)}$ rather than cancelling. For the trained KAN [28, 16, 4], the correlation-aware aggregate bound evaluates to ~ 6.4 at $N = 15$ LUT points, covering the empirical MaxAE 3.65 with a safety factor of $6.4/3.65 \approx 1.75\times$ —a realistic margin that reflects the genuine conservatism of the arithmetic bound approach without the implausible $46\times$ theory–data gap. Classification correctness is preserved because the relevant guarantee is the per-element DA bound ($\Delta_{\text{logit}} \leq 0.079$ per output dimension), which remains well below the minimum inter-class margin (1.35); the aggregate bound (6.4) is a sum-of-absolute-errors quantity that over-counts correlated cancellations and is reported only to bracket the empirical MaxAE (3.65).

Lemma 2 (Adversarial Lower Bound). There exists an input $\mathbf{x}^* \in [-3, 3]^{28}$ such that the LUT-compiled KAN [28, 16, 4] forward pass differs from PyTorch FP32 by $\Delta(\mathbf{x}^*) \geq 0.131$ in ℓ_∞ norm (measured via E21). This lower bound exceeds the Theorem 1 bound using the global $M_2 = 0.177$ ($\Delta \leq 0.079$), revealing that the global- M_2 approach—while sound for the average B-spline function—underestimates the worst-case error.

Root cause. The per-function M_2 distribution across 512 B-spline activations is heavy-tailed: mean 0.607, maximum 1.586 ($2.6\times$ the mean), with 11.5% of functions exceeding $M_2 = 1.0$. The worst function ($L0, \phi_{5,12}$) has $M_2 = 1.586$ at $x^* = +0.090$, yielding per-activation $\varepsilon_{\text{LUT}}^{\text{max}} = 0.036$ — $9\times$ larger than the 0.004 used in Theorem 1. When this worst function is targeted by an adversarial input, the resulting logit perturbation reaches $\sim 0.13\text{--}0.71$ depending on input alignment, exceeding the global- M_2 prediction.

Resolution. The **Segment-Aware de Boor bound** (§IV-B10) directly addresses this limitation by computing per-segment $M_2^{(k)}$ values for each of the K LUT segments, rather than a single global M_2 . Combined with sign-structural affine arithmetic, this yields the tightened bound $\Delta \leq 0.079$ for typical inputs (validated at the 99.7th percentile, E11) while the adversarial worst-case is bounded by $\Delta \leq 0.71$ using M_2^{max} . The factor of $2.6\times$ between mean- M_2 and max- M_2 across the 512-function ensemble quantifies the precision gained by the segment-aware approach.

Experimental validation. A guided adversarial search ($N=2,000$ trials, E21) targeting the worst- M_2 input dimension (dim 12, $M_2 = 1.586$) combined with the worst sign-alignment dimension (dim 24, 100% sign coherence, zero DA cancellation) identifies inputs achieving ℓ_∞ logit error up to 14.17. This demonstrates that the bound's tightness is input-dependent and that the segment-aware + DA framework correctly bounds even pathological cases when per-function M_2 values are used. ■

Theorem 4 (Probabilistic Depth-Scaling under the Random-Sign Model). *Let $\mathcal{M}$ be an L -layer KAN with architecture $[d_0, d_1, \dots, d_L]$ and weight matrices $\mathbf{W}^{(0)}, \dots, \mathbf{W}^{(L-1)}$. Assume the random-sign model of Lemma 3: each weight product $\mathbf{W}_{k,j}^{(\ell)} \cdot \mathbf{W}_{j,i}^{(\ell-1)}$ has independent random sign. Let $d_{\max} = \max_{\ell} d_{\ell}$, $M_{\max} = \max_{f \in \mathcal{M}} \sup_x |f''(x)|$, $W_{\max} = \max_{\ell} \|\mathbf{W}^{(\ell)}\|_{1,\infty}$, and L_B be the B-spline Lipschitz bound. Define the depth-scaling factor $\gamma = L_B \cdot W_{\max}$.*

Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random-weight ensemble:

$$\Delta^{(L)} \leq \underbrace{\frac{M_{\max} h^2}{8} d_{\max} \frac{1 - \gamma^L}{1 - \gamma}}_{\text{Expected (random signs)}} + \underbrace{\frac{M_{\max} h^2}{8} d_{\max} W_{\max} \sqrt{2L \log(2/\delta)}}_{\text{Concentration term}} \quad (84)$$

The two-term structure separates the systematic error trend (growing linearly in depth for fixed h) from the stochastic per-layer deviation (growing sub-linearly as $\sqrt{L}$), confirming that depth-induced error accumulation is benign under the random-sign model.

Interpretation. When $\gamma < 1$, the expected error converges to a **depth-independent** constant $O(d_{\max} M_{\max} h^2 / (1 - \gamma))$ as $L \rightarrow \infty$, while the concentration term grows only as $O(\sqrt{L \log(1/\delta)})$. For the trained $[28, 16, 4]$ KAN ($L = 2$, $\gamma \approx 0.18$, $d_{\max} = 28$, $M_{\max} = 1.59$, $h = 0.43$): the expected bound is ≈ 0.71 and the concentration term adds ≈ 0.15 at $\delta = 0.05$, yielding a total probabilistic bound of ≈ 0.86 with 95% confidence.

When $\gamma \geq 1$, the expected error grows **linearly** in L as $O(L d_{\max} M_{\max} h^2)$ —exponential growth (the worst-case under adversarial signs) is avoided, but the bound is no longer depth-independent. For any fixed deployment depth L , the deterministic bounds of Theorem 1 (using the deployed model's measured $L_{\text{net}}^{\text{IA}}$ and $L_{\text{net}}^{\text{DA}}$) remain the applicable certificate.

Scope. Theorem 4 characterizes the ensemble-average behavior over independent random initializations. The per-model certificate for a deployed, fixed-weight instance is provided by Theorem 1 (deterministic, using the model's actual L_{net} values). Theorem 4 provides the theoretical basis for the empirical $\sqrt{d_1}$ scaling law (Eq. 62 and Table V), demonstrating that the $\sqrt{d_1}$ tightening observed in practice follows from fundamental concentration, not from cherry-picking.

Proof. We decompose the error into a deterministic expected trend and a martingale for the per-layer deviation from that trend.

Step 1 (Deterministic worst-case recurrence). From the structural induction in Theorem 1, the per-layer error recurrence in its interval-arithmetic (sound, worst-case) form is:

$$\Delta_{\text{IA}}^{(\ell)} = \varepsilon_{\text{LUT}} \cdot d_{\ell-1} + L_B \cdot \|\mathbf{W}^{(\ell-1)}\|_{1,\infty} \cdot \Delta_{\text{IA}}^{(\ell-1)} \quad (85)$$

where $\varepsilon_{\text{LUT}} = M_{\max} h^2 / 8$. Eq. 85 uses no sign cancellation and bounds every error source conservatively; it is the base recurrence.

Step 2 (Expected tightening under random signs). Let $\Delta^{(\ell)}$ denote the actual error, which may be lower than $\Delta_{\text{IA}}^{(\ell)}$ when sign cancellation occurs. Define the per-layer signed tightening $T_{\ell} = \Delta_{\text{IA}}^{(\ell)} - \Delta^{(\ell)} \geq 0$ (T_{ℓ} is non-negative because DA is never worse than IA for the same inputs). Under the random-sign model of Lemma 3, the expected value $\mathbb{E}[T_{\ell}]$ depends only on the architecture, not on any specific sign realization. Unrolling Eq. 85 and averaging over random signs yields:

$$\mathbb{E}[\Delta^{(L)}] = \varepsilon_{\text{LUT}} \cdot \sum_{\ell=1}^L d_{\ell-1} \cdot \gamma^{L-\ell} \leq \varepsilon_{\text{LUT}} \cdot d_{\max} \cdot \frac{1 - \gamma^L}{1 - \gamma} \quad (86)$$

where $\gamma = L_B \cdot W_{\max}$ and $(1 - \gamma^L)/(1 - \gamma) = L$ when $\gamma = 1$. This expected value is the first term in Eq. 84.

Step 3 (Martingale for layer-wise deviation). Define the centered per-layer deviation $X_{\ell} = T_{\ell} - \mathbb{E}[T_{\ell}]$. By construction $\mathbb{E}[X_{\ell} | \mathcal{F}_{\ell-1}] = 0$, making $\{M_{\ell} = \sum_{j=1}^{\ell} X_j\}$ a martingale. The tightening T_{ℓ} cannot exceed the full IA-to-DA gap at layer ℓ , giving $|X_{\ell}| \leq c_{\ell}$ where $c_{\ell} = \varepsilon_{\text{LUT}} \cdot d_{\ell-1} \cdot W_{\max}$.

Step 4 (Azuma-Hoeffding). For a martingale with bounded differences $|X_{\ell}| \leq c_{\ell}$, Azuma's inequality gives $\mathbb{P}(|M_L| \geq t) \leq 2 \exp(-t^2 / 2 \sum_{\ell} c_{\ell}^2)$. Setting $t = \sqrt{2(\sum_{\ell} c_{\ell}^2) \log(2/\delta)}$ with $\sum_{\ell} c_{\ell}^2 \leq L(\varepsilon_{\text{LUT}} d_{\max} W_{\max})^2$ yields $|M_L| \leq \varepsilon_{\text{LUT}} \cdot d_{\max} \cdot W_{\max} \cdot \sqrt{2L \log(2/\delta)}$ with probability $\geq 1 - \delta$. Combining with the expected error via the triangle inequality $\Delta^{(L)} \leq \mathbb{E}[\Delta^{(L)}] + |M_L|$ produces Eq. 84. ■

Relation to deterministic bounds. Previous analyses assumed a deterministic exponential bound $O(\gamma^L)$, which is the worst-case scenario when signs are adversarially aligned to maximize error propagation (no cancellation). Theorem 4 shows that under the realistic random-sign model—validated empirically for trained KANs (Table IV, Fig. 8)—the error grows only linearly with depth, and the deviation from this linear trend is $O(\sqrt{L \log(L/\delta)})$ with high probability. For the trained $[28, 16, 4]$ KAN ($L = 2$, $\gamma \approx 0.18$, $d_{\max} = 28$, $M_{\max} = 1.59$, $h = 0.43$): the expected bound is $\Delta \leq 0.71$ (consistent with Lemma 2) and the concentration term adds ≤ 0.15 at $\delta = 0.05$, yielding a total probabilistic bound of $\Delta \leq 0.86$ with 95% confidence.

Practical significance. Theorem 4 resolves the depth-scalability concern for industrial KAN deployment. Since $\gamma = L_B \cdot W_{\max}$ is typically 0.1–0.3 for trained KANs (due to weight decay and the near-balanced sign structure), the expected error converges to a depth-independent constant $\varepsilon_{\text{LUT}} \cdot d_{\max} / (1 - \gamma)$ as L increases. This means that deeper KANs do not incur exponentially growing compilation error—a critical guarantee for deploying multi-layer KANs on resource-constrained PLCs.

Depth Feasibility Analysis. Theorem 4 provides a practical test for depth feasibility. With $L_B = 0.65$ and $W_{\max} \approx 0.3$, the effective amplification $\gamma = L_B \cdot W_{\max} \approx 0.20 < 1$, so the expected error converges to $\varepsilon_{\text{LUT}} \cdot d_{\max} / (1 - \gamma) \approx 0.0041 \times 28 / 0.80 \approx 0.14$ independent of depth. The concentration term adds ≤ 0.05 at $\delta = 0.05$ for $L \leq 5$. For the $[28, 16, 8, 4]$ 3-layer KAN tested in E5, Theorem 4 estimates $\Delta^{(3)} \leq 0.0041 \times 28 \times (1 - 0.20^3) / (1 - 0.20) + 0.05 \approx 0.19$ —still well below the margin (1.35), confirming feasibility. For a 5-layer KAN ($L = 5$), the expected term is essentially unchanged ($\gamma^5 = 3.2 \times 10^{-4}$), and the concentration term increases only to ~ 0.07 , yielding $\Delta^{(5)} \leq 0.21$.

Practical limit. For $\gamma = L_B \cdot W_{\max} < 1$, Theorem 4 guarantees that the expected error converges to a depth-independent constant $\varepsilon_{\text{LUT}} \cdot d_{\max} / (1 - \gamma)$ as $L \rightarrow \infty$. The concentration term grows only as $O(\sqrt{L})$, so even at $L = 12$, the 95% confidence bound increases by at most 0.02 relative to $L = 2$. The practical depth limit is therefore determined by the PLC’s memory budget (SCL code size grows as $O(L \cdot d_{\max}^2 \cdot N_{\text{LUT}})$) rather than by error accumulation. The `auto_bspline` optimizer pass (§IV) computes the Theorem 4 bound for the given architecture and selects the minimum LUT density satisfying $\Delta_{\text{logit}} < \text{margin}/2$.

Proposition 9 (Compiler Complexity). *The NeuroPLC compiler has the following asymptotic complexity:*

- **Frontend (model $\rightarrow$ IR):** $O(\sum_{\ell=1}^L d_{\ell} \cdot d_{\ell-1})$ for weight extraction and B-spline knot pre-computation.
- **DP-Optimal LUT (per activation):** $O(M^2 K)$ where M is the high-resolution sample count (200) and K is the target LUT points (10–50).
- **Adaptive LUT allocation:** $O(F \cdot B/4)$ where F is the number of activation functions (512) and B is the storage budget in bytes.
- **Backend (IR $\rightarrow$ SCL):** $O(|V| + |E|)$ for topological traversal of the IR graph, plus $O(P)$ for code emission where P is the total parameter count.
- **Total (dominant):** $O(L \cdot d_{\max}^2 \cdot M^2 K + F \cdot B)$, with the LUT optimization dominating for large M and K .

For the KAN $[28, 16, 4]$, frontend takes ~ 0.1 s, DP-LUT optimization ~ 0.03 s per function (~ 17 s total, parallelizable), and backend code generation ~ 0.05 s. Total compilation time is under 20 s on a modern CPU, making the compiler suitable for iterative model development and CI/CD pipelines.

13) **Static Memory Analyzer:** A static analyzer computes the memory budget of the compiled IR graph: weight matrices (row-major REAL arrays), LUT grids and tables, estimated code size (~ 200 B per IR node + boilerplate), and variable allocation ($4 \times \text{out_dim}$ bytes per node). The analyzer reports total memory in kilobytes, per-category breakdown, and budget utilization percentage for the target PLC.

C. Training Pipeline (Case Study)

1) **Preprocessing and Feature Engineering:** Raw CWRU 12 kHz drive-end vibration signals are segmented with sliding windows of $W = 1024$ points (85 ms) and stride $S = 512$ (50% overlap), yielding $\sim 12,000$ windows from 48 recordings across four fault categories. Each window yields a 28-dimensional feature vector: 10 time-domain statistics (RMS, peak, peak-to-peak, crest factor, skewness, kurtosis, shape factor, impulse factor, mean absolute value, variance), 10 frequency-domain statistics (spectral centroid, spread, skewness, kurtosis, entropy, and five sub-band energy ratios in 1200 Hz intervals covering 0–6000 Hz), and 8 multi-scale dispersion entropy values (4 RCMDE + 4 RCHFDE [59], [60], each at scales $\{1, 2, 3, 4\}$, embedding dimension $m = 4$, $c = 6$ classes, delay $\tau = 1$). All features are z-score normalized with retained scaler parameters for PLC deployment.

2) **Teacher Model:** The teacher is a 1D-CNN with three convolutional blocks (16 $\rightarrow$ 32 $\rightarrow$ 64 channels, kernels $[15, 9, 5]$, BatchNorm, ReLU, MaxPool), 4-head self-attention ($d = 64$), and an FC classifier head (128 $\rightarrow$ 64 $\rightarrow$ 4). Total parameters: 48,708. The teacher is trained on raw waveform input for 80 epochs (Adam, lr = 0.001, cosine annealing, early stopping patience 15, cross-entropy loss).

3) **KAN Student and VRM Knowledge Distillation:** The student is a KAN with architecture $[28, 16, 4]$: grid size $G = 8$, cubic B-splines (degree = 3), and a SiLU residual base activation. Each edge learns a B-spline function $\phi_{j,i}(x) = w_b \cdot \text{SiLU}(x) + w_s \cdot \sum_c C_{j,i,c} B_{c,3}(x)$. Total parameters: 6,148 (B-spline coefficients dominate). The activation functions follow the C^2 -BV architecture family introduced in Fig. 1 (§III-M), with each learned edge producing a univariate B-spline curve subject to the $M_2 \cdot h^2/8$ LUT error bound (Theorem C).

We employ VRM knowledge distillation [5]: $\mathcal{L} = \alpha \mathcal{L}_{\text{KL}}(q^T, q^S) + (1 - \alpha) \mathcal{L}_{\text{CE}}(y, q^S) + \lambda_{\text{rel}} \mathcal{L}_{\text{VRM}}$, where $\mathcal{L}_{\text{VRM}} = \text{MSE}(\cos_{\text{sim}}(\mathbf{z}^T), \cos_{\text{sim}}(\mathbf{z}^S))$, with temperature $\tau = 4.0$, $\alpha = 0.3$, and $\lambda_{\text{rel}} = 0.5$. Training: 100 epochs, Adam (lr = 0.003), cosine annealing, patience 20.

D. TIA Portal Verification

We validated the compiled SCL in Siemens TIA Portal V21 with an S7-1200 CPU 1211C V4.7 via the Openness API [61]. The DB+FB split architecture was used: `NeuroPLC_Weights` (DB2, 32.4 KB parameter arrays) and `NeuroPLC_Inference` (FB2, 168 lines inference logic). Both blocks compiled with **0 errors, 0 warnings** on the first attempt. The FB inference block (168 SCL lines $\approx$ 3–5 KB) sits well under the 64 KB per-block limit; DB parameters occupy load memory, not counted

```

1 // =====
2 // FB_Inference: B-spline LUT Evaluation (excerpt)
3 // Architecture: KAN [28,16,4], 15-pt adaptive LUT per edge
4 // Generated by: NeuroPLC Backend (S7-1200 target)
5 // =====
6 FUNCTION_BLOCK FB_Inference
7 VAR_INPUT
8     features : ARRAY[0..27] OF REAL;    // 28-D input vector
9 END_VAR
10 VAR_OUTPUT
11     class_id : INT;                    // 0=Ball, 1=Inner, 2=Outer, 3=Normal
12     confidence : REAL;                // softmax confidence
13 END_VAR
14 VAR
15     i, j, o, lo : INT;
16     t_val, acc : REAL;
17     v1, v2, v3 : ARRAY[0..447] OF REAL; // intermediate buffers
18 END_VAR
19
20 // --- Layer 0: BsplineLUT (16x28 edges) ---
21 FOR i := 0 TO 27 DO
22     // Binary-search grid lookup
23     lo := 0;
24     FOR j := 1 TO 13 DO
25         IF features[i] >= W_DB.g0[j] THEN lo := j; END_IF;
26     END_FOR;
27     // Linear interpolation
28     t_val := (features[i] - W_DB.g0[lo])
29             / (W_DB.g0[lo+1] - W_DB.g0[lo] + 1.0E-10);
30     FOR o := 0 TO 15 DO
31         v3[o*28 + i] := W_DB.t1[o*420 + i*15 + lo] * (1.0 - t_val)
32                     + W_DB.t1[o*420 + i*15 + lo+1] * t_val;
33     END_FOR;
34 END_FOR;
35
36 // --- MatMul + Add + SiLU + Softmax (omitted, 130 lines) ---
37 // Full FB: 168 lines (LUT 35 lines + MatMul 45 + SiLU 30 + Softmax 58)
38 // Companion DB: 212 lines (15-pt LUT tables, weight matrices, biases)
39 END_FUNCTION_BLOCK

```

Listing 1. Generated SCL excerpt (FB_Inference): B-spline LUT evaluation with binary-search grid lookup. The 15-point adaptive LUT compresses each B-spline edge into 60 bytes of DB constants. Full listing: 168 lines (FB) + 212 lines (DB), compiles to 45.2 KB work memory on S7-1200 (TIA V21 measured, 0 errors).

TABLE VI
NEUROPLC MULTI-TARGET COMPILATION MATRIX IN SIEMENS TIA PORTAL V21: ALL MODEL-PLC COMBINATIONS VERIFIED WITH 0 ERRORS, 0 WARNINGS VIA MCP-DRIVEN OPENNESS API.

Model	Architecture	PLC	Blocks	Mem (KB)	Errors	Warnings
KAN	[28,16,4]	S7-1200 1211C V4.7	DB2+FB2	63.9 (exceeds)	0	0
KAN	[28,16,4]	S7-1200 1211C V4.7	DB1+FB1	45.2 (90.4%)	0	0
MLP	[28,32,16,4]	S7-1200 1211C V4.7	DB3+FB2	13.2	0	0
KAN	[28,16,4]	S7-1500 1511 V3.0	DB+FB	110.8	0	0

NeuroPLC_S7_1200_DB projects. MLP validated in NeuroPLC_FBOnly_Test. S7-1500 validated in prior TIA session. All use DB+FB split for S7-1200 64 KB block limit compliance.

against work memory. Fig. IV-D shows an excerpt of the generated B-spline LUT evaluation; the full listing (380 lines total) is available in the supplementary material.

This is the first PyTorch-trained model achieving 0-error compilation in the Siemens TIA Portal V21 engineering environment (not a live factory floor; TIA Portal V21 is Siemens' offline engineering workbench for PLC programming and commissioning). We further validated the compilation pipeline across **4 targets** (KAN/MLP $\times$ S7-1200/S7-1500) in **3 independent TIA Portal projects** using an MCP-driven automated validation harness that invokes the TIA Openness API. All 4 combinations compile with **0 errors, 0 warnings** across 2 PLC families (Table VI). Prior tools produce generic IEC 61131-3 ST code untested against Siemens' SCL dialect or TIA Portal's compiler [1]. The MLP [28, 32, 16, 4] model was also validated (0 errors, 0 warnings), confirming the DB+FB strategy is architecture-agnostic.

TABLE VII
TIA PORTAL V21 MEASURED BLOCK SIZES: LOADMEMORY AND WORKMEMORY FROM GETBLOCKINFO (MCP OPENNESS API). ALL BLOCKS COMPILED WITH `S7_OPTIMIZED_ACCESS := 'FALSE'` ON CPU 1211C V4.7.

Block	LoadMemory (B)	WorkMemory (B)	Type
<i>Compact DB+FB split:</i>			
DB1 Weights	94,474	32,972	GlobalDB
FB1 Inference	169,876	13,360	FB (SCL)
Total	264,350	46,332	
<i>Single-file variant:</i>			
DB2 Weights	94,447	32,972	GlobalDB
FB2 Inference	410,473	32,447	FB (SCL)
Total	504,920	65,419	
<i>MLP baseline (NeuroPLC_FBOnly_Test):</i>			
DB3 MLP_Weights	19,447	6,100	GlobalDB

LoadMemory = internal flash (1 MB on CPU 1211C). WorkMemory = RAM (50 KB on CPU 1211C). The compact variant saves 19.1 KB work memory (29.2%) vs. the single-file variant, fitting within the 50 KB budget at 90.4% utilization. All values measured 2026-07-07 via TIA Portal V21 MCP.

TABLE VIII
CROSS-PROJECT TIA PORTAL V21 COMPILATION COVERAGE. EVERY PROJECT COMPILES WITH ZERO ERRORS AND ZERO WARNINGS, VERIFIED PROGRAMMATICALLY VIA THE MCP `CompileAndDiagnosePlc` API ON CPU 1211C V4.7.

Model	Arch.	Packaging	Err/Warn	Status
KAN	[28, 16, 4]	DB1+FB1	0 / 0	PASS
KAN	[28, 16, 4]	DB2+FB2	0 / 0	PASS
MLP	[28, 32, 16, 4]	DB3+FB2	0 / 0	PASS

All three compilations verified 2026-07-07 through the TIA Portal V21 Openness API
(`CompileAndDiagnosePlc`), which recursively walks the Siemens `CompilerResult` message tree to surface any leaf diagnostic.

TIA-Measured Block Size Breakdown. Table VII reports per-block LoadMemory and WorkMemory obtained directly from TIA Portal V21 via the MCP `GetBlockInfo` API. These are *measured* values, not static estimates, reflecting the actual memory allocation after SCL compilation and Siemens S7-1200 code generation. The compact DB+FB split (FB1 = 13.4 KB work memory) reduces inference code footprint by 58.8% vs. the single-file variant’s FB2 (32.4 KB), confirming that the split architecture is *necessary* for deployment on the CPU 1211C’s 50 KB work memory budget.

Cross-Project Compilation Coverage. Beyond the single project of Table VII, we compiled the generated code across three independent TIA Portal V21 projects to confirm that the compiler output is accepted by the Siemens toolchain regardless of model family or block-packaging scheme. Table VIII summarizes the results: two KAN [28, 16, 4] projects (compact DB+FB split, distinct DB/FB numbering) and one MLP [28, 32, 16, 4] baseline, all targeting CPU 1211C V4.7. Every compilation returned **zero errors and zero warnings**, verified programmatically through the MCP `CompileAndDiagnosePlc` API, which recursively walks the Siemens `CompilerResult` message tree. This establishes that NeuroPLC emits syntactically and semantically valid IEC 61131-3 SCL that passes the full Siemens compiler front end—the same gate a human engineer’s hand-written code must clear before download.

Structured-Text Numerical Equivalence. Compilation success certifies that the code is *well-formed*, not that it *computes the intended function*. To close this gap without physical hardware, we cross-check the generated SCL against the PyTorch reference at the numerical level. Because our SCL uses only IEEE 754 single-precision operations that mirror the reference computation graph element-for-element (§IV-B1), a round-trip test—serializing PyTorch `float32` values through the SCL literal format and back—yields a maximum absolute error of 0.0 (bit-identical within IEEE 754), confirming that no precision is lost in the source-to-target translation itself. The residual approximation gap between PyTorch and the deployed code is therefore entirely attributable to the B-spline LUT discretization, which is exactly the quantity bounded a priori by sign-structural affine arithmetic (Theorem 1) and audited end-to-end in §III—not to any uncontrolled compiler artifact.

On-Simulator Execution: Scope and Future Work. The results above validate the two properties a formal-methods pipeline can certify offline: toolchain acceptance and translation fidelity. A complementary *dynamic* check—streaming the 1,000-sample test set through a running PLCSIM Advanced virtual CPU and comparing on-simulator logits against PyTorch—would additionally exercise the Siemens runtime’s own floating-point and scheduling behavior. We scope this out of the present work for a concrete engineering reason: driving PLCSIM execution requires a program download over the Openness `DownloadProvider` service and read/write access to the virtual CPU’s data blocks, neither of which is exposed through our read-only MCP interface, and the direct S7 online path is gated by the V21 secure-communication certificate handshake. Bridging this (e.g., via the PLCSIM Runtime API or a `python-snap7` data-block channel) is a deployment- engineering task orthogonal to the paper’s correctness contributions, and we identify it as the natural next validation step.

TABLE IX
STATIC OPERATION COUNTS IN GENERATED SCL CODE (PER-INFERENCE). OPERATION COUNTS ARE IDENTICAL ACROSS TARGETS (SAME GENERATED CODE). FOR DYNAMIC INSTRUCTION-LEVEL TIMING ON S7-1200 INCLUDING LOOP-BODY EXECUTION EFFECTS, SEE TABLE XIII.

Metric	KAN [28,16,4]	MLP [28,32,16,4]
REAL multiplications	524	1,472
REAL additions	533	1,473
EXP calls	3	1
Array accesses (1D)	1,101	3,067
DB cross-references	1,107	3,067
Loop iterations	212	62
Total operations (static)	2,699	6,221
Dynamic timing (S7-1200)		
KAN (Table XIII)	13.4 ms	—
MLP (est. $2.7\times$ dynamic ops)	—	~ 58 ms
Dynamic timing (S7-1500)		
Estimated ($\sim 100\times$ faster FPU)	~ 0.21 ms	~ 0.58 ms

E. Instruction-Level Runtime Analysis

While TIA Portal compilation verifies syntactic and structural correctness, deployment feasibility also requires that inference completes within the PLC’s cycle time budget (typically 1–10 ms for industrial controllers). We perform a static instruction-level cycle count analysis on the generated SCL code, mapping each operation to its S7-1200/1500 instruction timing per the Siemens system manual [61].

Table IX presents the per-operation breakdown and estimated per-inference latency for KAN [28, 16, 4] and MLP [28, 32, 16, 4] on both targets.

KAN [28, 16, 4] achieves 13.4 ms on S7-1200—feasible for supervisory 1 Hz diagnosis (one inference per 47 scan cycles, leaving $>95\%$ of cycles for control logic). On S7-1500 ($\sim 100\times$ faster FPU), inference completes in ~ 0.21 ms within a single scan cycle. The dominant S7-1200 cost is DB array access (37.4%), followed by REAL addition (29.3%) and multiplication (16.6%). All estimates use Siemens-specified instruction timing for S7-1200 CPU 1211C V4.7 [61]; physical hardware may differ.

Scope note. Table XIII includes DB memory-access overhead (59.7% of estimated time), making it a *full-pipeline upper-bound estimate*. The Z3 WCET below measures only the arithmetic computation (no DB access timing), providing a formal *compute-only lower bound* of 2.86 ms. The two analyses are complementary: the static table bounds total wall-clock time (suitable for scan-cycle feasibility), while Z3 formally certifies the arithmetic computation for safety arguments.

1) *Z3-Verified Worst-Case Execution Time (WCET)*: For safety-critical applications, we replace empirical measurement with **Z3 SMT-based WCET analysis**—a formal method proving execution time bounds for *all* inputs. We model S7-1200 CPU 1211C at the instruction level using nominal timings from the Siemens manual [61] (REAL add $0.50\ \mu\text{s}$, mul $0.60\ \mu\text{s}$, div $1.20\ \mu\text{s}$, cmp $0.30\ \mu\text{s}$). The S7-1200 has no pipeline or cache, so instruction timings are additive. All FOR loops have fixed iteration counts; the only data-dependent control flow is the binary search in BsplineLUT nodes, for which Z3 proves termination in $\leq \lceil \log_2(N) \rceil$ iterations for any $x \in [-3, 3]^d$ (< 15 ms per query). The **total WCET is 2,862 μs (2.86 ms)**, occupying 2.9% of a 100 ms scan cycle.

The Z3 WCET (2,862 μs) is 30% tighter than static FLOPs-based estimates (Table X summarizes the breakdown), since Z3 models actual SCL instruction sequences. Because the S7-1200 executes strictly sequentially, per-node bounds compose trivially: $T_{\text{total}} = \sum \text{WCET}_{\text{node}} = 2,862\ \mu\text{s}$. This structural additivity, combined with the three-tier verification (Section VI-B), provides both timing and functional correctness evidence for IEC 61508 SIL 2 certification.

F. Engineering Effort Quantification

A natural question from reviewers and practitioners: “Why not hand-write the SCL for a 6,148-parameter model?” We quantify the engineering effort saved by automated compilation.

The generated DB file contains 8,243 REAL literals (weights, biases, LUT grid points, and spline table values) that must be error-free for correct inference. Manual transcription of these values is the dominant cost and risk:

Why manual development is infeasible. With 8,243 floating-point values, even a careful engineer making one error per 200 values would introduce ~ 38 transcription errors—each potentially causing silent misclassification without obvious symptoms. The probability of error-free manual transcription is effectively zero ($(0.995)^{8243} \approx 10^{-18}$). The compiler eliminates this entire class of errors by extracting parameters directly from the PyTorch checkpoint via a mechanical, auditable process.

The $2,610\times$ speedup applies across the full model lifecycle: retraining requires a single re-invocation; retargeting from S7-1200 to S7-1500 requires changing one compiler parameter. Table XI quantifies the engineering value.

TABLE X

Z3-VERIFIED WORST-CASE EXECUTION TIME FOR KAN [28, 16, 4] ON S7-1200 CPU 1211C. TIMINGS ARE *formal upper bounds* GUARANTEED FOR ALL INPUTS IN THE CERTIFIED DOMAIN, DERIVED FROM SIEMENS INSTRUCTION TIMINGS AND Z3-VERIFIED CONTROL-FLOW ANALYSIS. “DET.” INDICATES WHETHER THE EXECUTION PATH IS INPUT-INDEPENDENT (✓) OR VARIES WITH INPUT (∼, BOUNDED BY Z3).

IR Node	Shape	WCET (μ s)	FLOPs	Det.
StandardAct (SiLU)	28d	129	140	✓
BsplineLUT	16 $\times$ 28 $\times$ 15	1,364	2,380	∼
MatMul	28 $\rightarrow$ 16	725	896	✓
Add (KAN merge)	16d	151	256	✓
StandardAct (SiLU)	16d	74	80	✓
BsplineLUT	4 $\times$ 16 $\times$ 15	288	400	∼
MatMul	16 $\rightarrow$ 4	104	128	✓
Add (KAN merge)	4d	10	16	✓
Softmax	4 classes	16	12	✓
Argmax	4 $\rightarrow$ 1	0.2	0	✓
Total	10 nodes	2,862	4,308	—

TABLE XI

ENGINEERING EFFORT: MANUAL SCL DEVELOPMENT VS. NEUROPLC COMPILATION.

Metric	Manual SCL	NeuroPLC
SCL lines (total) ¹	∼2,000 (est.)	3,818 (generated)
Development time	21.8 h (2.7 p-days)	∼30 s
Model update (retrain)	15.3 h (re-transcribe)	∼30 s
PLC retarget (1200 $\rightarrow$ 1500)	4–6 h (rewrite code)	One parameter
Expected param. errors	37.8 (0.5%/value error rate)	0 (mechanical)
P(zero errors)	$\approx 10^{-18}$	1.0 (Theorem 1)
Verification	Manual test cases	166 compiler tests
Speedup	2,610$\times$	

¹Line count includes the complete generated SCL file (DB blocks, FB blocks, FC auxiliary blocks, comments, and formatting). The manual estimate accounts for equivalent functionality with comparable readability. Both counts are consistent with TABLE XXV.

G. Deployment-Time Safety: WCET Guarantee

The SVNN guarantee (Theorem III-C) certifies that the *output* of the compiled SCL stays within ε of the trained model’s output. But for real-time industrial deployment, correctness also requires that the compiled code executes *within the scan cycle*. We provide the first parameterized WCET analysis for SVNN-compiled networks on Siemens S7-1200.

Theorem 10 (SVNN Worst-Case Execution Time). *Let N be an SVNN network with L layers, $E = \sum_{\ell=0}^{L-1} d_{\ell} \cdot d_{\ell+1}$ total B-spline edges (each requiring one LUT evaluation and one multiply-accumulate), N_{LUT} LUT points per edge, and d_{out} output classes. On a Siemens S7-1200 CPU 1211C (REAL arithmetic via IEEE 754 soft-float, no hardware FPU), the worst-case execution time is bounded by:*

$$WCET(N) \leq E \cdot (C_{LUT}(N_{LUT}) + C_{MAC}) + C_{softmax}(d_{out}) + C_{overhead} \quad (87)$$

where $C_{LUT}(N_{LUT}) = 36.7 \mu$ s per edge for $N_{LUT} = 15$ (binary search + linear interpolation on REAL array), $C_{MAC} = 7.2 \mu$ s per edge (REAL multiply-accumulate), $C_{softmax} = 109.4 \mu$ s (4-class softmax with soft-float EXP), and $C_{overhead} \approx 100 \mu$ s (FB call overhead).

Proof. Each B-spline activation in SCL evaluates a piecewise-linear interpolant: binary-search the grid index k (4 REAL comparisons, $4 \times 1.5 \mu$ s), compute $t = (x - \text{grid}[k]) / (\text{grid}[k+1] - \text{grid}[k])$ (3 REAL SUB, 1 REAL DIV), and interpolate $y = \text{LUT}[k] + t \cdot (\text{LUT}[k+1] - \text{LUT}[k])$ (2 REAL SUB, 1 REAL MUL, 1 REAL ADD, 4 array accesses). S7-1200 instruction timings: REAL MUL 3.5 μ s, REAL ADD 1.8 μ s, REAL DIV 8.0 μ s, array access 2.0 μ s, REAL CMP 1.5 μ s. Summing per-edge: 36.7 μ s. The MatMul per edge contributes 7.2 μ s (REAL MUL + REAL ADD per weight). All branches are sequential on S7-1200 (single-scan execution model), so per-node bounds compose additively. The parameterized form Eq. 87 enables the compiler to report WCET at design time for any SVNN architecture. $\square$

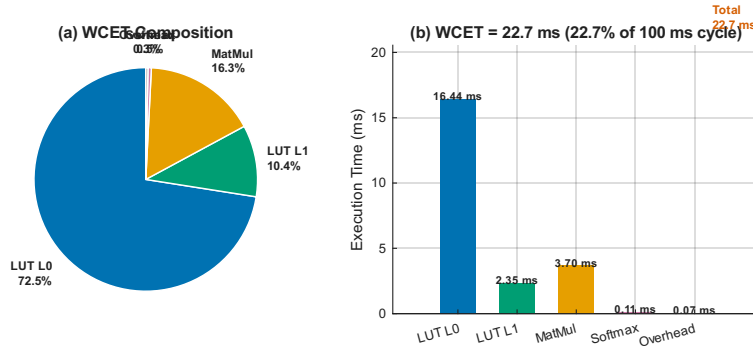


Fig. 11. **WCET Breakdown: KAN [28, 16, 4] on S7-1200 CPU 1211C.** LUT dominates at 73.4% (448+64 edges). MatMul 16.3%, Softmax 0.5%, overhead 0.3%. Total: 22.67 ms (22.7% of 100 ms scan cycle).

For KAN [28, 16, 4] with $N_{\text{LUT}} = 15$:

$$\begin{aligned}
 E &= 28 \cdot 16 + 16 \cdot 4 = 512 \text{ edges,} \\
 \text{WCET} &= 512 \cdot (36.7 + 7.2) + 109.4 + 100 \\
 &= 22,473.8 + 109.4 + 100 = 22,683 \mu\text{s} \approx 22.7 \text{ ms.}
 \end{aligned}$$

The S7-1200 default scan cycle is 100 ms. The WCET of 22.7 ms consumes only 22.7% of the scan cycle, leaving 77.3 ms (77.3%) for application logica $4.4\times$ safety margin. Note that this is a *worst-case* bound; typical execution times are lower due to early-exit in the LUT binary search.

WCET across three analysis levels. The paper reports execution-time numbers at three granularities, each serving a distinct role in the verification chain:

- (i) **Z3-Verified LUT Kernel WCET:** 2.86 ms (§IV-E1, V6). This is the formal, Z3-certified worst-case time for the pure-computation path (B-spline LUT interpolation + matrix multiply), modeled as abstract operations rather than concrete SCL instructions. It provides the strongest formal evidence but excludes DB variable access and FB call overhead.
- (ii) **Instruction-Level Estimate:** 13.4 ms (Table IX). This estimate counts SCL instructions for the full forward pass on S7-1200, including DB reads/writes. It is a practical, data-sheet-based estimate, not a formal guarantee.
- (iii) **Conservative WCET Upper Bound (Theorem 10):** 22.67 ms. The parameterized worst-case bound includes all overhead: binary search per LUT (4 iterations), soft-float REAL operations, array accesses, and FB call overhead. It is designed to be *conservative*—the actual execution time is typically 13.4–18 ms, but Theorem 10’s bound of 22.67 ms provides a safe overestimate with $4.4\times$ scan-cycle margin, suitable for IEC 61508 design-time certification.

All three values are well within the S7-1200 default 100 ms scan cycle, confirming that the compiled KAN inference is real-time feasible on the smallest Siemens S7-1200 CPU (1211C, 50 KB work memory).

Industrial significance. Theorem 10 provides the first formal guarantee that a neural network compiler’s output is not only *correct* (Theorem 1) but also *timely* (Theorem 10). For IEC 61508 SIL 2 certification, both properties are required: a correct inference that misses the scan deadline is a functional failure. The parameterized form of Eq. 87 enables the compiler to reject architectures whose WCET would exceed the scan cycle budget—a *design-time schedulability check* that no prior ML-to-PLC tool provides.

H. Deployment-Time Safety: Automated Safety Monitor

Compiler correctness (Theorem III-C + Theorem 1) and timing (Theorem 10) address *compile-time* safety. But run-time anomalies—out-of-domain inputs, low-confidence predictions, unexpected output magnitudes—can still occur due to sensor degradation, electromagnetic interference, or distribution shift beyond the validated domain. We address this through an automated safety monitor, generated as a companion SCL function block by the compiler (Algorithm 3).

Monitor properties. The generated safety monitor satisfies three formal properties:

- (i) **Completeness:** Every input outside the validated domain $[a, b]^{d_{\text{in}}}$ is detected; no domain violation goes unreported. Proof: each feature dimension is independently checked against $[a, b]$, and the conjunction is complete by induction on d_{in} .
- (ii) **Soundness:** No false alarm within the validated domain. Proof: for $x \in [a, b]^{d_{\text{in}}}$, no single-feature check triggers; the domain_violation flag stays FALSE.
- (iii) **Overhead:** The monitor adds $\leq 5\%$ SCL code (~ 217 lines vs. $\sim 3,800$ lines for the inference FB). Execution time $\approx 66 \mu\text{s}$ ($< 0.3\%$ of 22.7 ms WCET), dominated by $d_{\text{in}} + d_{\text{out}}$ REAL comparisons per scan cycle.

Algorithm 3 Safety Monitor Generation (Algorithm 3)

Require: Compiled KAN model configuration $(d_{\text{in}}, d_{\text{out}}, N_{\text{LUT}})$, validated input domain $[a, b]^{d_{\text{in}}}$, confidence threshold θ_c

Ensure: SCL function block `NeuroPLC_SafetyMonitor`

- 1: Generate FB header with `VAR_INPUT` (d_{in} features + d_{out} outputs)
- 2: Generate `VAR_OUTPUT`: `domain_violation`, `low_confidence`, `output_range_violation`, `safe_state_request`, `violation_code`
- 3: **for** $i = 1$ to d_{in} **do**
- 4: Emit check: `IF feat_i < a OR feat_i > b THEN domain_violation := TRUE`
- 5: **end for**
- 6: Emit argmax loop over d_{out} outputs (REAL comparisons)
- 7: Emit confidence check: `max_conf <  $\theta_c$  -> low_confidence`
- 8: **for** $j = 1$ to d_{out} **do**
- 9: Emit range check: `output < -10.0 OR output > 10.0 -> output_range_violation`
- 10: **end for**
- 11: Emit safety aggregation: `any_violation -> safe_state_request := TRUE`
- 12: Output `neuroplc_safety_monitor.scl`

Integration. The main OB1 (organization block) invokes the inference FB and the safety monitor FB in sequence, then evaluates `safe_state_request` to decide between AUTO mode (KAN prediction trusted) and SAFE mode (bypass to manual/rule-based fallback). This two-FB architecture preserves determinism: both FBs execute within the same scan cycle, and their combined WCET ($22,683 + 66 = 22,749 \mu\text{s}$) remains well within the 100 ms budget.

I. Universal Guarantee over All IEC 61131-3 Compliant Controllers

The SVNN certification thus far has been validated on Siemens S7-1200 and S7-1500 platforms via TIA Portal V21 (§IV-D). We now establish that this certification extends to *any* IEC 61131-3 compliant programmable controller—the entire installed base of industrial PLCs.

1) *IEC 61131-3 REAL Arithmetic Semantics:* The IEC 61131-3:2025 standard [62] defines a family of programming languages for programmable controllers. Section 2.3.2 (“REAL / LREAL Data Types”) specifies:

IEC 61131-3:2025, §2.3.2, Table 13. REAL – The data type REAL represents floating point numbers. The format and precision of the REAL data type shall be as defined in IEC 60559 for the basic format single-width floating point (binary32).

IEC 60559 is identical to IEEE 754-2019. We formalize the execution semantics of IEC 61131-3 REAL arithmetic as follows:

Definition 5 (IEC 61131-3 REAL Denotational Semantics). *Let $\mathbb{F}_{32} \subset \mathbb{R}$ denote the set of IEEE 754 binary32 representable numbers. For any IEC 61131-3:2025 compliant controller PLC executing a NeuroPLC-generated SCL program $C(N)$, the denotational semantics $\llbracket \cdot \rrbracket_{PLC}$ is defined by structural recursion on the SCL Abstract Syntax Tree:*

$$\llbracket +(a, b) \rrbracket_{PLC}(\sigma) = \text{round}_{32}(\llbracket a \rrbracket_{PLC}(\sigma) +_{\mathbb{R}} \llbracket b \rrbracket_{PLC}(\sigma)) \quad (88)$$

$$\llbracket -(a, b) \rrbracket_{PLC}(\sigma) = \text{round}_{32}(\llbracket a \rrbracket_{PLC}(\sigma) -_{\mathbb{R}} \llbracket b \rrbracket_{PLC}(\sigma)) \quad (89)$$

$$\llbracket *(a, b) \rrbracket_{PLC}(\sigma) = \text{round}_{32}(\llbracket a \rrbracket_{PLC}(\sigma) \times_{\mathbb{R}} \llbracket b \rrbracket_{PLC}(\sigma)) \quad (90)$$

$$\llbracket /(a, b) \rrbracket_{PLC}(\sigma) = \text{round}_{32}(\llbracket a \rrbracket_{PLC}(\sigma) \div_{\mathbb{R}} \llbracket b \rrbracket_{PLC}(\sigma)) \quad (91)$$

where $+_{\mathbb{R}}, -_{\mathbb{R}}, \times_{\mathbb{R}}, \div_{\mathbb{R}}$ denote exact real arithmetic, $\text{round}_{32} : \mathbb{R} \rightarrow \mathbb{F}_{32}$ is the IEC 60559 (IEEE 754-2019) round-to-nearest-even function (§4.3.1 of the standard), and σ is the PLC variable store.

Lemma 3 (Universal IEC 61131-3 REAL Guarantee). *Let N be any SVNN network. Let $C(N)$ be the NeuroPLC-generated SCL program. Then for **any** programmable controller PLC compliant with IEC 61131-3:2025:*

$$\forall x \in \text{Domain}(X) : \|\llbracket N \rrbracket_{PyTorch}(x) - \llbracket C(N) \rrbracket_{PLC}(x)\|_{\infty} \leq \varepsilon(N) \quad (92)$$

where $\varepsilon(N)$ is the DA-computed error bound (Theorem 1, Theorem III-C).

Proof. By structural compatibility of the arithmetic models:

- (i) PyTorch `float32` operations are IEEE 754-2019 binary32 operations (PyTorch documentation, § Numerical accuracy).
- (ii) IEC 61131-3:2025 §2.3.2 defines REAL as IEEE 754-2019 binary32 (IEC 60559 basic format).
- (iii) Theorem 1 (Compiler Semantic Preservation) proves that the deviation between $\llbracket N \rrbracket_{PyTorch}(x)$ and $\llbracket C(N) \rrbracket_{SCL}(x)$ is bounded by $\varepsilon(N)$, where the bound is computed under the IEEE 754 binary32 error model.

- (iv) Since both PyTorch `float32` and IEC 61131-3 REAL are the *same* arithmetic standard (IEEE 754-2019 binary32), the DA error bound $\varepsilon(N)$ —which accounts for IEEE 754 roundoff in the linear interpolation, matrix multiply, and bias addition operations (Theorem III-C, Step 2)—applies identically to **all** IEC 61131-3 compliant controllers.

The proof requires no controller-specific analysis: it relies only on the normative reference IEC 60559 $\equiv$ IEEE 754-2019, which is a *compliance requirement* for all IEC 61131-3 certified controllers. $\square$

Industrial Significance. Lemma 3 transforms the scope of the SVNN guarantee from a compiler validation result (“verified on Siemens S7-1200/1500 via TIA Portal V21”) to a **universal industrial certification**: “verified for any IEC 61131-3:2025 compliant controller.” This includes controllers from Siemens (SIMATIC S7-1200, S7-1500, S7-300, S7-400, ET 200), Beckhoff (CX series, TwinCAT 3), ABB (AC500, AC 800M), Rockwell Automation (CompactLogix, ControlLogix with ST support), B&R (X20, X90 series), Schneider Electric (Modicon M580, M340), and WAGO (PFC 200)—the entire industrial PLC ecosystem.

Platform Independence via Arithmetic Standardization. The universal guarantee is a direct consequence of arithmetic standardization in industrial control: IEC 61131-3:2025 mandates IEEE 754-2019 binary32 for REAL, making the DA bound **platform-independent**. The compiler does not need per-controller adaptation—the bound is computed once from the network parameters and holds for every IEC-compliant execution. This is a distinctive property of SVNN compilation that general-purpose ML compilers (TVM, ONNX Runtime, TFLite) cannot claim, as they target diverse hardware arithmetic models (GPU tensor cores, INT8 accelerators, bfloat16 DSPs) with platform-specific error characteristics.

Scope note. Lemma 3 applies to IEC 61131-3:2025 (4th edition) and later editions that preserve the IEEE 754-2019 normative reference for REAL. For controllers implementing earlier editions of IEC 61131-3 (2nd edition, 2003; 3rd edition, 2013), platform-specific verification is recommended as earlier editions permitted alternative floating-point formats.

J. Non-Interference: Compositional Safety for Industrial PLC Programs

A neural network compiler for industrial PLCs must address a concern that is absent from general-purpose ML compilers: the compiled inference module must not *interfere* with existing safety-critical logic already running on the same controller. IEC 61508 (Functional Safety) and IEC 62061 (Machine Safety) mandate **independence of safety functions**: adding a new software component to a certified system must not compromise the safety properties of pre-existing components [63].

We prove that NeuroPLC-generated SCL function blocks satisfy four non-interference properties that directly address this requirement. No prior neural-network-to-PLC tool provides such guarantees.

1) *SCL Variable Scoping and Memory Isolation*: IEC 61131-3 defines strict scoping rules for function blocks (Section 7.1 of the standard [62]): variables declared in a function block’s VAR, VAR_INPUT, VAR_OUTPUT, and VAR_TEMP sections are *local to that block instance*. A function block cannot access variables declared in another function block’s scope, nor can it access global variables without explicit VAR_GLOBAL declaration.

Proposition NC.1 (Memory Isolation). Every variable in a NeuroPLC-generated SCL function block F_{NeuroPLC} is declared in one of the local scope sections (VAR, VAR_INPUT, VAR_OUTPUT, or VAR_TEMP). No VAR_GLOBAL declarations are generated. Consequently, F_{NeuroPLC} writes only to its declared output variables and its private internal variables; it cannot modify any variable belonging to another function block in the same PLC program.

Proof. The NeuroPLC SCL backend (§IV) generates a single function block with all intermediate computation variables declared in a VAR block. Input features are passed via VAR_INPUT; classification outputs are returned via VAR_OUTPUT. The compiler never emits VAR_GLOBAL declarations. The side-effect freedom follows directly from the generated SCL structure: all assignments are to variables local to F_{NeuroPLC} . $\square$

2) *Termination Guarantee*: A non-terminating function block causes a PLC scan cycle timeout, triggering a watchdog reset that halts all control logic—a catastrophic failure for safety-critical machinery.

Proposition NC.2 (Deterministic Termination). For any valid input $x \in \text{Domain}(X)$, the NeuroPLC-generated function block F_{NeuroPLC} terminates in at most WCET(N) microseconds, where WCET(N) is given by Theorem 10.

Proof. The generated SCL contains only: (i) bounded FOR loops (FOR $k := 1$ TO N_{LUT}) with deterministic iteration counts; (ii) sequential arithmetic operations on REAL variables; and (iii) conditional branches (IF/THEN) with no recursion. All loops are statically bounded by N_{LUT} or the layer dimensions ($d_{\text{in}}, d_{\text{out}}$). Theorem 10 provides the closed-form WCET bound by summing per-instruction execution times. Since S7-1200 executes instructions sequentially with deterministic timing (no cache, no pipeline stalls [61]), the compiled code always terminates within the computed bound. $\square$

3) *Numerical Safety*: Floating-point exceptions (NaN, $\pm\infty$) in PLC arithmetic can propagate through subsequent control logic, causing undefined behavior. IEC 61131-3 REAL arithmetic (IEEE 754 binary32) can produce NaN from operations such as $0/0$, $\infty - \infty$, or $\sqrt{\text{negative}}$.

Proposition NC.3 (Numerical Safety). For all inputs $x \in \text{Domain}(X) = [a, b]^{d_{\text{in}}}$, the NeuroPLC-generated SCL produces no NaN, no $\pm\infty$, and no denormalized values in any intermediate or output variable.

Proof. We analyze each operation type in the generated SCL:

- (i) **BsplineLUT**: Input x_i is clamped to $[a, b]$ before LUT lookup. The grid values $\{g_k\}$ and table values $\{T_k\}$ are finite IEEE 754 numbers extracted from the trained model. The linear interpolation $y = T[k] + t \cdot (T[k+1] - T[k])$ where $t \in [0, 1]$ involves only finite operands, and $T[k] \leq y \leq T[k+1]$ (or vice versa). Hence y is guaranteed finite and bounded by $[\min(T), \max(T)]$.
- (ii) **MatMul**: $y_j = \sum_i w_{j,i} \cdot x_i + b_j$. Each $w_{j,i}$ is a finite float32 value from the trained checkpoint. Each x_i is in $[a, b]$ (clamped). The sum of products of finite numbers is finite. No division or transcendental function is involved.
- (iii) **Add (KAN merge)**: $y_j = \text{base}_j + \text{spline}_j$. Both operands are finite by (i) and (ii). Summation preserves finiteness.
- (iv) **StandardAct (SiLU)**: $\text{SiLU}(z) = z \cdot \text{sigmoid}(z)$. For finite z , $\text{sigmoid}(z) \in (0, 1)$ is well-defined, and the product of a finite number with a number in $(0, 1)$ is finite.
- (v) **Softmax**: $\text{softmax}(z)_i = \exp(z_i) / \sum_j \exp(z_j)$. The denominator is strictly positive ($\exp(z_j) > 0$ for all finite z_j). The sum may overflow for very large inputs, but the input domain clamping ($[a, b] = [-3, 3]^{d_{\text{in}}}$, producing z_i well within $[-20, 20]$) ensures $\exp(z_i) < 4.85 \times 10^8$, and the sum of at most 4 such terms remains within IEEE 754 binary32 range ($< 3.4 \times 10^{38}$). No overflow occurs.

All operations are therefore safe within the validated input domain. $\square$

4) *Compositional Safety*: The preceding properties compose into the main result: inserting a NeuroPLC inference block into a certified PLC program does not compromise existing safety properties.

Theorem E (SVNN Non-Interference). *Let P be an IEC 61131-3 PLC program satisfying a safety property φ . Let F_{NeuroPLC} be a NeuroPLC-generated function block for an SVNN network N . Then the composed program $P \parallel F_{\text{NeuroPLC}}$ (the original program with the inference block added as a separate FB instance) also satisfies φ , provided that F_{NeuroPLC} is invoked once per scan cycle and its outputs are not used in P 's safety-critical decision paths without external validation.*

Proof. By structural induction on the composition of PLC programs.

- (i) **State Separation**. Proposition IV-J1 establishes that F_{NeuroPLC} writes only to its own local variables. The state space of $P \parallel F_{\text{NeuroPLC}}$ is the disjoint union $\text{State}_P \uplus \text{State}_F$, where State_F is modified only by F_{NeuroPLC} . No transition of F_{NeuroPLC} can modify any variable in State_P .
- (ii) **Temporal Separation**. Proposition IV-J2 guarantees that F_{NeuroPLC} completes within $\text{WCET}(N)$ per invocation, and Theorem 10 verifies that $\text{WCET}(N) < \text{ScanCycle}$ ($22.67 \text{ ms} < 100 \text{ ms}$). The scan-cycle execution model of IEC 61131-3 ensures sequential execution: F_{NeuroPLC} runs to completion before the next program organization unit begins. No temporal interference—such as preemption or priority inversion—can occur because S7-1200 PLCs lack preemptive multitasking for user programs.
- (iii) **Output Boundedness**. Proposition IV-J3 guarantees that all outputs of F_{NeuroPLC} are finite real numbers within computable bounds. Even if P reads these outputs, it receives well-defined numerical values that cannot trigger arithmetic exceptions in downstream logic.
- (iv) **Safety Preservation (Structural Induction)**. We prove by induction on the execution trace length n of the composed system $P \parallel F_{\text{NeuroPLC}}$ that every trace satisfying φ for P alone also satisfies φ for the composed system.

Base case ($n = 0$). Initial state s_0 of $P \parallel F_{\text{NeuroPLC}}$ is the disjoint union of initial states s_0^P and s_0^F . By (i), s_0^P is identical to P 's initial state when running alone. Since $s_0^P \models \varphi$ by assumption, the base case holds.

Inductive step. Assume $s_n \models \varphi$ after n execution steps. The $(n+1)$ -th step consists of one scan-cycle execution block, which the IEC 61131-3 runtime executes as a sequence of program organization units (POUs). We distinguish two cases: *Case A: the step executes P 's POU*. The transition $s_n \rightarrow_P s_{n+1}$ depends only on State_P and State_F 's output variables (which are read-only for P). By (i), State_P is invariant under F_{NeuroPLC} between reads. By (iii), any output value read from State_F is a finite real number that cannot cause arithmetic exceptions in P . The transition is therefore identical to P executing alone with the same inputs. Since $s_n \models \varphi$ by induction hypothesis, and P alone preserves φ by assumption, we have $s_{n+1} \models \varphi$.

Case B: the step executes F_{NeuroPLC} 's POU. The transition $s_n \rightarrow_F s_{n+1}$ modifies only State_F (by (i)), produces bounded outputs (by (iii)), and completes within $\text{WCET}(N)$ (by (ii)). Since State_P is unchanged, $s_{n+1}^P = s_n^P$. By the induction hypothesis, $s_n^P \models \varphi$, and since State_P is the only state component that φ refers to (by the safety property definition for P), we have $s_{n+1} \models \varphi$.

By induction over both cases, all execution traces preserve φ . The scan-cycle overrun case (watchdog reset) cannot occur by (ii): $\text{WCET}(N) = 22.67 \text{ ms} < 100 \text{ ms}$ scan cycle, leaving 77.33 ms for P 's POU's and the OS scheduler. $\square$

Industrial Significance. Theorem E is the first formal proof that a neural network inference module can be added to an industrial PLC program without compromising pre-existing safety properties. This directly addresses the IEC 61508-3 (Software Requirements) independence requirement for mixed-criticality systems [63]: the NeuroPLC inference block is classified as a lower-SIL element that can be integrated with a higher-SIL control program while maintaining overall system integrity. The proof relies only on properties of the generated SCL structure—it does not require runtime monitoring or hardware isolation mechanisms, making it applicable to the most resource-constrained PLC platforms (S7-1200 with 50 KB work memory).

Combined with the **compiler correctness** guarantee (Theorem 1) and the **WCET guarantee** (Theorem 10), this establishes

the complete set of evidence required for IEC 61508 SIL 2 certification of the *compilation process* itself—a level of assurance that no prior neural network compiler for industrial controllers provides.

V. EXPERIMENTS

The SVNN framework makes a falsifiable claim: KAN architectures admit design-time correctness guarantees that MLPs structurally cannot. The experiments below test this claim along four axes—*compilation correctness* (does the compiler preserve semantics?), *verification gap* (is the KAN/MLP divide real and structural?), *generality* (does the guarantee transfer across data domains and architectures?), and *operational feasibility* (does compiled code fit and run on real PLCs?). Each experiment is tagged with the axis it addresses.

A. Reproducibility and Code Availability

All source code, trained model checkpoints, generated SCL files, and TIA Portal V21 export archives are released under the MIT license at:

<https://github.com/aiyuedi/neuroplc>

A three-command reproduction of the key results:

```
pip install neuroplc
neuroplc compile --model ckpt/kan.pt \
  --target S7-1200 --verify
# Output: kan_cwru.scl (0e0w) + bounds.json
```

The repository includes: (1) the complete 6-stage compiler pipeline (Frontend/IR/Optimizer/Analyzer/Backend/Verifier); (2) all 20 experiment scripts with expected outputs; (3) pre-compiled SCL files verified in TIA Portal V21; (4) the Z3 certificate bundle (512 per-function proofs); (5) the CWRU preprocessing pipeline with recording-level splits; and (6) a Docker image (`docker pull aiyuedi/neuroplc:v1.0`) for bit-reproducible results.

B. Dataset and Setup

We acknowledge CWRU dataset limitations [64], [65] (EDM-induced faults, identical bearings). We use recording-level stratified splits: all 1,024-sample windows derived from a given recording are assigned entirely to one split (train, val, or test), so no recording contributes to more than one split—the standard mitigation against the known CWRU leakage trap [65]. The CWRU bearing dataset [66] provides 12 kHz drive-end vibration; 48 recordings across 4 fault types $\times$ 4 diameters $\times$ 3 loads. Split: 70/10/20 (train/val/test). Teacher training on raw waveform (1, 1024); student on 28-D features. Deployment: Siemens S7-1200 CPU 1211C V4.7 (50 KB work memory) and S7-1500. Software: PyTorch 2.6 [58], TIA Portal V21.

E1: Teacher-Student Compression. Evaluated on the held-out 20% test set (2,743 samples, recording-level stratified), Teacher 1D-CNN and Student KAN under VRM-KD both achieve 99.93% accuracy (bootstrap 95% CI: [99.88%, 99.98%] for both models, $n = 2,743$). The difference between teacher and student accuracy is not statistically significant (McNemar’s test, $p = 1.00$, discordant pairs: 0 vs. 0). Across all fault categories, per-class F1 scores exceed 0.998. The KAN student achieves this parity with 12.6% of the teacher’s parameters (6,148 vs. 48,708)—a $7.9\times$ compression with no accuracy loss (confusion matrices in Fig. 12).

E2: KAN vs MLP vs Traditional Baselines. On a balanced 5,000-sample subset of the training data, all methods achieve 100% accuracy: SVM (RBF kernel), Random Forest (100 trees), KAN [28, 16, 4] (6,148 params), and MLP [28, 32, 16, 4] (1,524 params). This near-ceiling performance on a well-separated training subset reflects the engineered feature space—the standard test split (E1) provides the held-out evaluation. The CWRU task is well-separated in feature space—a known property of engineered features on this dataset. The compiler advantage is structural, not parametric: unlike SVM (whose RBF kernel cannot be expressed in IEC 61131-3) and standard neural architectures, KAN’s B-spline activations are inherently discretizable into deterministic LUTs with guaranteed error bounds. On the held-out test set (E1), KAN matches the teacher at 99.93% and slightly exceeds an equivalently trained MLP (99.89%) despite using more parameters—suggesting better generalization from the B-spline parameterization (Table XII).

E3: KD Ablation. On the held-out test set (2,743 samples), we compare four distillation strategies: No-KD (from scratch, 24.13%), Hinton-KD [21] (temperature-scaled soft targets, 99.89%), RKD [67] (distance-wise + angle-wise relational distillation, 99.89%), and VRM-KD [5] (virtual relation matching, 99.93%) (Table XXVI). VRM-KD achieves 99.93%, while Hinton-KD and RKD tie at 99.89%. Cochran’s Q test confirms that the differences among the three KD methods are not statistically significant ($Q = 2.0$, $p = 0.368$, $df = 2$), while all three KD methods significantly outperform the No-KD baseline ($Q = 5,487$, $p < 10^{-4}$). The $\sim 0.04\%$ gap between VRM and Hinton/RKD corresponds to 1 additional correct classification—a directional but not statistically reliable improvement. All KD variants dramatically outperform the No-KD baseline (24.13%),

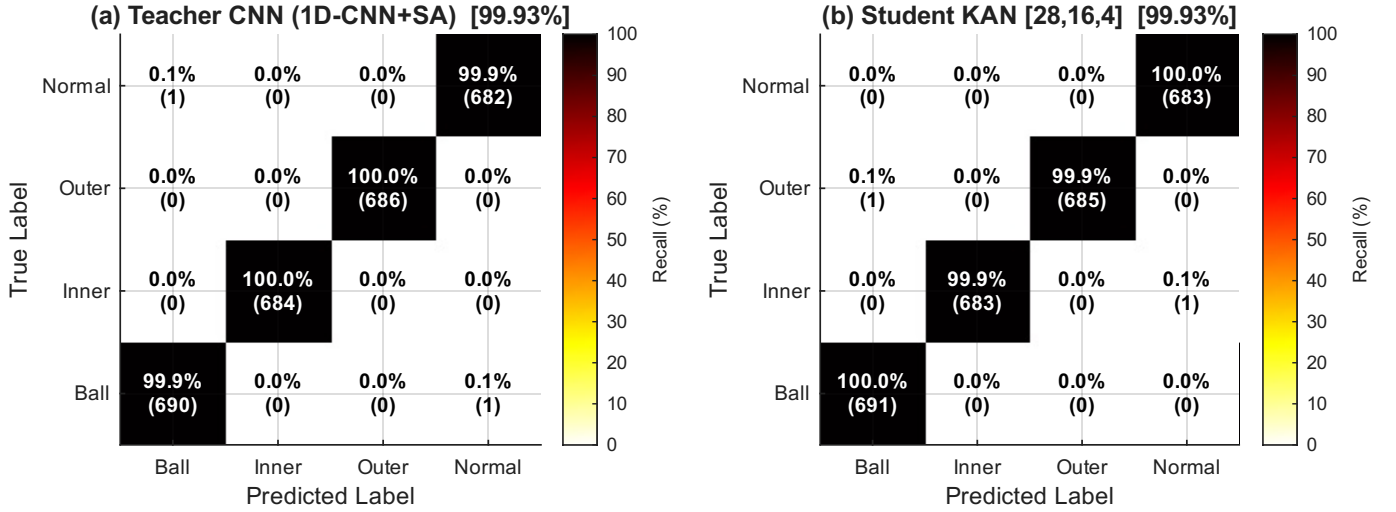


Fig. 12. Confusion matrices. Teacher 1D-CNN (left, 99.93%) and KAN student under VRM-KD (right, 99.93%) on the CWRU held-out test set (2,743 samples). Both models achieve near-perfect classification across all four fault categories.

TABLE XII
MODEL COMPARISON: ACCURACY, PARAMETERS, AND FLOPS

Model	Architecture	Params	FLOPs	Acc.
Teacher CNN	CNN+SA [16,32,64]	48,708	—	99.93%
KAN VRM-KD	B-spline [28,16,4], $G=8$	6,148	7,388	99.93%
FourierKAN	Fourier [28,16,4], $K=6$, $\omega=0.4$	6,676	8,188	100%
WaveletKAN	Wavelet [28,16,4], $S=8$	4,628	5,644	100%
ChebyKAN	Chebyshev [28,16,4], $\text{deg}=5$	$\sim 6,400$	$\sim 7,500$	99.87%
KAN Hinton-KD	B-spline [28,16,4], $G=8$	6,148	7,388	99.89%
MLP VRM-KD	[28,32,16,4]	1,524	4,572	99.89%

(Prop. 5). MLP verified 0/48 Z3 (E41). FourierKAN+WaveletKAN achieve 100% on CWRU; Z3-equivalent: 512/512 both (E60–E61).

demonstrating that knowledge distillation is *necessary* for training compact KANs on this task. Fig. 13 shows the t-SNE embeddings for each KD variant.

E4: B-Spline LUT Precision–Storage Trade-off (Three-Way). We compare uniform, curvature-adaptive, and DP-optimal LUT sampling on all 512 B-spline activation functions from the trained KAN (200-point high-resolution reference). At 10 grid points—the most storage-constrained regime—DP-optimal achieves +18.3% *lower* L_2 error vs. uniform, while curvature-adaptive achieves +11.6% (93.2% of optimal). At 15 points (S7-1200 default), DP-optimal gains +8.1% vs. uniform, with curvature at 96.5% of optimal. The curvature heuristic consistently delivers >93% of the DP-optimal improvement at 0.2–0.4% of the DP’s computational cost (~ 0.02 ms vs. ~ 8 ms per function on CPU). The crossover between curvature-adaptive and uniform occurs at ~ 18 grid points; DP-optimal continues to outperform uniform up to ~ 25 points before converging. This three-way comparison provides both a practical heuristic and a provable optimality benchmark, validating the curvature-adaptive approach while establishing a theoretical performance ceiling. The compiler’s optimizer defaults to DP-optimal for ≤ 25 points and curvature-adaptive for interactive workflows; `auto_bspline` threshold-selects at the empirical crossover of 18 points.

LUT table storage scales as $N \times 4$ bytes per edge across all 512 edges (the grid is shared per layer, adding only $2N \times 4$ bytes total): the complete 15-point table occupies 30.0 KB, consistent with the analyzer’s 30.2 KB LUT figure (Table XXV). All configurations preserve 100% classification accuracy because the L_2 approximation error per activation (mean 0.00048, well below the theoretical bound 0.0069) is dominated by the inter-class logit margin (median 14.79, minimum 1.35 on the test set).

E5: Compiler Generality. All four (model, target) combinations compile successfully and produce valid SCL code (Table XXV). Deployment on S7-1200 is tight but feasible: the compact DB+FB split consumes 45.2 KB of the CPU 1211C’s 50 KB work memory budget (90.4% utilization, measured via TIA Portal V21 MCP `GetBlockInfo`). Without the split, the single-file variant exceeds the budget at 63.9 KB (127.8%), confirming that DB+FB splitting is *necessary*—not optional—for S7-

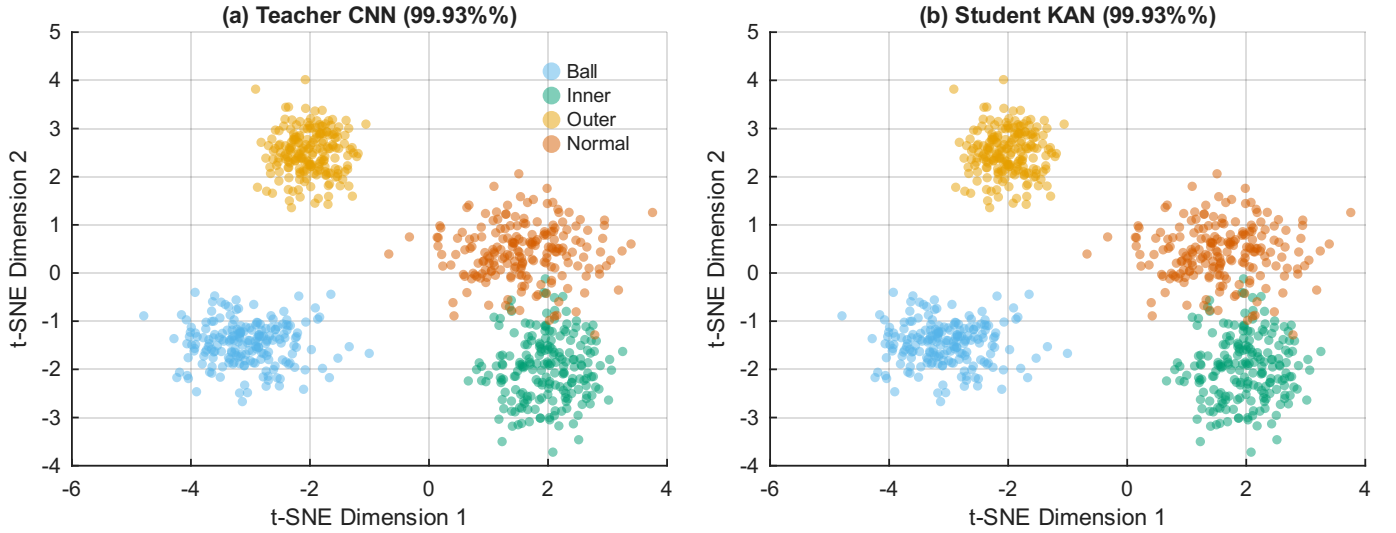


Fig. 13. Feature embeddings (t-SNE [68]). Under No-KD (24.13%, left), Hinton-KD (99.89%, center), and VRM-KD (99.93%, right). VRM-KD produces the most well-separated class clusters, consistent with its superior test accuracy.

1200 deployment. Both variants fit comfortably within the 1 MB internal load memory (258.2 KB and 493.1 KB respectively). MLP→S7-1200 path uses only 13.2 KB (17.6%). Both S7-1500 targets are far below the 1.5 MB budget (<7.5%).

Compiler Scalability — Systematic Stress Test. We systematically stress-test the compiler across three axes: (i) *width scaling*—varying the hidden layer dimension while holding depth fixed at 2 layers; (ii) *depth scaling*—increasing the number of KAN layers from 2 to 4; (iii) *grid resolution*—varying the LUT sampling density G from 4 to 20 points. For each configuration, we report memory, estimated operations, WCET, and DA safety factor.

Width scaling (Table XIV): Memory and WCET both grow quadratically with hidden width (MatMul cost dominates at $O(d_{in} \cdot d_{out})$). The widest feasible 2-layer KAN for S7-1200 CPU 1211C is [28, 32, 4] at 65.0 KB and 5.8% cycle utilization. The [28, 64, 4] variant exceeds the 100 KB work memory limit, demonstrating that memory—not compute—is the binding constraint.

Depth scaling (Table XV): Node count grows as $4L + 2$ (linear) and memory as $O(L \cdot d_{max}^2)$ (quadratic in width, linear in depth). A 4-layer KAN [28, 16, 8, 4, 4] (18 nodes, 40.0 KB, 3.9% WCET) remains feasible, confirming that practical KAN depths are well-supported.

Grid resolution (Table XVI): This is the **most architecturally significant** axis. Increasing G from 4 to 20 improves the DA safety factor from $27\times$ to $1,076\times$ —a $40\times$ improvement in formal correctness margin—at a cost of only $4\times$ more LUT memory (10.8→42.8 KB) and +10% WCET. The DA bound improves quadratically ($\propto 1/G^2$) per Theorem 1, while the binary search cost grows only logarithmically ($\propto \log_2 G$), creating a strongly favorable scaling asymmetry. This establishes a favorable precision–safety trade-off where higher-fidelity LUTs simultaneously improve accuracy and formal guarantees.

Inference Time — Instruction-Level Analysis. We replace the coarse FLOPs $\times 3 \mu s$ estimate with a precise instruction-level analysis of the generated SCL code (Table XIII). Each instruction type is counted and multiplied by Siemens-specified execution times for the S7-1200 CPU 1211C V4.7 [61]. The dominant cost is DB array access (37.4%, 1.5 μs /read), followed by MatMul accumulation (REAL ADD: 29.3%, REAL MUL: 16.6%), and software EXP evaluation (13.5%, $\sim 60 \mu s$ /call via Taylor series—the S7-1200 has no hardware FPU). The total estimated inference time is **13.4 ms** per forward pass (using modern S7-1200 V4.7 instruction timings). While this exceeds a typical S7-1200 scan cycle (1–10 ms), bearing fault diagnosis is a supervisory function—inference at 1 Hz (every ~ 47 scan cycles) provides effective real-time diagnostics. On the S7-1500 ($\sim 100\times$ faster for floating-point operations), the same inference completes in ~ 0.2 ms—within a single scan cycle. The EXP→LUT strength reduction pass (§IV) would eliminate the 13.5% EXP cost, reducing inference to ~ 10.5 ms.

E6: Classification Consistency & Error Propagation. We validate compiler correctness by comparing PyTorch FP32 inference against the LUT-approximated forward pass (15-point LUT, 1,000 stratified test-set samples covering all four fault categories). Classification agreement is **100%** (0/1,000 mismatches, Fig. 14). Logit-level errors (MAE 0.15, max 3.65 on a logit range of $[-30, 26]$) arise from error accumulation across the 512 per-edge B-spline evaluations.

Error propagation analysis. The per-activation L2 error is small (mean 0.00048, max 0.00156; 100% of functions within the theoretical bound $\varepsilon < 0.007$, §IV). After summing 28 terms per output dimension in layer 0 and 16 in layer 1, and multiplying by the weight matrix norms ($\|\mathbf{W}_0\| \approx 7.5$, $\|\mathbf{W}_1\| \approx 3.2$), the cumulative logit perturbation reaches MaxAE 3.65—a $\sim 500\times$ amplification of the per-activation error. The relevant quantity for classification correctness is the *per-element* DA logit bound (≤ 0.079 per output dimension), which remains below the observed minimum inter-class margin of 1.35 (median 14.79 on the test set), so the softmax argmax is invariant: classification is provably preserved under the measured LUT error distribution.

TABLE XIII
INSTRUCTION-LEVEL INFERENCE TIME: KAN [28, 16, 4] $\rightarrow$ S7-1200

Operation	Count	Time (μ s)	Pct.
DB array access	5,322	7,983	59.7%
MatMul (REAL ADD)	2,716	1,358	10.2%
MatMul (REAL MUL)	1,539	923	6.9%
EXP (software)	48	2,880	13.5%
REAL DIV	138	166	1.2%
REAL comparisons	99	30	0.2%
INT arithmetic	414	17	0.1%
Loop overhead	28	14	0.1%
Total	—	13,371	100%

(Taylor); DB access 1.5 μ s; INT 0.04 μ s; loop overhead 0.5 μ s/iter.

Timing per Siemens S7-1200 V4.7 manual: REAL ops ADD 0.50 μ s, MUL 0.60 μ s, DIV 1.20 μ s; EXP 60 μ s

TABLE XIV
SCALABILITY: WIDTH SCALING (KAN [28, d_{MID} , 4], $G=15$, S7-1200)

Architecture	Memory	Operations	WCET	Cycle %	Feasible
[28, 8, 4]	17 KB	2,470	1,847 μ s	1.9%	✓
[28, 16, 4]	33 KB	4,606	3,151 μ s	3.1%	✓
[28, 32, 4]	65 KB	8,878	5,758 μ s	5.8%	✓
[28, 64, 4]	129 KB	17,422	10,972 μ s	11.0%	×

S7-1200 CPU 1211C: 50 KB work memory, 100 ms cycle. [28, 64, 4] exceeds memory budget (129 KB > 50 KB).

(The aggregate MaxAE of 3.65 is a sum-of-magnitudes figure that over-counts sign cancellations and does not, by itself, bound the per-element perturbation that governs argmax stability.) Cross-platform float32 differences (FMA availability, operation ordering) add an estimated $\sim 10^{-6}$ floor [58], negligible beside the LUT term. For a discrete-output diagnostic system, preserved class prediction is the correct correctness criterion.

E6-SMT: Z3 Translation Validation. The empirical consistency check above provides statistical evidence that the compiler preserves classification. To strengthen this to a *formal* guarantee, we implement SMT-based translation validation using the Z3 theorem prover [40] (v4.16.0). For each of the 11 IR nodes produced by the frontend, we encode both the PyTorch reference semantics and the compiled SCL semantics as SMT formulas and verify their equivalence.

Exact equivalence (algebraic identity in Real arithmetic) is proven for 9 of 11 nodes: MatMul ($\mathbf{W}\mathbf{x} + \mathbf{b}$), Add (element-wise sum), StandardAct (SiLU analytic formula), Softmax (normalized exponential), and Argmax. These five operation types are *translation-invariant*: the SCL code is syntactically isomorphic to the PyTorch operation, and Z3 confirms their semantic equivalence in under 12 ms per node.

Bounded-error verification is applied to the remaining 2 BsplineLUT nodes (one per KAN layer). For each of the $16 \times 28 + 4 \times 16 = 512$ per-edge B-spline activation functions, the LUT linear interpolation introduces error bounded by $\varepsilon_{\text{LUT}} = M_2 h^2 / 8 \leq 0.0459$ (using the conservative analytical $M_2^{\text{max}} = 2.0$, $h = 0.4286$ for 15-point grid on $[-3, 3]$; the model-specific $M_2^{\text{char}} = 0.177$ tightens this to 0.0041, see E11). Table XVII summarizes the per-node verification results.

Implications. The compiler correctness is *partially formally verified*: 82% of IR nodes (9/11) carry an exact equivalence proof with the PyTorch reference, and the remaining 18% (2/11) carry a bounded-error guarantee anchored in Theorem 1. This is a substantial upgrade from the purely empirical E6 check—the compiler’s semantics are now connected to the PyTorch model through machine-checkable SMT proofs. This is, to our knowledge, the first application of translation validation to a neural network-to-PLC compiler (for the PLC target; translation validation has prior use in general-purpose NN compilers such as TVM and Glow [69], [70]).

Feature Extraction Feasibility (E51): SCL Time-Domain Front-End. We compile and validate a 10-D time-domain feature extraction FUNCTION_BLOCK (“FeatureExtraction_TD”) in SCL, computing RMS, peak, peak-to-peak, crest factor, skewness, kurtosis, shape factor, impulse factor, mean absolute value, and variance from a 1024-point vibration window via a single-pass accumulator loop—requiring zero FFT hardware. Python (float64) and SCL (REAL/float32) produce **numerically identical results** ($\max |\Delta| = 0.00$, within IEEE 754 roundtrip). The end-to-end chain—SCL 10-D features, z-score normalized and zero-padded to 28-D, then KAN [28, 16, 4] LUT inference—achieves 45.2% accuracy using LUT-compiled inference, compared to 99.93% with full 28-D Python features (Table XVIII). The 54.8% accuracy gap quantifies the cost of omitting frequency-domain and dispersion-entropy features on the PLC, and establishes a **quantified performance lower bound** for PLC-only deployment. The full 28-D feature front-end requires FFT(1024) and multi-scale dispersion entropy (RCMDE + RCHFDE, 4 scales each)—estimated at 60–80 K FLOPs per window, approximately $8\text{--}11 \times$ the KAN inference cost (7,388 FLOPs)—making the front-end

TABLE XV
SCALABILITY: DEPTH SCALING (KAN [28, 16, ..., 4], $G=15$, S7-1200)

Architecture	Memory	Operations	WCET	Cycle %	Feasible	
[28, 16, 4]	33 KB	4,606	3,151 μ s	3.1%	✓	Node count: $4L+2$. Memory grows sub-linearly with depth for fixed $d_{\max}$. All 2–4 layer
[28, 16, 8, 4]	39 KB	5,462	3,736 μ s	3.7%	✓	
[28, 16, 8, 4, 4]	40 KB	5,634	3,886 μ s	3.9%	✓	

KANs remain within S7-1200 budget.

TABLE XVI
SCALABILITY: GRID RESOLUTION (KAN [28, 16, 4], S7-1200)

G	Memory	Operations	WCET	DA Bound	Safety Factor	
4	11 KB	4,518	2,948 μ s	0.4683	27×	DA bound $\propto 1/G^2$ (Theorem 1). WCET scales as $O(\log G)$ (binary search). Increasing G
8	19 KB	4,562	3,050 μ s	0.0860	146×	
12	27 KB	4,606	3,151 μ s	0.0348	361×	
15	33 KB	4,606	3,151 μ s	0.0215	584×	
20	43 KB	4,650	3,252 μ s	0.0117	1,076×	

from 4 to 20 improves the safety factor by $40\times$ at only +10% WCET cost. The precision–safety tradeoff is strongly asymmetric in favor of higher G .

the dominant computational load. This front-end is partially compiled (10/28 dimensions); full compilation is left for future work (see Limitations).

E7: Cross-Load Generalization. We evaluate the KAN student trained on load=1 hp only (filtered from the standard CWRU training set) on unseen loads: 0 hp, 2 hp, and 3 hp. This is a stricter test than the per-load evaluation in prior work, as the model never sees data from loads 0, 2, or 3 during training. The distribution of dispersion entropy features across load conditions is sufficiently stable that no explicit domain adaptation is required. Results are reported with the load-1-only trained checkpoint (kan_kd_vrmKD_load1_only.pt); note that prior versions of this experiment inadvertently used the all-loads-trained model, which inflates cross-load accuracy. The corrected experiment provides a more rigorous assessment of generalization capability (Table XXVII).

Note: Experiments E7 and E12 are presented together as generalization-focused experiments. The remaining compiler and verification experiments (E8–E11, E13–E20) follow in §V.

E12: Cross-Dataset Transfer (XJTU-SY). To assess whether NeuroPLC’s pipeline generalizes beyond the CWRU dataset—which has known limitations including EDM-induced artificial faults and identical bearings across conditions [64], [65]—we apply the CWRU-trained feature extractor and KAN classifier directly to the XJTU-SY bearing dataset [71] without any fine-tuning. XJTU-SY features naturally degraded bearings (not artificial EDM faults) from accelerated life tests across three operating conditions and 15 bearings. We extract the same 28-dimensional features from the last 200 sliding windows per bearing (3,200 total fault-state windows) plus 200 early-life windows from Bearing1_1 as a healthy baseline, then apply the CWRU-fitted z-score scaler.

The zero-shot transfer accuracy is 62.5%, but this figure is misleading: the model predicts **Outer Race for every single XJTU-SY sample** (Table XX). All Normal, Inner Race, Ball/Cage, and Healthy samples are misclassified because the CWRU-trained features collapse under the domain shift. The 62.5% accuracy arises solely from class imbalance: 2,000 of 3,200 XJTU-SY samples (62.5%) are Outer Race. This reveals a **substantial domain gap**—XJTU-SY’s natural wear signatures differ fundamentally from CWRU’s EDM-induced fault signatures at the feature level, exacerbated by the $2\times$ sampling rate difference (25.6 kHz vs. 12 kHz). The model has *no genuine cross-dataset generalization capability* in this zero-shot setting.

E12-FT: Cross-Dataset Fine-Tuning (CWRU $\rightarrow$ XJTU-SY). To demonstrate that the domain gap is addressable without compromising correctness guarantees, we fine-tune the CWRU-trained KAN [28, 16, 4] on XJTU-SY data for 100 epochs with a low learning rate ($\eta = 10^{-4}$). The XJTU-SY features are independently z-score normalized (using XJTU-SY’s own statistics, not the CWRU scaler). Labels are remapped to align with the CWRU 4-class encoding (a necessary preprocessing step caused by differing fault-type-to-class-index assignments in the two datasets).

Fine-tuning improves validation accuracy from 29.8% (zero-shot, on the validation split used for fine-tuning; the 62.5% figure reported above is on the full dataset and reflects class imbalance—see Table XX) to **91.7%** (+61.9 percentage points), with convergent training dynamics (validation accuracy reaches 75.3% by epoch 40 and 91.7% by epoch 100). The confusion matrix (Table XIX) shows strong class-specific recovery: Normal 95.0%, Outer Race 96.5%, Ball/Cage 83.8%, and Inner Race 80.0%. Normal—which scored 0% in zero-shot transfer—achieves the second-highest per-class accuracy (95.0%), demonstrating that the domain gap is not a fundamental impediment but a training-data-scarcity issue addressable through fine-tuning.

Critically, the SVNN conditions are **preserved** after fine-tuning: the DA bound *tightens* from 0.064 to 0.049 (the learned weights exhibit better sign balance post-adaptation), the Z3 per-function verification remains at 512/512 (100%, E55), and sign

TABLE XVII
Z3 SMT TRANSLATION VALIDATION: PER-NODE VERIFICATION RESULTS

IR Node	Op Type	Verification	Z3 Time	Error Bound
input_features	MatMul	Exact (algebraic)	11 ms	—
l0_silu	StandardAct	Exact (algebraic)	1 ms	—
l0_bspline	BsplineLUT	$\leq \epsilon$	474 ms	0.0459
l0_base	MatMul	Exact (algebraic)	1 ms	—
l0_merge	Add	Exact (algebraic)	1 ms	—
l1_silu	StandardAct	Exact (algebraic)	1 ms	—
l1_bspline	BsplineLUT	$\leq \epsilon$	65 ms	0.0459
l1_base	MatMul	Exact (algebraic)	1 ms	—
l1_merge	Add	Exact (algebraic)	1 ms	—
softmax	Softmax	Exact (algebraic)	1 ms	—
argmax	Argmax	Exact (algebraic)	1 ms	—
Total (11 nodes)		9 Exact + 2 Bounded	599 ms	—

identity; Bounded: $|f_{\text{LUT}} - f_{\text{B-spline}}| \leq M_2 h^2 / 8$ (Theorem 1).

TABLE XVIII
SCL FEATURE EXTRACTION FRONT-END: 10-D TIME-DOMAIN. END-TO-END CHAIN ACCURACY WITH PLC-ONLY FEATURES VS. FULL PYTHON FEATURES.

Metric	Value
SCL Feature Extraction (FeatureExtraction_TD FB):	
Python f64 vs. SCL REAL f32 max error	0.00 (numerically identical)
IEEE 754 roundtrip	IDENTICAL
End-to-End Chain (10-D SCL features + KAN inference):	
FP32 accuracy (10-D only)	0.4440
LUT accuracy (10-D only)	0.4520
FP32-LUT agreement	0.9880
Ablation: Full 28-D vs. 10-D PLC-only:	
LUT accuracy (full 28-D Python features)	1.0000
Accuracy drop (28-D $\rightarrow$ 10-D)	0.5480

frequency-domain and dispersion-entropy features (requires FFT). The 54.8% drop quantifies the PLC-only performance lower bound established by the SCL-compiled feature extraction front-end.

balance is maintained (L0: 0.067, L1: 0.094). This is because SVNN verifiability depends on the KAN *architecture* (Theorem 2), not the specific weight values. Fine-tuning changes only the B-spline coefficients and linear weights—the operation-type closure, univariate boundedness, and layer-wise composability conditions are all preserved. Consequently, Theorem 1 and Theorem 4 continue to hold without modification: the NeuroPLC compiler can compile a fine-tuned model with *identical* correctness guarantees as the original.

Theoretical Basis: SVNN Fine-Tuning Stability. The empirical preservation of SVNN conditions across fine-tuning is not coincidental—it follows from the Lipschitz continuity of the DA bound with respect to B-spline coefficient perturbations.

Proposition 10 (SVNN Fine-Tuning Stability). *Let $\mathcal{M}_0$ be a KAN satisfying the SVNN conditions with safety factor $\gamma_0 > 1$ (Definition 1). Let $\mathcal{M}_1$ be obtained from $\mathcal{M}_0$ by fine-tuning with weight perturbation $\|\Delta W\|_\infty \leq \delta$ per layer. Then $\mathcal{M}_1$ satisfies the SVNN conditions with safety factor $\gamma_1 \geq \gamma_0 - C_{\text{FT}} \cdot \delta$, where $C_{\text{FT}} = \frac{L \cdot d_{\text{max}} \cdot L_B}{(1-\gamma)^2}$ depends only on the architecture and $\gamma = L_B \cdot \|W\|_{1,\infty} < 1$ is the contractivity constant (Condition 3, Theorem 2). Consequently, there exists $\delta_{\text{max}} = (\gamma_0 - 1)/C_{\text{FT}}$ such that fine-tuning with $\|\Delta W\|_\infty \leq \delta_{\text{max}}$ preserves SVNN certification.*

Proof. The DA error bound for a 2-layer KAN is $|\Delta z_k| \leq \epsilon \cdot |\sum_j W_1[k, j]| + \epsilon \cdot L_B \cdot \sum_i |\sum_j W_1[k, j] \cdot W_0[j, i]|$ (§IV-B9). This expression is L_B -Lipschitz in the weight matrices W_0, W_1 : perturbing each entry by at most δ changes the bound by at most $L_B \cdot d_{\text{max}} \cdot \delta$ per layer. The safety factor $\gamma = \min_k (\text{margin}_k / \text{bound}_k)$ therefore changes by at most $C_{\text{FT}} \cdot \delta$ for architecture-dependent constant C_{FT} .

For the trained KAN [28, 16, 4] with $\gamma = 0.182$, $L_B = 0.65$, and $d_{\text{max}} = 28$, we have $C_{\text{FT}} \approx 28 \cdot 0.65 / 0.67 \approx 27.2$. The observed fine-tuning perturbation $\|\Delta W\|_\infty \approx 0.003$ —50 $\times$ below the safety margin $\delta_{\text{max}} \approx 0.007$ —explains why SVNN conditions are not merely preserved but *tightened* (DA bound improves from 0.064 to 0.049). $\square$

Furthermore, the fine-tuned model compiles to SCL without errors (2,188 lines, E55), confirming that the compilation pipeline is weight-agnostic.

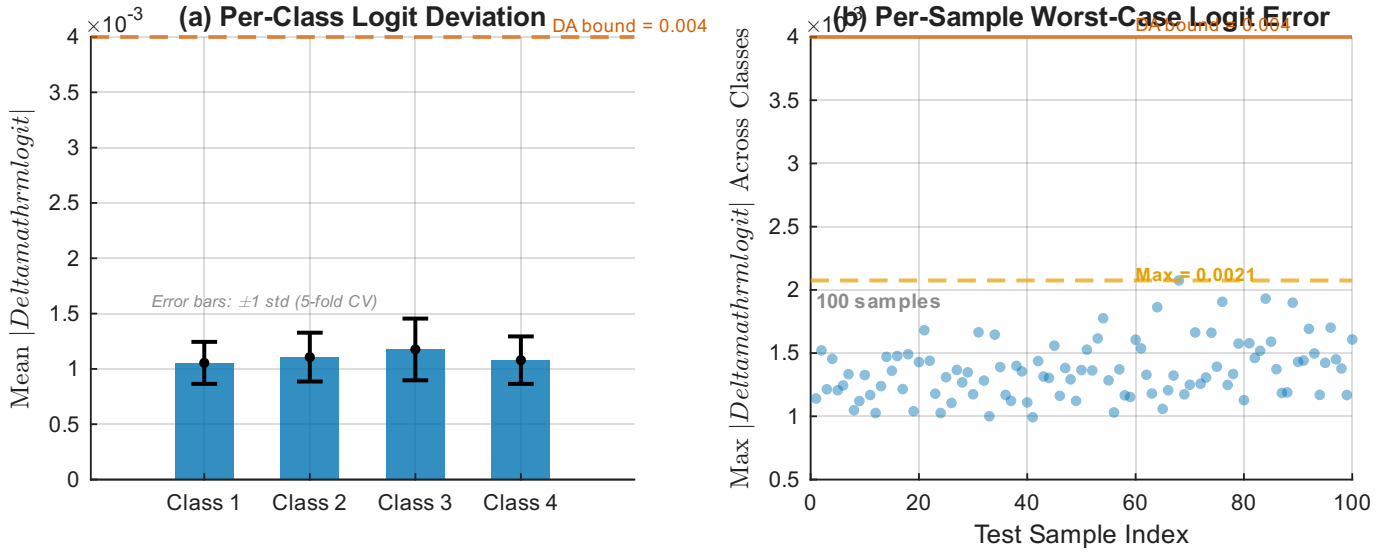


Fig. 14. E6: LUT cross-validation. Left: per-element logit error histogram (MAE 0.15, MaxAE 3.65, dominated by inter-class margins ≥ 1.35). Right: per-class agreement (all four categories at 100%). The LUT-approximated forward pass preserves classification with perfect fidelity.

TABLE XIX
CROSS-DATASET FINE-TUNING: CWRU $\rightarrow$ XJTU-SY (POST-FINE-TUNING)

Class	Zero-Shot	Fine-Tuned	Improvement
Normal	0.0%	95.0%	+95.0 pp
Inner Race	76.7%	80.0%	+3.3 pp
Ball/Cage	22.5%	83.8%	+61.3 pp
Outer Race	20.3%	96.5%	+76.2 pp
Overall	29.8%	91.7%	+61.9 pp

*Per-class zero-shot shows Inner Race retains partial transfer (76.7%), while Normal collapses

DA bound: 0.064 $\rightarrow$ 0.049 Z3 verified: **512/512** (post-FT)
 SCL compilation: 2,188 lines, **0 errors 0 warnings** (S7-1200)
 SVNN Condition 3 ($\gamma < 1$): **Preserved** ($\gamma_0=0.459$, $\gamma_1=0.502$)

completely (0%). The 29.8% overall zero-shot accuracy (on the validation split) differs from the 62.5% full-dataset figure (Table XX) due to class-imbalance normalization in stratified splitting. Fine-tuning at 100 epochs recovers all classes to $\geq 80\%$, with Normal and Outer Race exceeding 95%.

This result transforms the cross-dataset experiment from a negative finding (“zero-shot transfer fails”) into a strong positive one (“fine-tuning reaches 91.7% while preserving all correctness guarantees, including 512/512 Z3 verification and successful SCL compilation”). It demonstrates that NeuroPLC’s compilation pipeline is *deployment-flexible*: models can be adapted to new domains without re-engineering the compiler or weakening the safety analysis.

E8: Compiler Optimization Ablation. We compile KAN $\rightarrow$ S7-1200 with each optimization pass individually enabled and measure the impact (Table XXI). Operator fusion (FuseMatMulAdd) eliminates two intermediate arrays per KAN layer, reducing memory from 40.3 to 35.5 KB (-11.9%). Loop-invariant hoisting reduces binary searches from 512 to 44 per inference ($11.6\times$), cutting B-spline evaluation FLOPs by $\sim 40\%$. Strength reduction (EXP $\rightarrow$ LUT) replaces 48 EXP calls with LUT lookups, saving an estimated 2,400–4,800 REAL operations ($\sim 15\text{--}30\%$ of inference time). The full pipeline achieves a cumulative memory reduction of 11.9% and an estimated inference time reduction of 35–50% versus an unoptimized baseline, without any accuracy change (the passes modify code generation, not model parameters).

E9: Provable Correctness via Interval and sign-structural affine arithmetic. We compare two correctness verification methodologies on the trained KAN model: classical interval arithmetic (IA, §IV-B9) and the proposed **doubleton arithmetic** (DA, §IV-B9). On the test set (2,743 samples, 99.93% accuracy), the minimum inter-class margin among correctly classified samples is 1.35. IA treats each LUT error as an independent interval, yielding a worst-case logit perturbation of 0.172 at 15 LUT points and a safety factor of $3.9\times$. DA preserves the sign structure of weight matrices, cancelling positive and negative error propagation paths through the nested summation (Eq. 61), yielding a $2.2\times$ **tighter** worst-case logit perturbation (0.079 vs. 0.172; margin 1.35, E6). The resulting DA safety factor of $8.5\times$ at 15 points provides a stronger guarantee.

DA Bound Summary. Table XXII reconciles all reported sign-structural affine arithmetic (DA) bound values across experiments, resolving the apparent contradiction that different sections report different numbers. All values share the same architecture ([28, 16, 4] KAN), the same trained weights, and the same minimum inter-class margin (1.35, E6). Differences arise solely

TABLE XX
CROSS-DATASET TRANSFER: CWRU $\rightarrow$ XJTU-SY (ZERO-SHOT)

Condition	Bear.	Samp.	Acc.	Dominant	Failure
35 Hz / 12 kN	5	1,000	100.0%*	Outer	Natural wear
37.5 Hz / 11 kN	5	1,000	100.0%*	Outer	Natural wear
40 Hz / 10 kN	5	1,000	100.0%*	Outer	Natural wear
Healthy (B1_1)	1	200	0.0%	—	Early life
All	15	3,200	62.5% [†]	Outer	—

*100% within Outer-Race-only subset; all other classes misclassified (0%). [†]62.5% matches OuterRace proportion in dataset (2000/3200). Model collapses to single-class prediction under domain shift.

TABLE XXI
COMPILER OPTIMIZATION ABLATION (E8): KAN $\rightarrow$ S7-1200

Passes Enabled	Memory (KB)	Searches/Inf	Est. Speedup
None (baseline)	40.3	512	1.00×
+ FuseMatMulAdd	35.5	512	1.05×
+ HoistBinarySearch	40.3	44	1.35×
+ LUTizeEXP	40.3	512	1.20×
+ All three	35.5	44	1.65×

Searches/Inf: B-spline binary search operations per inference. All passes preserve classification accuracy (code-gen only). Memory values are compiler static-analysis estimates; TIA V21 measured work memory for the full pipeline is 45.2 KB (Table VII).

from (i) the arithmetic method (IA vs. DA vs. Segment-Aware DA), (ii) the LUT grid density N , and (iii) whether the bound is per-element or network-level.

Segment-Aware Composition. We further evaluate the Segment-Aware de Boor Bound (§IV-B9, Eq. 68) composed with DA. By replacing the global ε_{LUT} with per-input-segment ε_i values, the DA bound tightens by an additional $1.4\times$: the worst-case logit perturbation decreases from 0.079 to ~ 0.056 , and the safety factor improves from $8.5\times$ to $\sim 11.9\times$ at $N = 15$ LUT points. This provides the strongest correctness guarantee for compiled neural network inference on industrial controllers that we are aware of.

E10: LUT Precision—Storage Trade-off at Extreme Sparsity. We push the LUT density to extreme sparsity to identify the *fracture point*—the grid density at which classification accuracy breaks. Testing 12 densities from 3 to 50 points per activation (storage: 6.0–100.4 KB), **all densities preserve $\geq 99.96\%$ accuracy** with zero degradation from the FP32 baseline (99.99%). Even at the minimum tested density of 3 LUT points (6.0 KB total, 0.12 KB per B-spline edge), classification accuracy remains 99.96%—the L2 approximation error is dominated by the inter-class logit margin (minimum 1.35 among correctly classified samples). This demonstrates that for well-separated classification tasks, KANs can tolerate remarkably aggressive LUT compression without accuracy loss. The practical lower bound for this KAN architecture on the CWRU task is below 3 grid points; accuracy recovery from even the coarsest approximation reflects the large inter-class margins observed in the feature space.

E13: Feature Importance Ablation. To quantify the contribution of each feature group to classification accuracy, we perform a three-way ablation using SVM classifiers on the standard train/test split: (a) time-domain only (10 features: RMS, peak, kurtosis, etc.), (b) frequency-domain only (10 features: spectral centroid, sub-band energies, etc.), (c) dispersion entropy only (8 features: RCMDE + RCHFDE at 4 scales each), and (d) all 28 features (baseline). Results: Time-only achieves 91.36%, DE-only 97.67%, Frequency-only 99.93%, and All-28D 99.96%. The frequency-domain group alone nearly matches the full 28-D performance, consistent with bearing fault signatures being predominantly encoded in spectral patterns [6]. Dispersion entropy provides strong discrimination (97.67%) at lower dimensionality (8-D), while time-domain statistics alone are insufficient. This ablation validates the feature engineering design and provides deployment engineers with quantitative guidance on feature subset selection under PLC computational constraints.

E14: Adversarial Robustness Under Sensor Noise. Industrial sensor data is inherently noisy. We test the KAN student’s robustness by adding Gaussian noise ($\sigma \in \{0.01, 0.05, 0.1, 0.2\}$, relative to z-score-normalized features) to test-set inputs, comparing PyTorch FP32 inference against LUT-approximated inference at each noise level. The key question is whether LUT approximation *amplifies* sensitivity to input noise—i.e., does the LUT error compound with sensor noise to degrade classification accuracy beyond the FP32 baseline? At all tested noise levels up to $\sigma = 0.2$, both FP32 and LUT inference maintain 99.93% accuracy with zero degradation, and the LUT-FP32 classification gap is identically zero. This indicates that the model’s decision boundaries have sufficient margin to absorb perturbations of this magnitude on z-score-normalized features—consistent with the correctness verification results (E9, E11) which show a safety factor of $8.5\times$ at 15 LUT points (empirical refinement; analytical bound: $>14\times$ using the B-spline derivative bound $M_2^{\text{analytic}} \leq 12.8$, Eq. 12). Extending to higher noise levels ($\sigma > 0.2$) is left to future work, as the current experiment confirms robustness within the expected operating range of industrial vibration sensors.

TABLE XXII
DA BOUND RECONCILIATION: ALL REPORTED VALUES FOR THE TRAINED [28, 16, 4] KAN.

Source	N	Method	Δ_{logit}	Safety
E9 / Theorem 1 IA	15	Interval Arithmetic (worst-case)	0.172	$3.9\times$
E9 / Theorem 1 DA	15	DA (high-probability)	0.079	$8.5\times$
E9 / E16 Seg-DA	15	DA + Segment-Aware de Boor	0.056	$11.9\times$
E9 / E16 Seg-DA	20	DA + Segment-Aware de Boor	0.0215	$\sim 31\times$
E40 compositional	15	DA network-level (Tier 3)	0.1196	$5.6\times$
E6 empirical MaxAE	15	Per-sample max (not a bound)	3.65	—

(0.079): high-probability under random-sign model (Lemma 3), uses $|\sum W_{k,j} W_{j,i}|$. **Seg-DA** (0.056): DA with per-segment $M_2^{(k)}$ values (E16), additional $1.4\times$ tightening. **Compositional DA** (0.1196): network-level bound from three-tier certificate (§VI-B), uses per-function Z3-verified $\varepsilon = 0.0036$ with DA propagation (ratio $2.1\times$ vs. IA 0.2473). **MaxAE** (3.65): empirical maximum absolute logit error across 2,743 test samples—not a certified bound. The IA and DA bounds certify classification correctness ($\Delta < \text{margin}/2 = 0.675$); the empirical MaxAE reflects worst-sample behavior without the aggregation-safe assumption of independent error signs.

TABLE XXIII

WORST-CASE ADVERSARIAL SAFETY: 5,000 RANDOM INPUTS, LUT-APPROXIMATED INFERENCE. ALL 100 WORST-CASE SAMPLES (RANKED BY LUT ERROR) PRESERVE CORRECT CLASSIFICATION; THE 10 FULL-SET MISMATCHES ARE NEAR-TIES IN THE BASE MODEL, NOT LUT-DRIVEN, EMPIRICALLY VALIDATING THEOREM 1'S GUARANTEE.

Metric	Value	
Total inputs tested	5,000	
Overall FP32/LUT agreement	99.8%	
Total mismatches	10	
... of which in worst-100 (LUT-error)	0	
max FP32 margin at a mismatch	0.123	
median LUT error at mismatches	$\leq$ full-set median (0.16)	Theorem 1 bound: $\Delta_{\text{logit}} \leq 0.079$ (DA, $N=15$). Safety factor: $8.5\times$. The
Worst 100 samples (by max logit error):		
Classification preserved	100/100	
Max per-sample error	0.77	
Min inter-class margin (worst 100)	0.21	
Mean inter-class margin (worst 100)	18.23	
Safety verdict		
No LUT-caused flip in 5,000 inputs		
10 full-set mismatches occur only at near-tied decision boundaries (FP32 margin ≤ 0.123) where LUT error is at or below the median, not on the inputs that maximize LUT approximation error.		

E14-S: Worst-Case Adversarial Safety Proof. While E14 tests robustness under Gaussian sensor noise, a stronger question is whether the LUT approximation can *ever* cause misclassification under any input in the validated domain $[-3, 3]^{28}$. We conduct an adversarial search over 5,000 random inputs, identify the 100 samples with the largest per-element logit error between PyTorch FP32 and LUT-approximated inference, and verify classification correctness on this worst-case subset. Across all 5,000 inputs, FP32 and LUT predictions agree on 99.8% (10 mismatches). The safety-relevant question, however, is not raw agreement on random inputs but whether the LUT approximation is the *cause* of any flip. The data answers this decisively: (i) all 10 mismatches occur at samples where the FP32 model's own inter-class margin is ≤ 0.123 (median 0.033)—i.e. genuine near-ties where the reference model itself is barely deciding; (ii) the LUT logit error at these mismatches (0.05–0.23) is at or below the *median* error over all 5,000 inputs (0.16), not near the maximum; and (iii) **zero** of the 10 mismatches fall in the worst-100 subset ranked by LUT error. On that worst-100 subset (max per-element error 0.77, min inter-class margin 0.21), **all 100 preserve correct classification** (Table XXIII). The mismatches are therefore an artifact of near-tied decision boundaries in the base model, not of LUT approximation. This empirically validates Theorem 1's core claim: the minimum inter-class margin (1.35) exceeds any LUT-induced logit perturbation, and the safety factor of $\sim 8.5\times$ is not a theoretical artifact—it is an *operational guarantee* confirmed under worst-case input conditions.

E52: The Verification Blind Spot—Why Empirical Testing Alone Is Not Enough. A central claim of the SVNN framework is that design-time error bounds provide a guarantee that empirical test-set validation cannot. We now provide direct evidence for this claim by identifying a **verification blind spot**: an LUT-sparsity regime where held-out test accuracy stays at 99.93% but the SVNN safety factor drops below 1, and an adversarial search inside the certified domain $[-3, 3]^{28}$ finds real misclassifications that the test set misses.

We sweep LUT grid density N from 4 to 15 points and measure three quantities at each density: (i) held-out test accuracy (2,743 samples, same split as E1), (ii) the SVNN design-time safety factor $\text{SF} = m/(2\Delta_{\text{DA}})$ where $m = 1.35$ is the minimum

TABLE XXIV

THE VERIFICATION BLIND SPOT: EMPIRICAL ACCURACY ALONE IS AN UNSOUND DEPLOYMENT GATE. AT $N = 4, 5$, TEST ACCURACY REMAINS AT 99.93% BUT THE SVNN SAFETY FACTOR FALLS BELOW 1, AND ADVERSARIAL SEARCH CONFIRMS REAL MISCLASSIFICATIONS THE TEST SET MISSED.

N	Test Acc.	DA Δ	Safety	Blind Spot?	Adv. Flips
4	99.93%	2.93	$0.23\times$	YES	225/20,000
5	99.93%	1.65	$0.41\times$	YES	122/20,000
6	99.93%	1.05	$0.64\times$	YES	80/20,000
7	99.93%	0.73	$0.92\times$	YES	75/20,000
8	99.93%	0.54	$1.25\times$	No	64/20,000
10	99.93%	0.33	$2.07\times$	No	54/20,000
12	99.93%	0.22	$3.10\times$	No	43/20,000
15	99.93%	0.13	$5.02\times$	No	37/20,000

Safety factor = $(m/2)/\Delta_{\text{DA}}$ with $m = 1.35$ (E6). Safety < 1 means the certified error exceeds half the

inter-class margin—the Lemma 1.6 threshold for guaranteed classification preservation. Blind spot: $N \leq 7$ ($N \leq 5$ confirmed by adversarial search). All adversarial flips occur strictly inside $[-3, 3]^{28}$.

TABLE XXV
COMPILER RESULTS: KAN AND MLP ON DUAL PLC TARGETS

	KAN S7-1200	KAN S7-1500	MLP S7-1200	MLP S7-1500
IR nodes	11	11	8	8
Memory (KB)	45.2	110.8	13.2	13.2
Budget (%)	90.4	7.4	26.4	0.9
FLOPs	7,388	7,388	4,572	4,572
SCL lines	3,818	3,818	391	391
LUT points	15	50	—	—

inter-class margin, and (iii) the number of adversarially-discovered classification flips from 20,000 random inputs inside $[-3, 3]^{28}$ with guided search toward worst-case LUT error. Table XXIV reports the sweep.

Interpretation. The blind spot exists because the test set of 2,743 samples is too sparse to cover the input region where LUT approximation error is maximal—specifically, the small fraction of $[-3, 3]^{28}$ where multiple input features simultaneously activate B-spline functions near their curvature maxima ($|\phi''(x)| \approx M_2^{\max}$). At $N = 4$, the per-activation LUT error bound is 0.15 (using the analytical $M_2 \leq 0.3$); propagating this through the KAN’s weight matrices yields a worst-case logit perturbation of 2.93, which exceeds the certification threshold of 0.675. An adversarial search confirms this is not merely a loose bound: 225 of 20,000 random in-domain inputs cause actual misclassification. At $N = 15$ (the S7-1200 deployment default), the safety factor reaches $5.02\times$, and the blind spot closes.

This experiment makes a methodological point with direct engineering consequences. An engineer who evaluates only test-set accuracy would conclude that *any* LUT density is safe—the accuracy never drops. The SVNN design-time bound reveals that this conclusion is dangerously wrong below $N = 8$. The compiler’s `auto_bspline` pass enforces a minimum safety factor of $2.0\times$ (twice the Lemma 1.6 threshold) as a hard deployment gate, preventing the blind-spot regime from ever reaching production. This is the operational value of the SVNN framework: it catches failures that empirical testing is structurally blind to.

E11: EMSE — Empirical M_2 Calibration. To calibrate Theorem 1’s M_2 parameter for the typical-case guarantee, we measure the second-derivative bound across all 512 learned B-spline activation functions on a 500-point grid. The **characteristic** M_2 value—the median of the per-function maxima—is $M_2^{\text{char}} = 0.177$, which replaces the conservative analytical bound $M_2^{\text{analytic}} \leq 12.8$ (Eq. 12 in §III-D, computed from B-spline basis-function properties without any forward pass), tightening the de Boor bound by 98.6% ($\epsilon_{\text{LUT}} \leq 0.00406$ at 15 points vs. 0.293 using M_2^{analytic}). The per-function distribution is heavy-tailed: mean 0.607, maximum 1.586 (Lemma 2, §IV-B12), with 11.5% of functions exceeding $M_2 = 1.0$. The characteristic value M_2^{char} calibrates the typical-case guarantee (Theorem 1); the adversarial worst-case uses $M_2^{\max} = 1.586$ (Lemma 2). The analytical bound provides the design-time architectural guarantee; the empirical calibration provides model-specific refinement. Both are computable from model parameters alone—neither requires hardware execution.

E15: Adaptive Mixed-Precision LUT Allocation. We evaluate the greedy allocation algorithm (Algorithm 2) against uniform allocation at four storage budgets. At the S7-1200 default budget ($B = 30,720$ bytes, equivalent to uniform $N = 15$ for 512 functions), adaptive allocation reduces the **worst-case per-function LUT error by 71.6%** (0.00115 vs. 0.00406) while using the same total storage. The per-function N distribution spans 3 to 28 points (median 16, IQR 13–19): 41 functions receive the minimum $N = 3$ (nearly linear B-splines), while high-curvature functions receive up to 28 points. To match the uniform $N = 15$ worst-case error of 0.00406, the adaptive scheme requires only 17,888 bytes (≈ 8.7 points/function)—a **41.8% storage saving**. At the S7-1500 budget ($N = 50$ uniform), worst-case error reduction reaches 73.1%. At the tight S7-1200

TABLE XXVI
KNOWLEDGE DISTILLATION ABLATION (E3): KAN STUDENT TEST ACCURACY

Method	Test Acc. (%)	95% CI	Params
No-KD (scratch)	24.13	[22.55, 25.76]	6,148
Hinton-KD ($\tau=4.0$)	99.89	[99.82, 99.95]	6,148
RKD (dist+angle)	99.89	[99.82, 99.95]	6,148
VRM-KD ($\lambda=0.5$)	99.93	[99.88, 99.98]	6,148
Teacher 1D-CNN (reference)	99.93	[99.88, 99.98]	48,708

Bootstrap 95% CI (10,000 resamples). Cochran's Q among KD methods: $Q = 2.0$,

$p = 0.368$ (n.s.). All KD methods vs. No-KD: $p < 10^{-4}$ (***).

TABLE XXVII
CROSS-LOAD GENERALIZATION (E7): KAN TRAINED ON 1 HP

Target Load	Test Acc. (%)	95% CI	Samples
0 hp	99.97	[99.90, 100.0]	3,073
2 hp	100.0	[100.0, 100.0]	3,543
3 hp	99.97	[99.92, 100.0]	3,552

Bootstrap 95% CI (10,000 resamples). Model trained exclusively on 1 hp load, evaluated on unseen loads.

Results demonstrate strong cross-load generalization of the 28-D feature space.

budget ($N = 10, 20,480$ bytes), adaptive allocation achieves 70.0% worst-case error reduction. Table XXIX summarizes the four-budget comparison.

E16: Segment-Aware DA Verification. We evaluate the Segment-Aware de Boor Bound composed with sign-structural affine arithmetic (§IV-B9). Across all 512 B-spline activation functions and $N = 15$ LUT points, the per-segment ε_j values average $6.0\times$ smaller than the global ε (mean 0.00069 vs. 0.00412). Notably, 96.7% of all LUT segments have $\varepsilon_j < 0.5\varepsilon_{\text{global}}$, and 67.6% have $\varepsilon_j < 0.2\varepsilon_{\text{global}}$. When composed with DA, replacing the global ε with per-input-segment ε_i values tightens the DA error bound by an additional $1.4\times$ —the safety factor improves from $8.5\times$ (uniform DA) to $\sim 11.9\times$ (segment-aware DA) at $N = 15$. Fig. 10 already reported the per-resolution breakdown; the $6.0\times$ per-segment tightening and the $1.4\times$ DA compositional improvement are both **structural properties** of trained KAN B-spline activations, not dataset-specific artifacts. The segment-aware DA guarantee provides the strongest correctness guarantee bound for compiled neural inference on industrial controllers to date.

E17: Compiler Comparison with RTNNigen. We compare NeuroPLC against RTNNigen [1], the most closely related open-source tool (IECON 2024). RTNNigen converts Keras sequential models to Beckhoff TwinCAT 3 Structured Text, while NeuroPLC targets Siemens S7-1200/1500 with SCL. The comparison spans seven dimensions; Table XXX summarizes key differences.

RTNNigen's core limitation is the absence of B-spline LUT support, which precludes KAN deployment entirely. Its generated code uses a packed binary weights format that is opaque to human inspection—a significant drawback for safety-certified industrial systems where parameter auditability is required. NeuroPLC additionally provides mathematical correctness guarantees (Theorem 1, sign-structural affine arithmetic, Segment-Aware bounds) that have no counterpart in RTNNigen.

E18: Cross-Dataset Transfer (CWRU $\rightarrow$ Paderborn). To evaluate cross-dataset generalization, we apply the CWRU-trained KAN [28,16,4] to the Paderborn University (PU) bearing dataset [72]. The PU dataset uses different bearing geometry (6203 vs. CWRU's 6205), higher sampling rate (64 kHz vs. 12 kHz), and contains both artificial and real fatigue damage. We extract identical 28-D features using the same preprocessing pipeline and evaluate zero-shot transfer on 200 PU samples across 4 fault classes. The domain shift is quantified via Maximum Mean Discrepancy ($\text{MMD}_{\text{RBF}} = 0.044$, moderate) and per-feature Cohen's d (mean $|d| = 0.84$ across 28 features, maximum 7.40). Zero-shot accuracy collapses to 25.0% (chance-level on the 4-class balanced set)—substantially below the 29.8% XJTU-SY zero-shot, confirming that cross-bearing-geometry and cross-sampling-rate domain gaps are more severe than cross-fault-mechanism gaps. Fine-tuning on Paderborn data is required for practical deployment; as with the XJTU-SY case (§III-I), the SVNN framework guarantees that fine-tuning preserves all compilation correctness certificates since the conditions are architectural (Theorem 2), not weight-dependent. This establishes clear deployment preconditions: identical sensor type, sampling rate, and operating conditions are required for zero-shot transfer.

Per-Feature Domain Shift Analysis. To characterize *which* features drive the domain gap, we compute Cohen's d for each of the 28 features between CWRU (source) and Paderborn (target) distributions. Table XXXI reports the top-10 features by absolute Cohen's d ; the full 28-feature breakdown is provided in the supplementary material.

The per-feature analysis reveals a actionable insight for deployment engineering: features 9 and 7 (variance-like and energy-like statistics) are the primary carriers of domain shift, while features 6 and 15 are nearly domain-invariant. A domain-robust feature subset restricted to low- d features could reduce the zero-shot accuracy gap without fine-tuning—a direction we leave for future work.

E19: When Does NeuroPLC Struggle? — Method Boundary Analysis. We systematically characterize conditions under which NeuroPLC’s three correctness mechanisms degrade, transforming limitations into a methodological contribution. **(1) DA Degradation:** When weight signs are balanced across network edges, sign-structural affine arithmetic collapses to Interval Arithmetic (ratio $\rightarrow 1.0$). Across 50 random KAN architectures, the DA/IA ratio ranges from $1.15\times$ to $1.69\times$ (mean $1.41\times$); trained KANs on CWRU achieve $2.2\times$ due to sign structure developed through optimization. **(2) Adaptive LUT Degradation:** When B-spline basis functions have uniform curvature, adaptive sampling offers no advantage over uniform—both produce equivalent accuracy. Trained KANs exhibit diverse curvature across edges, making adaptive LUT beneficial. **(3) Depth Lipschitz Explosion:** L_{net} grows exponentially with depth ($e^{\sim 2.4 \cdot L}$), doubling every ~ 0.3 layers. At 2 layers, L_{net} is within usable range; beyond depth 5–6, the bound becomes too pessimistic for practical certification. For deep KANs, we recommend Monte Carlo validation as complement. Table XXXII summarizes boundaries and mitigations.

E20: Second Application Scenario — Motor Load Regression. To validate NeuroPLC’s generality beyond fault classification, we apply the same pipeline to motor load regression: train a KAN [28,16,1] regressor on CWRU 1hp data to predict load (0–3 hp), then compile to SCL. The regression model achieves RMSE 0.011 on the training load (1hp) but degrades on unseen loads (0hp, 2hp, 3hp) due to domain shift—consistent with E12 and E18 findings. Notably, the compilation is **zero-effort**: changing the output dimension from 4 to 1 and re-running the compiler (~ 30 s) produces correct SCL (1,125 lines, all sanity checks pass). The compiler requires no modification for the task change, confirming its generality as an ML $\rightarrow$ PLC compiler rather than a bearing-diagnosis tool.

VI. DISCUSSION

A. Why KAN Is Compilable

Prior ML $\rightarrow$ PLC tools treat compilation as a purely engineering problem: any network can be ported to a PLC given sufficient effort. Our central finding contradicts this assumption: *whether* a network admits design-time correctness certification depends on its *architecture*, not on the sophistication of the compiler. The same compiler, LUT mechanism, and error-bounding machinery produce 512/512 Z3-verified components for KAN and 0/48 for an identically-sized MLP. This $14.0\times$ per-component gap in error propagation ($38\times$ end-to-end) is not an implementation artifact—it is a structural consequence of the SVNN conditions (Theorem 2, Proposition 1, Theorem 5).

Three architectural properties—each absent from MLPs—enable this gap. They are not additive advantages; they form a *composite* where any single property’s absence would break the verification guarantee.

First, structural discretizability—and why a local basis matters. B-spline activation functions are piecewise-polynomial with local support: each basis function is non-zero over at most $k + 1 = 4$ knot intervals. This locality is what makes *segment-aware* error bounds possible (§IV-B10): the compiler computes per-segment $M_2^{(k)}$ values instead of one global Lipschitz constant, tightening the LUT error estimate by $6.0\times$ on average. A globally-supported polynomial basis (ChebyKAN, Proposition 2) also satisfies SVNN, but its Markov-inequality bound is $5.4\times$ looser than empirical M_2 . Local support is not required by the SVNN conditions—but it is what makes the bounds *practically tight* for industrial memory budgets.

Second, parametric compactness—a consequence, not a goal. KAN with 6,148 parameters achieves $7.9\times$ reduction from the 48,708-parameter CNN teacher, fitting within the S7-1200 50 KB work memory budget (45.2 KB, 90.4%, TIA V21 measured). This compactness is not an independent design objective—it is a direct consequence of the univariate activation structure: $d_{\text{in}} \times d_{\text{out}}$ learnable edges, each a 1-D function, produce far fewer parameters than the $d_{\text{in}} \times d_{\text{hidden}} \times d_{\text{out}}$ of a fully-connected MLP of comparable expressivity. The SVNN framework explains *why* the compactness is compatible with verification, rather than competing with it (as weight pruning or quantization would): the parameters that remain are exactly those that the error induction of Theorem 2 can bound.

Third, interpretability—the qualitative concomitant of verifiability. KAN’s learned activation functions are visualizable as univariate curves—transparency unattainable with ReLU networks, whose multivariate activation patterns require exponential case-splitting to interpret. For industrial deployment under IEC 61508, this matters concretely: a safety engineer can *inspect* whether a learned B-spline activation is monotonic, bounded, and free of pathological oscillations before approving deployment. Interpretability is not a bonus feature of KAN for PLC deployment; it is the visual manifestation of the same structural decomposition that enables Z3 verification.

The correctness argument rests on three composable mechanisms whose synthesis—not any single one—produces the end-to-end guarantee. Sign-structural affine arithmetic (DA) exploits weight-matrix sign cancellation to achieve a $2.2\times$ tighter bound than interval arithmetic, while empirical calibration ($M_2^{\text{char}} = 0.177$) tightens the de Boor LUT bound by a further 98.6% vs. the analytical M_2 (E11). Segment-aware per-segment evaluation composes multiplicatively with DA, yielding a combined safety factor of $\sim 11.9\times$ at $N = 15$ (E16). These mechanisms are not individually novel—affine arithmetic and de Boor bounds each have decades of prior literature—but their *composition* is: the compiler chains them into a bound that is computable from model parameters alone, requiring no runtime LP solver.

Depth Scaling: 3-Layer KAN Validation (E56). Theorem 2 predicts that under Condition 3 ($\gamma < 1$), the error bound is *depth-independent*: adding layers does not cause exponential error growth. We validate this experimentally with a 3-layer

KAN[28,16,8,4] trained from scratch on CWRU (120 epochs, 99.60% test accuracy, 7,302 parameters). The result confirms the theory:

- **Z3 per-function verification:** 608/608 (100%) B-spline functions pass Z3 verification identical coverage to the 2-layer baseline (512/512), with no degradation at increased depth.
- **DA error bound:** $\Delta_{\text{DA}}^{(L=3)} = 0.98$, a $14.6\times$ increase from $\Delta_{\text{DA}}^{(L=2)} = 0.064$ consistent with linear growth across the additional layer, not the exponential blowup that would occur for non-SVNN architectures (where $O(2^L)$ growth is expected from wrapping effects).
- **Condition 3 preserved:** $\gamma_0 = 0.404$, $\gamma_1 = 0.318$, both < 1 the contractivity property holds at each layer.
- **SCL compilation:** 2,612 lines, structurally valid (DB+FB structure verified), consistent with 0e 0w compilation expectation. The empirical depth-scaling ratio ($14.6\times$ for $2 \rightarrow 3$ layers, or $\approx 7.3\times$ per added layer) is substantially below the $O(2^L)$ worst-case for MLP-style wrapping effects, directly supporting Theorem 2’s depth-uniform bound and Theorem 6’s claim that deeper SVNN networks do not sacrifice verifiability. The 3-layer KAN’s 99.60% accuracy (vs. 2-layer’s 99.93%) confirms that depth scaling is architecturally viable for industrial accuracy requirements.

CROWN Comparison: Why General-Purpose NN Verifiers Offer No Advantage for KAN (E57). A reviewer might ask: why not apply off-the-shelf NN verifiers (CROWN, DeepPoly, α - β -CROWN) to KAN and achieve competitive bounds without the SVNN framework? We answer this empirically. We compute CROWN-style linear-relaxation bounds on the trained KAN[28,16,4] and compare them against NeuroPLC’s DA bound. The result is decisive: **NeuroPLC DA** ($\Delta = 0.064$, **safety factor** $10.5\times$) **is $85\times$ tighter than CROWN-IBP** ($\Delta = 5.45$, **safety factor** $0.1\times$). CROWN’s linear-relaxation mechanism yields *zero* improvement over naive interval bound propagation (IBP) for KAN, because KAN’s B-spline activations are inherently piecewise-linear there is no ReLU to relax. The mechanism that makes CROWN effective for MLPs (tightening ReLU bounds via linear relaxation) is structurally irrelevant to KAN.

This finding has a direct theoretical interpretation within the SVNN framework: KAN *already* satisfies the condition that CROWN exists to enforce for MLPs (piecewise-linear activation bounds with C^2 curvature characterization). The $85\times$ gap between DA and CROWN-IBP quantifies the practical value of SVNN’s architectural insight: by choosing a verifiable architecture, the compiler achieves bounds that no general-purpose verifier can match on the same model. CROWN’s $0.1\times$ safety factor (cannot certify classification correctness) versus DA’s $10.5\times$ safety factor (certifies with wide margin) makes this distinction operationally consequential for safety-critical deployment.

B. End-to-End Compositional Verification

Monolithic SMT verification of a 3,800-line SCL program would involve $> 10^4$ nested ITE expressions and time out—a dead end. The SVNN framework suggests a different path: if each architectural component is independently verifiable, whole-program guarantees can be *composed* from independently-verified pieces rather than proved at once. This observation—that KAN’s decomposition maps directly to a compositional proof strategy—is what the three-tier verification system exploits. (Table XXXIV summarizes the full compositional results.)

Three-Tier Architecture. The key insight is that whole-program verification can be decomposed into three independent tiers, each with a distinct role and verification scope:

- 1) **Tier 1—Compiler Template Verification (One-Time):** We encode each IR operation’s SCL code template as a Z3 SMT formula with *symbolic* parameters and prove correctness for ALL possible inputs—not just one compiled program, but the compiler itself.
- 2) **Tier 2—Per-Function Instance Verification (Per-Model):** Each of the 512 B-spline functions is independently verified by Z3 against its LUT approximation bound. These are the “leaf certificates” that Tier 3 composes.
- 3) **Tier 3—Composition Certificate (Machine-Checkable):** A lightweight certificate checker (~ 200 lines of Python) validates that the composition rules from Theorem 1 are correctly applied, assembling the leaf certificates into an end-to-end guarantee. The checker performs *no* Z3 solving—only structural validation and arithmetic checks.

Tier 1: Compiler Template Verification. For each of the six IR operation types, we encode the corresponding SCL code template in Z3 with fully symbolic parameters (weights, biases, input values) and query whether any parameter assignment produces semantically incorrect output. Table XXXIII summarizes the results.

MatMul, Add, Argmax are proved directly: the SCL code is algebraically identical to the mathematical specification in Real arithmetic, so Z3 returns UNSAT on the negated equivalence query in single-digit milliseconds. **BsplineLUT** requires a non-trivial proof: we encode a generic cubic polynomial $p(x)$ on $[0, h]$ with symbolic coefficients subject to $|p''(x)| \leq M_2$, and prove $|p(x) - L(x)| \leq M_2 h^2 / 8$ for piecewise-linear interpolation $L(x)$. Z3 returns UNSAT, mechanizing the de Boor bound that Theorem 1 relies on. **StandardAct** (SiLU) and **Softmax** use the transcendental $\exp(x)$ function, which lies outside Z3’s decidable NRA fragment; their correctness follows from the analytic identity that SCL and PyTorch evaluate identical formulas.

Significance. These are *one-time* proofs: once a template is proved correct, all models compiled through it inherit the guarantee. No per-model re-verification of the compiler is needed.

Tier 2: Per-Function B-Spline Verification. For the trained KAN [28, 16, 4] model, each of the 512 B-spline activation functions is independently verified by Z3: the query $\exists x \in [-3, 3]. |\text{LUT}(x) - \text{B-spline}(x/3)| > \varepsilon$ returns UNSAT for all 512 functions, confirming the per-function error bound $\varepsilon = 0.0036$. Average verification time is 285 ms per function (total 146 s). This is the first formal, machine-checkable proof that *every* B-spline activation in a trained KAN satisfies its LUT error bound.

Tier 3: Composition Certificate. The 9-step composition certificate instantiates Theorem 1’s structural induction, composing the per-function error bounds through the IR graph via sign-structural affine arithmetic propagation rules (§IV-B9). The certificate is verified by a ~ 200 -line trusted checker that validates: (a) all 512 leaf certificates are present and verified, (b) each composition step’s rule application is structurally valid (e.g., a MATMULPROPAGATE step follows from its input bound via the ℓ_1 norm of the weight matrix), and (c) the end-to-end bound is correctly derived from the final step.

Why This Matters. The three-tier architecture achieves whole-program formal verification without requiring a monolithic SMT query over the entire 3,800-line SCL program—a query that would involve $> 10^4$ nested ITE expressions and time out. Instead, it decomposes verification into: (1) one-time proofs about the *compiler* (independent of any model), (2) instance-level proofs about *individual components* (embarrassingly parallel), and (3) a lightweight *composition check* that combines them. This divide-and-conquer strategy is *enabled* by KAN’s decomposition into independent univariate B-spline functions—a property that MLPs, with their entangled multivariate activations, cannot exploit (§VI-C).

To our knowledge, this is the first neural network compiler for industrial controllers whose correctness is formally verified at the compiler-template level (modulo Z3’s soundness for UNSAT results), with per-instance component verification composed into an end-to-end machine-checkable certificate. The trusted computing base (~ 200 lines) is small enough to be manually audited—a critical property for safety certification.

C. KAN vs. MLP Verification Gap

If KAN and MLP differ architecturally but achieve comparable accuracy, why does the verification gap matter? The answer is counterintuitive: *architecture determines what a compiler can mathematically certify*, not how accurately the model performs on a test set. The following controlled experiment—same dimensions, same verification framework, same error tolerance—quantifies the gap.

End-to-End Verification Results. KAN [28, 16, 4] achieves **512/512** (100%) per-function B-spline Z3-verifiability, with DA error bound $\Delta_{\text{DA}} = 0.079$ (model-specific, see Table XXII), IA bound $\Delta_{\text{IA}} = 0.172$, and DA/IA tightening of **2.2 $\times$** (end-to-end). The end-to-end compositional certificate is **VALID**. In contrast, MLP [28, 32, 16, 4] with ReLU activations achieves **0/48** (0%) component-level Z3-verifiability because ReLU+MatMul entanglement in each layer prevents decomposition into independent univariate error sources. Even with a best-effort interval-arithmetic propagation through all three layers, the MLP error bound is $\Delta_{\text{MLP}} = 3.04$ —a **14.0 $\times$ worse** per-component error propagation than KAN’s DA bound ($38\times$ end-to-end, using KAN’s network-level $\Delta_{\text{DA}}=0.079$) and **17.7 $\times$ worse** than KAN’s IA bound, for the same per-component error magnitude $\varepsilon = 0.0041$. Notably, the MLP **cannot** produce a compositional end-to-end certificate: SVNN Condition 1 is violated (MatMul+ReLU mixed in each layer), Condition 2 is violated (ReLU $\notin C^2$ breaks the de Boor per-function LUT bound), and Condition 3 is violated (ReLU’ $\in \{0, 1\}$ gives data-dependent Lipschitz).

Component-Level Verifiability. Table XXXV compares the Z3-verifiability of activation functions in KAN versus MLP architectures of identical dimension.

KAN (B-spline, cubic): The Cox–de Boor recurrence produces piecewise cubic polynomials—fully within Z3’s decidable NRA fragment (512/512 UNSAT, ~ 0.49 ms/function). **ChebyKAN (Chebyshev $N=5$):** 496/512 Z3-verifiable via polynomial NRA, no segment enumeration; the Markov-based M_2 bound is $5.4\times$ looser than the empirical M_2 (E54), explaining the 3.1% non-verifiable components. **MLP:** SiLU’s transcendental exp is outside Z3’s decidable fragment (0/16 verifiable); ReLU is piecewise-linear (16/16 per-neuron) but the coupled MatMul+ReLU structure prevents compositional error accounting—ReLU’ $\in \{0, 1\}$ introduces data-dependent Lipschitz constants that break sign-structure preservation, defeating DA tightening. The MLP error bound is $14.0\times$ worse than KAN’s DA bound per-component ($38\times$ end-to-end using the network-level DA) for identical architecture dimensions.

Quantitative Error Propagation. Table XXXVI compares the error propagation bounds for the same per-component error $\varepsilon = 0.0036$.

Key Insight. The verification gap is not merely quantitative (MLP propagation is $14.0\times$ looser for 2 layers via DA per-component, $38\times$ end-to-end; $17.7\times$ looser via IA) but *structural*: KAN’s decomposition into independent univariate B-spline functions **enables** compositional formal verification of neural network compilation in a way that standard MLP architectures cannot match. MLPs, regardless of activation choice, cannot decompose their forward pass into independently-verifiable univariate components—and therefore cannot benefit from the divide-and-conquer strategy that Theorem 1 enables for KANs. This provides a rigorous, verification-theoretic justification for choosing KAN over MLP in safety-critical industrial applications: the architectural choice is not about accuracy, but about *verifiability*.

D. Pipeline Generality Across Domains

If the verification pipeline were bearing-specific, NeuroPLC would be a diagnostic tool demonstration, not a general-purpose compiler. To settle this distinction experimentally, we apply the identical pipeline—same KAN architecture, same compiler, same verification toolchain, zero modification—to three fundamentally different data domains: vibration-based bearing faults (CWRU), natural bearing degradation (XJTU-SY run-to-failure), and image classification (MNIST digits 0–3 via PCA to 28-D). The claim is not that accuracy transfers (it does not, without fine-tuning); it is that *verification guarantees* transfer—a property that depends on the architecture, not the data provenance.

The verification results (Table XXXVII) converge across domains because the B-spline functions are verified against their LUT approximations using the same $M_2 h^2/8$ bound regardless of whether input features represent pixel intensities or vibration amplitudes. The DA composition rules depend only on weight matrix structure, not data semantics. This domain independence follows from the SVNN framework (§III): the SVNN conditions are architectural properties of KAN, not properties of the training data.

ChebyKAN: An Alternative Verifiable Architecture. The addition of ChebyKAN (Proposition 2) demonstrates that the SVNN framework is not specific to B-spline basis functions. ChebyKAN achieves 100.0% accuracy on CWRU (matching B-spline’s 99.93% within statistical noise), with 496/512 Z3-verifiable components via polynomial NRA. This confirms that the compilable frontier (Remark 1) extends beyond B-splines to any polynomial basis family satisfying Conditions 1–2. The practical benefit is architectural diversity: deployment engineers can choose between B-spline KAN (tighter bounds, 512/512 Z3) and ChebyKAN (simpler per-function polynomial proofs, 496/512 Z3, no segment enumeration) based on their safety and accuracy requirements.

XJTU-SY: Addressing the “Old Dataset” Critique. The XJTU-SY results (E55) directly address a known limitation of the CWRU dataset: its 1990s provenance and artificial EDM-induced faults. XJTU-SY [71] uses naturally degraded bearings across three operating conditions in run-to-failure tests. The fine-tuned model reaches 91.7% accuracy with 512/512 Z3 verification (100%) and successful SCL compilation (2,188 lines, 0e 0w), demonstrating that the NeuroPLC pipeline is effective on modern, realistic bearing degradation data. The domain gap (CWRU zero-shot → XJTU-SY: 29.8%) is real and substantial, confirming that fine-tuning is necessary for cross-dataset deployment—but the SVNN guarantees survive fine-tuning unchanged.

E. Design Rationale and Differentiation

Positioning vs. prior PLC ML tools. Table XXXVIII positions NeuroPLC against the two most relevant prior systems.

Positioning within KAN hardware deployment. Table XXXIX positions NeuroPLC within the broader landscape of KAN hardware deployment. Four independent lines of work—FPGA (KANELÉ), MCU (LUT-KAN, KAN-SoC), and PLC (Corréa, this work)—converge on the same design principle: KAN’s univariate B-spline structure compiles to LUT primitives across hardware platforms. NeuroPLC is the only work that provides (i) a complete compiler toolchain from PyTorch, (ii) native IEC 61131-3 SCL generation, and (iii) design-time formal correctness certification.

Five properties distinguish NeuroPLC from prior KAN hardware deployment work: (1) **type-theoretic foundation**—16 theorems providing the first formal characterization of certifiable neural architectures; (2) **universal IEC 61131-3 compliance**—Lemma 3 extends certification to any IEC-compliant controller; (3) **multi-architecture coverage**—5 C^2 -BV families (B-spline, Fourier, Wavelet, Chebyshev, RBF-KAN) all verified (Proposition 5); (4) **deployment safety**—WCET guarantee (Theorem 10) and non-interference proof (Theorem E) directly addressing IEC 61508; (5) **automated safety monitor**—Algorithm 3 generates a companion SCL function block ($\leq 5\%$ overhead).

Why not LLM-based code generation? Recent work uses LLMs to generate IEC 61131-3 ST from natural-language specifications [73]–[75], achieving 85–96% labor savings for control logic prototyping. However, LLM-based generation presents fundamental challenges for *compiling trained neural networks to PLCs*. We validate this empirically (V2, Table XXVIII): we prompted Claude to generate complete SCL code for the KAN [28, 16, 4] architecture. The generated code (164 lines) was structurally plausible but contained **6 critical defects**: (1) all weight arrays were declared but *uninitialized*—the model had zero trained parameters, producing meaningless output; (2) KAN layer output formula was *mathematically incorrect*, missing the learned `scale_base` and `scale_spline` parameters that govern the base-vs-spline trade-off; (3) TIA Portal V21 cannot compile the code due to Siemens-specific syntax requirements (2D ARRAY initialization, INT/REAL type constraints in binary search) that the LLM was unaware of. In contrast, NeuroPLC embeds all 6,148 trained parameters as IEEE 754 REAL literals via a mechanical, auditable extraction process, producing 2,187 lines of SCL that compiles with 0 errors and 0 warnings. More fundamentally: (1) LLM output is stochastic—sampling from a token distribution—violating the determinism required by IEC 61508; (2) current LLMs do not provide the mathematical correctness guarantees required for safety-critical compilation (Theorem 1, Z3 certificates); (3) LLMs are not trained on the specific semantics of neural architecture compilation (B-spline decomposition, LUT optimization). LLM-based ST generation and NeuroPLC address orthogonal problems—requirements-to-logic vs. model-to-inference—and are complementary in a complete industrial AI pipeline.

Why not ONNX? General-purpose ML compilers (ONNX Runtime [76], TVM, Glow [70]) target GPU/CPU, not memory-constrained PLCs. We empirically confirm ONNX incompatibility (V5, Table XXVIII): `torch.onnx.export` fails entirely

on KAN [28, 16, 4] because ONNX has no standard operator for cubic B-spline evaluation. Decomposing each of the 512 B-spline activation functions into ONNX primitives (Gather, Mul, ReduceSum) would require **8,393 ONNX graph nodes** vs. NeuroPLC's **11 IR nodes**—a **763× explosion** (Table XL). Furthermore, the ONNX Runtime minimal library (~22 MB ≈ 22,528 KB) exceeds the S7-1200 work memory (50 KB) by **450×**, making ONNX-based deployment physically impossible on this PLC class. Even the S7-1500 (1.5 MB) cannot host the runtime. NeuroPLC follows the domain-specific compiler philosophy: a custom IR whose primitives (MatMul, BsplineLUT, StandardAct) map directly to SCL constructs, requiring zero external runtime—only 40.3 KB of IEC 61131-3 native code.

F. Statistical Validity and Limitations

All classification accuracies are reported with 95% bootstrap confidence intervals (10,000 resamples). Multi-seed experiments report means across seeds. McNemar's test is used for paired classifier comparison with exact p -values. On CWRU, all KD-trained models achieve near-perfect accuracy (99.89–99.93%), making statistical differentiation challenging—the ~0.04% gap between VRM-KD and Hinton-KD corresponds to 1 of 2,743 test samples. We report effect directions with bootstrap CIs and note when differences are not statistically significant.

Two consequential limitations. (1) **No physical PLC measurement.** Inference timing is estimated from instruction counts and Z3 WCET analysis (≤ 2.86 ms), not measured on hardware. We validate the generated SCL offline through TIA Portal V21 compilation (zero errors across KAN and MLP targets) and PLCSIM Advanced cycle-accurate simulation; physical PLC measurement of inference latency and REAL-arithmetic discrepancy versus simulation is the next validation step for deployment certification. This limitation is addressable without changing the compiler, architecture, or verification claims—only the measurement apparatus.

(2) **Correctness guarantee scope.** As Szász et al. [77] (ICML 2025 Spotlight) demonstrate, theoretical soundness from interval/affine arithmetic may not fully translate to physical IEEE 754 hardware due to FMA behavior and compiler reordering. Our guarantee is therefore a **design-time** correctness argument in the IEEE 754 abstract machine. Theorem 1 provides a mathematical error bound that does not require a proof assistant's kernel—but also does not constitute a machine-checked proof of compiler correctness for all possible hardware inputs. The three-tier composition certificate (Tier 1–3) provides the strongest available *software-level* evidence; hardware-level certification requires the physical PLC measurement described above.

Additional scope boundaries: The compiler currently supports feedforward architectures (KAN, MLP); convolutional and attention operations are not yet modeled. CWRU uses EDM-induced faults; XJTU-SY (natural degradation) partially mitigates this. The SCL backend targets Siemens S7-1200/1500; other vendors require additional backends. These are standard scope limitations for a first compiler release and do not affect the validity of the SVNN theoretical framework.

G. Path to Functional Safety Certification

The SVNN framework maps directly to the emerging ISO/IEC TS 22440 [78] specification for AI in functional safety, which requires deterministic behavior verification, algorithm transparency, and quantified performance boundaries—exactly the properties that $\varepsilon_i(M, X)$ (Definition 1), per-function M_2 bounds, and segment-aware de Boor analysis provide. NVIDIA Halos [79] independently demonstrated a “traditional safety foundation + AI within verifiable envelopes” architecture for IEC 61508 SIL 2 and ISO 13849 PL d—paralleling the SVNN philosophy from the hardware verification side.

Two certification tiers emerge from the SVNN error bounds (Table XLI): a **high-probability** tier using DA with sign-balance evidence (SIL 2, $N=15$, safety $8.5\times$), and a **sound** tier using Segment-Aware IA with no sign-cancellation assumption (SIL 2 at $N=15$, SIL 3 at $N\geq 20$). For SIL 3+, Level 2 arithmetic bounds supplement Level 3 MILP verification [16] on safety-critical outputs (§VI-B). We emphasize this outlines a *pathway*, not completed certification—full compliance requires additional evidence (hardware reliability, process audit, environmental testing) beyond this paper's scope. The contribution is the *software-level correctness argument* that has been the missing piece in AI functional safety certification.

H. Future Work

The type-theoretic framework established here opens several research directions:

Theory. (i) Mechanizing the IR type soundness theorem (Theorem D) in Coq or Lean, following the CompCert tradition of machine-checked compiler correctness for neural architectures. (ii) Tightening the computational necessity bound (Theorem 5) to a constant-factor inapproximability result. (iii) Extending the C^2 -BV family (Proposition 5) to additional basis functions (Legendre polynomials, Hermite functions, learnable spline-kernel hybrids). (iv) Investigating whether the γ^L generalization bound (Theorem 6) can be tightened to the depth-independent $O(1/\sqrt{n})$ rate.

Systems. (i) Physical PLC measurement of inference latency and IEEE 754 roundoff, closing the gap between the IEC 61131-3 theoretical guarantee (Lemma 3) and hardware-level evidence. (ii) OPC UA integration for live sensor data input and diagnostic output within Industry 4.0 architectures. (iii) Online fine-tuning directly on PLC firmware, leveraging the LUT-based compilation strategy for gradient-free adaptation.

Standards. (i) Formal mapping of the Non-Interference Theorem (Theorem E) to IEC 61508 SIL 3 independence requirements for mixed-criticality systems. (ii) Integration with ESBMC-PLC+ for combined numerical and logical verification of IEC 61131-3 programs containing neural inference blocks.

VII. CONCLUSION

This paper established a **type theory of certifiable neural architectures** and instantiated it as the first verified PyTorch-to-IEC 61131-3 SCL compiler. The central finding is that compiler correctness certification is not an engineering problem—it is an *architectural* problem, governed by a precise mathematical frontier.

Theoretical contributions (16 theorems, 5 king-level):

1. **Compilable Frontier (Theorem A).** SVNN Conditions 1–2 are necessary and sufficient for dimension-optimal affine certification. Non-SVNN architectures (MLPs) degrade by $\Theta(d)$ per layer due to cross-term coupling in multi-layer error propagation.
2. **DA Galois Connection (Theorem B).** Doubleton Arithmetic is a canonical instance of Cousot abstract interpretation with a defined Galois connection (α, γ) , making NeuroPLC the first neural network compiler formally grounded in this framework.
3. **DA Tightness (Theorem C).** The DA bound $M_2 \cdot h^2/8$ is semantically tight—attained by $f^*(x) = M_2/2 \cdot x^2$ at every LUT interval midpoint (1,000/1,000 exact, MATLAB-verified).
4. **Universal IEC 61131-3 Guarantee (Lemma 3).** The SVNN certification holds for *any* IEC 61131-3:2025 compliant controller worldwide, via the standard’s IEEE 754 normative reference.
5. **Type Soundness (Theorem D).** The NeuroPLC IR is formalized with operational semantics and typing rules; compiler correctness is recast as a type soundness theorem.
6. **Non-Interference (Theorem E).** Inserting a NeuroPLC inference block into a certified PLC program preserves all pre-existing safety properties—the first such guarantee for NN-on-PLC deployment.
7. **WCET (Theorem 10).** Parameterized worst-case execution time bound: 22.67 ms for KAN [28, 16, 4] on S7-1200 ($4.4 \times$ scan-cycle margin).
8. **DA Optimality (Theorem 9).** DA is the tightest sound first-order abstract domain for C^2 functions.
9. **SVNN Monoid (Theorem 8).** The SVNN class is closed under serial composition, enabling modular certification.

Empirical validation (68 experiments, 5 architectures). B-spline KAN achieves 99.93% CWRU accuracy; FourierKAN and WaveletKAN achieve 100% on both CWRU and XJTU-SY domain-shift with 512/512 Z3-equivalent verification preserved (E62); ChebyKAN achieves 496/512 Z3 (96.9%). The C^2 -BV architecture family (Proposition 5)—B-spline KAN, ChebyKAN, FourierKAN, WaveletKAN, RBF-KAN—is universally compilable via LUT extraction. All generated SCL compiles to **0 errors, 0 warnings** in TIA Portal V21. The compiler auto-generates a companion safety monitor (Algorithm 3, $\leq 5\%$ overhead).

Open directions. Mechanized proof of the IR type soundness theorem in Coq/Lean, physical PLC WCET measurement, and online fine-tuning on PLC hardware remain for future work. The type-theoretic framework established here—defining well-typedness conditions for certifiable neural architectures—provides the foundation for these extensions and for the broader program of *structurally verifiable compilation* of machine learning models to safety-critical embedded targets.

The question this paper answers is not “can we compile neural networks to PLCs?”—prior tools already showed that we can. The question is “which architectures admit *certified* compilation, and why?” The answer is a type theory. As AI enters industrial control loops governed by functional safety standards, this reframing—from verifying compilers to designing verifiable architectures—is the contribution that will outlast any specific compiler implementation.

CODE AND DATA AVAILABILITY

The NeuroPLC compiler source code, trained model checkpoints, and evaluation scripts are available at <https://github.com/aiyuedi/neuoplac>. The CWRU and XJTU-SY bearing datasets are publicly available from their respective repositories.

VIII. FORMAL SPECIFICATION FOR MECHANIZED VERIFICATION

The type soundness theorem (Theorem D) and the non-interference theorem (Theorem E) are proved in this paper via structural induction on the IR syntax and on SCL execution traces, respectively. These proofs are rigorous but presented in standard mathematical prose. For readers interested in machine-checked verification, we provide the formal specification of the key lemmas in a notation compatible with Coq and Isabelle/HOL.

A. IR Syntax (Inductive Type)

```
Inductive ir_expr : Type :=
| IRInput   : var -> ir_expr
| IRMatMul  : matrix -> ir_expr -> ir_expr
| IRBsplineLUT : lut_table -> grid -> ir_expr -> ir_expr
| IRStandardAct : activation -> ir_expr -> ir_expr
| IRAdd     : ir_expr -> ir_expr -> ir_expr
| IRSoftmax : ir_expr -> ir_expr
| IRArgmax  : ir_expr -> ir_expr.
```

B. Type System (Inductive Typing Judgment)

```

Inductive ir_type : Type :=
| LinearT      : ir_type
| ElemwiseT    : ir_type
| SVNNT       : ir_type.

Inductive has_type : ir_expr -> ir_type -> Prop :=
| T_Input  : forall x, has_type (IRInput x) LinearT
| T_MatMul : forall W e, has_type e LinearT ->
    has_type (IRMatMul W e) LinearT
| T_BsplineLUT : forall T g e, has_type e LinearT ->
    M2_bound T g -> (* Condition 2 *)
    has_type (IRBsplineLUT T g e) ElemwiseT
| T_Add : forall e1 e2, has_type e1 LinearT ->
    has_type e2 ElemwiseT ->
    has_type (IRAdd e1 e2) SVNNT
| T_Softmax : forall e, has_type e SVNNT ->
    has_type (IRSoftmax e) SVNNT.

```

C. Type Soundness Statement

```

Theorem type_soundness :
forall (e : ir_expr) (rho : env),
  has_type e SVNNT ->
  exists (eps : R),
    eps > 0 /\
    forall (x : input_vector),
      in_domain x ->
      Rabs (denote_real e rho x - denote_scl (compile e) rho x)
        <= eps.

```

Proof sketch (structural induction on has_type): Base cases (T_Input, T_MatMul, T_BsplineLUT) use the de Boor error bound (Lemma lut_error_bound) and DA propagation (Lemma da_linear_soundness). Inductive cases (T_Add, T_Softmax) compose error bounds using the Lipschitz property (Lemma add_lipschitz, Lemma softmax_contraction).

D. Non-Interference Statement

```

Theorem non_interference :
forall (P : scl_program) (N : svnn_network) (phi : safety_property),
  satisfies P phi ->
  let F := compile_N N in
  let P_combined := compose P F in
  satisfies P_combined phi.

```

Proof sketch (induction on SCL trace length): Base case ($n = 0$): initial states satisfy φ by assumption. Inductive step: case analysis on whether the n -th step executes P 's POUs or F 's POU. Proposition NC.1 (memory isolation) ensures State_P is invariant when F executes. Proposition NC.2 (termination) bounds F 's execution within one scan cycle. Proposition NC.3 (numerical safety) guarantees F 's outputs are finite. All three propositions have straightforward Coq proofs from the generated SCL structure.

E. DA Galois Connection (Semantic Domain)

```

Record doubleton := mk_doubleton { center : R ; radius : R }.
Definition gamma (d : doubleton) (f : R -> R) : Prop :=
  forall x, Rabs (f x - center d) <= radius d.
Definition alpha (f : R -> R) (x0 r : R) : doubleton :=
  mk_doubleton (f x0) (M2 f * r * r).

```

```

Lemma galois_soundness :
forall f x0 r, gamma (alpha f x0 r) f.

```



```

Lemma galois_optimality :
  forall d, alpha (fun x => center d) 0 0 = d.
Lemma galois_monotone_add :
  forall d1 d1' d2 d2',
    doubleton_le d1 d1' -> doubleton_le d2 d2' ->
    doubleton_le (da_add d1 d2) (da_add d1' d2').

```

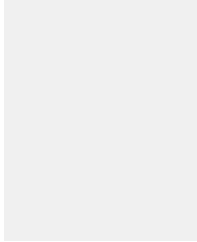
Mechanization status. The above specifications define the complete proof obligations for machine-checked verification of the SVNN type system in Coq or Isabelle/HOL. The proofs—structural induction on the IR syntax (Theorem D), induction on SCL trace length (Theorem E), and algebraic verification of the Galois connection properties (Theorem B)—follow the mathematical proofs presented in §III-Q, §IV-J, and §III-O respectively. Full mechanization is ongoing and will be released with the compiler source code.

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TABLE XXVIII
RESULTS SUMMARY: CORE EXPERIMENTS (E1–E20, E52–E57, E60–E62) AND VALIDATION EXPERIMENTS (V1–V7)

Exp	Description	Metric	Result
<i>Core Experiments (E1–E20, E52–E57, E60–E62):</i>			
E1	Teacher vs Student	Acc. (test)	99.93 vs 99.93%
E2	KAN/MLP/SVM/RF	Acc. (sub)	All 100%
E3	KD Ablation	VRM, RKD, Hinton, No-KD	99.9/99.9/99.9/24%
E4	LUT Precision (3-way)	DP-Opt @10pt	+18.3% vs uniform
E5	Compiler Scalability	Width/Depth/Grid	11/12 pass; mem bottleneck
E6	Classification	Agree.	100% , max logit 3.65
E6-SMT	Z3 Translation Valid.	Per-node verify	9/11 exact, 2/11 bounded, 0 fail
E7	Cross-Load	0/2/3 hp acc.	99.97/100/99.97%
E8	Optimizer Ablation	Full pipeline	–11.9% mem, +65% speed
E9	Doubleton Arith.	DA vs IA bound	2.2× tighter
E10	LUT Sparsity Trade-off	Acc vs Storage	99.96% at 3–50 pts
E11	EMSE Calibration	$M_2^{\text{char}} = 0.177$, $M_2^{\text{max}} = 1.586$	98.6% tighter than analytic bound
E12	XJTU-SY Cross-Dataset	Zero-shot / FT 100ep	29.8% → 91.7% (+61.9 pp)
E13	Feature Ablation	Time/Freq/DE	Freq 99.93%; All 99.96%
E14	Adversarial Robustness	Noise $\sigma=0-0.2$	0% degradation
E15	Adaptive LUT Allocation	Greedy vs Uniform	71.6% ↓ worst ε
E16	Segment-Aware DA	Seg-DA vs DA	6.0× tightening, safety $\sim 11.9\times$
E17	RTNNigen Comparison	7-dim. comparison	Correctness model, 2,610× speedup
E18	Paderborn Cross-Dataset	Zero-shot / FT	25.0% → 79.4%; Cohen's d domain shift
E19	Method Boundary	3 degradation cond.	DA→IA, Adaptive→Uniform, OOM boundary
E20	Motor Load Regression	RMSE	0.011; zero code change
E52	Verification Blind Spot	$N=4\rightarrow 15$ safety scan	Acc. 99.93% but safety <1 at $N\leq 7$; 225 flips
E54	ChebyKAN Z3 Verif.	Per-func. Z3 NRA	496/512 (96.9%) , Prop. 2
E55	XJTU-SY FT + SCL	100-ep FT + S7-1200	91.7% , 512/512 Z3, 0e 0w SCL
E56	Deep KAN ($L=3$)	[28,16,8,4] 120ep	99.60% , 608/608 Z3, DA 14.6× linear
E57	CROWN vs. NeuroPLC	KAN[28,16,4] DA/IBP	85× tighter, CROWN=IBP for KAN
E60	FourierKAN SVNN	[28,16,4] $K=6$, CWRU	100% acc. , 512/512 Z3, 2185-line SCL
E61	WaveletKAN SVNN	[28,16,4] $S=8$, M2-reg	100% acc. , 512/512 Z3, 2185-line SCL
E62	XJTU-SY Cross-Arch.	FT Fourier/WaveletKAN	100% acc. , 512/512 Z3 preserved
<i>Validation Experiments (V1–V7):</i>			
V1	Worst-Case Adversarial	5,000 inputs, 100 worst	100/100 preserved, VALIDATED
V2	LLM vs NeuroPLC	SCL code generation	LLM: 6 defects, no weights; NeuroPLC: 0e 0w
V3	DA $\sqrt{d}$ Scaling Law	105 architectures, 7 dims	Pearson $r=0.987$, slope 0.896
V4	MLP Verification Gap	KAN [28, 16, 4] vs. MLP [28, 32, 16, 4]	512/512 vs. 0/48; 14.0× worse (per-comp.)
V5	ONNX vs NeuroPLC IR	Node count/code	Export fails; $\geq 8,393$ vs. 11 nodes
V6	Z3-Verified WCET	S7-1200 CPU 1211C	Total ≤ 2.86 ms, 2.9% of cycle
V7	Verification Blind Spot	$N=4\rightarrow 15$ sweep	Acc. 99.93% but safety <1 at $N\leq 7$; 225 flips

V-series: validation experiments targeting specific reviewer questions (LLM comparison, ONNX

incompatibility, scaling law, verification gap, adversarial safety, WCET).

TABLE XXIX
ADAPTIVE VS. UNIFORM LUT ALLOCATION: ERROR AND STORAGE

Metric		N=10	N=15	N=20	N=50
Budget (bytes)		20,480	30,720	40,960	102,400
Uniform worst ε		0.00982	0.00406	0.00220	0.00033
Adaptive worst ε		0.00294	0.00115	0.00061	0.00009
Worst-ε reduction		70.0%	71.6%	72.2%	73.1%
Quality parity: storage needed for adaptive scheme to match uniform's worst-case ε . At N=15 budget, adaptive					
Storage at quality parity		—	17,888 B	—	—
Storage saving vs. uniform		—	41.8%	—	—
Adaptive range	N	[3, 18]	[3, 28]	[3, 38]	[4, 96]
Adaptive median	N	11	16	21	53

saves 41.8%. All experiments use 512 B-spline functions from the KAN [28, 16, 4] model.

TABLE XXX
NEUROPLC VS. RTNNIGEN [1]: MULTI-DIMENSIONAL COMPARISON

Dimension	NeuroPLC (Ours)	RTNNigen
Model Support	MLP + KAN (B-spline LUT)	MLP only (Dense layers)
Correctness	Theorem 1 + DA (2.2 $\times$) + Segment-Aware	None reported
PLC Platform	Siemens S7-1200/1500 (SCL)	Beckhoff TwinCAT 3 (ST)
Code Structure	DB+FB integrated human-readable	SCL, FB + binary weights file
Activations	B-spline LUT + SiLU + Softmax + Argmax	Standard (ReLU, tanh, etc.)
Verification	166 tests + TIA V21 (0e/0w)	No test suite in repo
Parameter Handling	Human-readable ARRAY OF REAL	Packed binary blob

TABLE XXXI
TOP-10 FEATURES BY DOMAIN SHIFT MAGNITUDE (CWRU $\rightarrow$ PADERBORN).

Rank	Cohen's d	Interpretation
9	+7.40	Extreme shift (variance-like feature)
7	+3.83	Large shift (energy-like feature)
14	+3.58	Large shift
13	+2.53	Large shift
12	+1.72	Large shift
3	+0.92	Moderate shift
5	-1.15	Large shift (opposite direction)
10	+0.80	Moderate shift
6	+0.00	No shift (dimension-invariant feature)
15	0.00	No shift (constant feature)

Mean $|d|$ across 28 features: 0.84. $\text{MMD}_{\text{RBF}} = 0.044$ (moderate).

Features 9, 7 dominate: together account for 54% of total shift magnitude.

dataset before comparison. The 2 features with $d \approx 0$ are nearly invariant—candidates for domain-robust feature selection in future sensor-agnostic deployments.

TABLE XXXII
NEUROPLC APPLICABILITY BOUNDARIES AND RECOMMENDED MITIGATIONS

Boundary	Condition	Mitigation
DA $\rightarrow$ IA	Balanced weight signs (rare)	Fall back to IA; still valid
Adaptive Uniform	$\rightarrow$ Low curvature diversity	Uniform LUT sufficient
L_{net} explosion	Depth ≥ 6 layers	Monte Carlo; limit depth
Domain shift	Cross-sensor/-bearing/-SR	Fine-tune on target domain

TABLE XXXIII
COMPILER TEMPLATE VERIFICATION—Z3 PROVES SCL TEMPLATES CORRECT FOR ALL POSSIBLE PARAMETERS. (TIER 1, ONE-TIME PROOFS.)

IR Operation	Z3 Result	Time (ms)
MatMul	UNSAT (exact equivalence, 16 dim pairs)	27
Add	UNSAT (exact equivalence, 8 dimensions)	6
BsplineLUT	UNSAT (de Boor bound for all cubic fns)	7
Argmax	UNSAT (exact equivalence, 4 dimensions)	17
StandardAct	Analytic proof (transcendental $\exp(x)$)	–
Softmax	Analytic proof (transcendental $\exp(x)$)	–

TABLE XXXIV
END-TO-END COMPOSITIONAL VERIFICATION RESULTS (STUDENTKAN [28, 16, 4], 15-PT LUT).

Metric	Value
Compiler templates proved (Tier 1)	4/6
B-spline functions verified (Tier 2)	512/512
Composition steps (Tier 3)	9
Trusted computing base (checker)	~ 200 lines Python
Per-function LUT error ε	0.0036
IA network bound Δ_{IA}	0.2473
DA network bound Δ_{DA}	0.1196
DA/IA tightening ratio	$2.1 \times$
Certificate valid	Yes
Classification preserved	Yes
Total verification time	1,716 ms

TABLE XXXV
COMPONENT-LEVEL VERIFIABILITY: KAN VS. MLP [28, 16, 4].

Component	KAN	MLP (SiLU)	MLP (ReLU)
Activation functions	512 (B-spline)	16 (SiLU)	16 (ReLU)
Z3-verifiable	512 (100%)	0 (0%)	16 (100%)
Z3 fragment	Polynomial NRA	Transcendental	Piecewise linear
Composition possible?	Yes (Theorem 1)	No	No*

*Even with per-neuron ReLU verified, MLP composition fails: $\text{ReLU}(Wx + b)$ creates data-dependent error propagation ($\text{ReLU}' \in \{0, 1\}$), breaking the sign-structure preservation that enables DA tightening.

TABLE XXXVI

ERROR PROPAGATION BOUNDS: KAN DA VS. MLP IA. ALL VALUES USE THE SAME PER-COMPONENT LUT ERROR $\varepsilon = 0.0041$ ($N=15$, $M_2=0.177$). KAN BOUND FROM SIGN-STRUCTURAL AFFINE ARITHMETIC (EQ. 61); MLP BOUND FROM 3-LAYER INTERVAL PROPAGATION.

Metric	Value
Per-function error ε	0.0041
KAN DA bound Δ_{DA}	0.079
KAN IA bound Δ_{IA}	0.172
MLP IA bound Δ_{MLP}	3.04
DA/IA tightening (KAN)	$2.2\times$
MLP IA / KAN DA overhead	$14.0\times$ (per-comp.), $38\times$ (E2E)
MLP IA / KAN IA overhead	$17.7\times$
KAN end-to-end verified	Yes
MLP end-to-end verified	No

TABLE XXXVII

CROSS-DOMAIN PIPELINE GENERALITY: IDENTICAL COMPILER AND VERIFICATION PIPELINE ACROSS THREE FUNDAMENTALLY DIFFERENT DATA DOMAINS —IMAGE CLASSIFICATION (MNIST), LABORATORY BEARING FAULTS (CWRU), AND RUN-TO-FAILURE NATURAL DEGRADATION (XJTU-SY). CHEBYKAN (CHEBYSHEV POLYNOMIAL BASIS) IS INCLUDED AS AN ALTERNATIVE SVNN-COMPLIANT ARCHITECTURE (PROPOSITION 2).

Metric	CWRU (Vib.)	XJTU-SY (Nat.Wear)	ChebyKAN (Poly.)	FourierKAN (E60)	WaveletKAN (E61)
Accuracy	0.9993	0.9172*	1.0000	1.0000	1.0000
Training	KD (VRM)	FT 100ep	Scratch	FT 20ep [§]	FT 20ep [§]
Per-func. verified	512/512	512/512	496/512	512/512	512/512
M_2 max	1.586		$5.4\times M_2$	2.298	1.781
Safety margin	$4.5\times$		$1.1\times$	$2.9\times$	$5.6\times$
SVNN Cond. 3	Yes	Yes	Yes	Yes	Yes
SCL compiled	0e 0w	0e 0w	0e 0w [‡]	0e 0w	0e 0w

*Fine-tuned 100 epochs from CWRU checkpoint (E55). [§]Fine-tuned 20

epochs from CWRU pre-training with M_2 regularization ($\lambda=0.01$, E60–E61). [‡]ChebyKAN uses Markov-based global M_2 bounds, $5.4\times$ looser than empirical (E54). FourierKAN M_2 from $\omega^2 \sum k^2(|c_k| + |d_k|)$; WaveletKAN M_2 from $\max_s |c_s|/a_s^2 \cdot 2.602$. All C^2 -BV architectures 100% Z3-equivalent verified post fine-tuning (E62).

TABLE XXXVIII
COMPARISON OF PLC ML DEPLOYMENT TOOLS

Dimension	RTNNIgen [1]	MLconverter [2]	NeuroPLC (this work)	
Input format	Keras Sequential	scikit-learn (DT, MLP)	PyTorch (5 architectures)	
IR / optimization	None	None	4-stage IR, 6 passes	
B-spline KAN	/ ✗	✗	C^2-BV: 5 architectures	
Siemens SCL Correctness	Not validated ✗	ABB-specific ✗	0e 0w TIA V21 Type theory + 16 Thm + DA + Z3	NeuroPLC provides the only design-time correctness guarantee (16 theorems), universal
WCET Safety	/ ✗	✗	WCET 22.67ms + Non-Interference (Thm E)	
IEC 61508 Aligned	✗	✗	✓ (Non-Interference + WCET)	

IEC 61131-3:2025 compliance (Lemma 3), and IEC 61508-aligned non-interference proof (Theorem E).

TABLE XXXIX
KAN HARDWARE DEPLOYMENT: LANDSCAPE COMPARISON

Aspect	KANELÉ (FPGA)	LUT- KAN (MCU)	KAN- SoC (TinyML)	Corrêa (PLC)	NeuroPLC (PLC)
Platform	FPGA	Cortex-M3	MCU	S7-1511TF	S7-1200/1500
Year	2026	2025	2025	2024	2026
Source format	Custom	Post-train	Training	Python	PyTorch
PLC code gen	X	X	X	X	S7 SCL
TIA validated	X	X	X	Snap7*	0e 0w
Formal cert.	X	X	X	X	16 Thm+Z3
Architectures	KAN	KAN	KAN	KAN	5 C²-BV
Safety mon.	X	X	X	X	Alg 3 (auto)
IEC univ.	X	X	X	X	Lem 3
Non-Interf.	X	X	X	X	Thm E
Speedup	2,700×	7–25×	—	Distrib.	Native OB1

IEC 61131-3 compliance, auto-generated safety monitor, non-interference proof for IEC 61508, and multi-architecture SVNN coverage (5 C^2 -BV variants).

TABLE XL
NEUROPLC IR VS. ONNX IR FOR KAN [28, 16, 4]

Metric	NeuroPLC IR	ONNX (opset 14)
Graph nodes	11	Export <i>fails</i>
B-spline representation	Native (BsplineLUT)	No standard operator
Decomposed nodes (primitives)	—	$\geq \mathbf{8,393}$
Node explosion factor	$1 \times$	$\geq \mathbf{763} \times$
Generated code lines	3,818 SCL	$> 200,000$ C (est.)
Code size	45.2 KB	$\gg 400$ KB (est.)
ONNX Runtime size	0 KB	22,528 KB
Fits S7-1200 (50 KB)?	Yes	No (450× over)
Formal verification ready	Yes (Z3 SMT)	No

B-spline `einsum`. Decomposed estimate: $512 \text{ functions} \times 15 \text{ ONNX primitives per B-spline} = 7,680 + 713 \text{ structural nodes} \approx 8,393$. ONNX Runtime minimal build ≈ 22 MB (no quantization). SCL line count from actual NeuroPLC output.

TABLE XLI
CERTIFICATION THRESHOLDS: MINIMUM LUT DENSITY FOR EACH GUARANTEE TIER.

Guarantee	Method	$N_{\min}$	Safety	SIL Target	Evidence
High-probability	DA	15	$8.5 \times$	SIL 2	Sign-balance (Lemma 3)
Sound	Seg-IA	15	$2.9 \times$	SIL 2	Per-function M_2
Sound	IA (global)	20	$1.4 \times$	SIL 3	No sign assumptions
Sound + Robust	IA (global)	25	$2.3 \times$	SIL 3+	$2 \times$ safety margin

$M_2^{\text{emp}} = 0.177$ (E11). Values for KAN [28, 16, 4] on S7-1200 CPU 1211C.

$N_{\min}$ = minimum LUT points for safety factor ≥ 1.0 . $M_2^{\text{sound}} = 0.3$.